The Perceived Monetary Policy Rule: Identification and Macroeconomic Implications

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Abstract

This paper shows that estimating perceptions about the monetary policy rule from interest rate forecasts could understate the true belief due to the uncertainty about future policy rules. In contrast, interest rate nowcasts, the prediction of the current period's interest rate, can recover the perceived rule without bias. Empirically, I find that forecasters' perceptions are consistent with the Taylor principle, which is rejected under the forecast-based measures of perceptions. Using a Subjective Perception Markov-Switching New Keynesian model, I show that equilibrium determinacy depends not on whether the central bank satisfies the Taylor principle, but on whether the public *believes* it does. Counterfactual exercises demonstrate that managing and accurately measuring policy perceptions is crucial for effective monetary policy.

Keywords: Perceived Monetary Policy Rule, Perception Channel, Indeterminacy, Perceived Taylor Principle, Markov-Switching New Keynesian Model

JEL Classification: E31, E52, E58, D84

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1 Introduction

Central to modern macroeconomic models is the role of agents' expectations, in which perceptions of the policy rule shape equilibrium dynamics. If agents perceive the Fed as hawkish, their inflation expectations become anchored, which in turn keeps realized inflation more stable. Yet, the Full Information Rational Expectations (FIRE) assumption underlying modern macroeconomic models presumes that agents perfectly observe the central bank's reaction function (e.g., the Taylor rule). Such a restrictive coincidence rendered the "perceived" rule by the public a secondary object of study, leaving its identification and macroeconomic implications largely underexplored. Motivated by this gap, this paper asks two questions. First, how can the perceived monetary policy rule be correctly identified from the survey data? Second, what are the macroeconomic implications when the perceived rule departs from the actual rule?

In answering these questions, this paper establishes that the *perceived monetary policy rule* is what ultimately governs equilibrium dynamics. The key contributions are threefold. First, I show that the conventional practice of estimating the perceived rule by running a Taylor-rule regression using interest rate *forecasts* is structurally biased due to uncertainty about the future policy rule within the forecast horizon. As a resolution, I propose using interest rate nowcasts, which do not suffer from this bias. Empirically, I find that the Taylor principle is satisfied under nowcast-based perception estimates from the survey data, but rejected by forecast-based estimates.

Second, I develop a *Subjective Perception Markov-Switching New Keynesian (SP-MSNK)* model in which the perceived Taylor rule differs from the actual rule implemented by the central bank. The model shows that equilibrium determinacy depends not on whether the central bank satisfies the Taylor principle, but on whether agents *believe* it does, referred to as the *perceived Taylor principle*. Third, the model shows that both managing and measuring perception matter for policy. I show that the Great Moderation, a period of reduced macroeconomic volatility in 80s and 90s, which is traditionally attributed to a shift in monetary policy regime, would not have been possible without an accompanying shift in perception. Furthermore, using the post-2020 inflation episode, I show that a central bank that measures the perceived rule from interest rate forecasts may misjudge the earlier and optimal timing of policy rate lift-off to be counterproductive.

The identification of a perceived policy rule is a non-trivial challenge. When agents understand both the general equilibrium feedback of interest rates and the possibility of a shift in the monetary policy rule (a.k.a., a regime shift), their expectations of future interest rates incorporate not only their expectations about future policy stance and the targeted variable (e.g., inflation) but also a joint, perceived covariance between the two. However, when an econometrician estimates a Taylor-rule-type regression using agents' forecasts of interest rates

and target variables, this perceived covariance creates an omitted variable bias. Using a textbook Markov-Switching New Keynesian model, I show that regressing interest rate *forecasts* on inflation forecasts yields a downward-biased estimate of the perceived policy stance as the omitted covariance and regressor (inflation expectation) are negatively correlated.[1] Intuitively, if an agent has inflation expectation above the target (positive inflation gap), he anticipates a hawkish regime shift would bring inflation down (negative covariance). Conversely, if an agent has inflation expectation below the target (negative inflation gap), he anticipates a more hawkish regime would bring inflation up to the target (positive covariance). As a resolution, I demonstrate that this bias vanishes in the limit of the nowcast, where the covariance disappears and allows for identification of the currently operative perceived rule.

Critically, the bias result is borne out in the data. Applying the TVP estimator to interest rate *forecasts* (rather than nowcasts) yields systematically attenuated coefficients, with the implied long-run inflation response falling below unity over the large portion of the sample period. Taken at face value, *forecast*-based estimates would lead the econometrician to conclude that the public perceives the Fed as being inconsistent with the Taylor principle, which would leave the anchoring of long-run inflation expectations, the stability of the Great Moderation, and the recent soft landing from the post-pandemic inflation surge unexplained. By contrast, the *nowcast*-based estimates recover an active perceived rule whose long-run inflation response is consistent with the Taylor principle, providing a unified account of all three patterns.[2]

The nowcast-based perceived rule estimates also reveal significant time variation and distinct divergences from the actual rule implemented by the Fed, as estimated from Greenbook/Tealbook forecasts (Greenbook or GB onward). Notably, during the mid-1990s, a period characterized by the Federal Reserve's "preemptive tightening" and "risk management" under Chairman Greenspan (Greenspan, 2004; Bernanke, 2022), the actual rule departed from the conventional Taylor rule. Interest rate decisions responded to tail risk in inflation and the Fed tightened interest rates despite stable Greenbook inflation expectations, producing statistically insignificant inflation response. In contrast, the perceived inflation rule stayed intact, as professional forecasters continued to believe in a strong and stable response to the mean inflation forecast.

To understand the macroeconomic implications of the perceived rule, I build a Subjective Perception Markov-Switching New Keynesian model (SP-MSNK). In this environment, agents form expectations based on a perceived monetary policy rule that may differ from the central bank's true rule. The model reorients the source of equilibrium determinacy, showing that the economy possesses a unique stable equilibrium only if the agents *believe* the monetary authority

[1] Unlike the simultaneity bias driven by monetary policy shocks, which can be addressed using instrumental variables, the covariance bias identified here is structural and endogenous. Consequently, standard instruments satisfying the exclusion restriction are unavailable. See Section 3 for a detailed discussion.

[2] Diercks, J. Kim, and Swanson (2026) use the Survey of Market Expectations and the bond market data and reach the same conclusion that the perceived rule remained consistent with the Taylor principle.

as satisfying the Taylor principle. This insight carries three important implications. First, concerns regarding the central bank's violation of the Taylor principle is overstated, provided the perception remains anchored. Second, the primary role of monetary policy to stabilize the economy is not to implement the Taylor principle per se, but to anchor public perceptions to it.

Finally, counterfactual simulations quantify the historical importance of perceptions and the value of measuring them accurately. I show that the Great Moderation (1986–1999) was underpinned by a strongly hawkish perception of monetary policy. Had the public perceived the Fed as dovish, both the level and volatility of inflation would have remained elevated despite the Fed's actual anti-inflationary stance. I next turn to the consequences of mismeasuring these perceptions, using the post-pandemic tightening cycle as an example. In a counterfactual in which the Fed lifted off four quarters earlier than the realized 2022Q1, calibrating the public's inflation perception from interest rate forecasts would overstate the cost of disinflation, leading to a misjudgment of the optimal timing of intervention. These findings suggest that the efficacy of monetary policy depends crucially on the public's perceptions of the monetary policy rule and on accurately measuring those perceptions.

Related Literature This paper contributes to multiple strands of literature. First, it contributes to the study of general equilibrium consequences of monetary policy perceptions. While Bauer, Pflueger, and Sunderam (2024) document the divergence between financial market perceptions and historical policy rules, their analysis is primarily restricted to asset pricing implications in a partial equilibrium setting. In contrast, this paper evaluates the general equilibrium consequences for inflation and output dynamics. Schorfheide (2005) and Cogley, Matthes, and Sbordone (2015) study a setup where private sector learns about monetary policy rule, but restrict the perceived object to the inflation target or to a particular form of the Taylor rule. Instead, this paper allows agents to hold beliefs about the full reaction function, including the output gap response and inertia, but abstracts from the learning process to focus on the consequences of perception, that is, what happens to equilibrium outcomes once a given perception is in place.[3] Hamilton, Pruitt, and Borger (2011) and Bocola et al. (2024) estimate perceptions about the monetary policy rule but analyze implications through a model in which actual and perceived rules coincide. This work departs from assuming such a coincidence and proposes a framework where agents' perceived rules differ from the actual rules.

Second, the paper advances the econometric identification of perceived monetary policy

[3]Other works have also investigated the implications of private sector learning for monetary policy rules (Evans and Honkapohja, 2003a; Evans and Honkapohja, 2003b; Bullard and Mitra, 2002). These papers mainly ask whether a given rational expectations equilibrium (REE) is learnable in a stable way. The REE under consideration equates the perceived and actual rules, attributing determinacy to the actual rule. This paper shows that the perceived rule is the ultimate source of equilibrium determinacy.

rules, clarifying a distinct source of bias from that of actual rule. A vast literature estimates the central bank's actual reaction function using historical realizations or Greenbook forecasts (Clarida, Gali, and Gertler, 2000; Orphanides, 2001; Boivin, 2005; C.-J. Kim and Nelson, 2006; Carvalho, Nechio, and Tristao, 2021). A different strand utilizes survey or asset-price data to infer market-implied rules (Hamilton, Pruitt, and Borger, 2011; Pierdzioch, Rülke, and Stadtmann, 2012; Bundick, 2015; Bauer, Pflueger, and Sunderam, 2024; Jia, Shen, and Zheng, 2023; Bocola et al., 2024). However, these latter studies typically run Taylor-rule type regressions, relying on interest rate forecasts under the implicit assumption of a constant rule or abstract from general equilibrium feedback of monetary policy into macroeconomic variables. I demonstrate that in an environment where agents internalize possible time-variation of policy rule and its feedback effect into the macroeconomy, standard forecast-based regressions are subject to a structural covariance bias. I provide a solution using nowcast data that properly identifies the real-time perceived reaction function.

Third, this work revisits the sources of equilibrium determinacy and the Great Moderation. A major view attributes the stabilization of the post-Volcker era to a shift in the *actual* conduct of monetary policy within a Full Information Rational Expectations (FIRE) framework (Clarida, Gali, and Gertler, 2000; Lubik and Schorfheide, 2004; Coibion and Gorodnichenko, 2011; Bianchi, 2013).[4] I reframe this narrative by showing that equilibrium determinacy is governed by the *perceived* policy rule. Consequently, the Great Moderation is reinterpreted not merely as a consequence of changes in policy actions, but also as a result of *"Good Perceived Policy"*.

Fourth, the paper extends the literature on regime-switching DSGE models. Previous work has extensively analyzed the implications of recurring regime shifts under full information rational expectations (FIRE) (Davig and Leeper, 2007; Bianchi, 2013; Barthélemy and Marx, 2017; Bianchi, Lettau, and Ludvigson, 2022). I contribute to this literature by relaxing the FIRE assumption and its strict coincidence between the actual and perceived Taylor rules and isolating the distinct channel through which perceived rule propagates.

Finally, this paper connects to recent works on the post-pandemic monetary policy and central bank credibility. Nakamura, Riblier, and Steinsson (2025) propose a *"minimalist Taylor principle"* in which the central bank responds in a shock-contingent manner while retaining freedom to look through supply shocks under anchored long-run expectations. Bocola et al. (2025) show that in an imperfect information environment, low-credibility central banks must respond more aggressively to inflationary shocks to prevent de-anchoring, while high-credibility central banks enjoy a *"reputation dividend"* permitting more flexible responses. This paper offers a complementary insight through a different angle: When the public's beliefs about the monetary policy rule are firmly anchored, the central bank's scope for deviating from the Taylor

[4]There are other papers such as Sims and Zha (2006) that have pushed forward an alternative view of "Good Luck" in this period rather than "Good Policy".

principle is expanded, as determinacy depends on the *"perceived Taylor principle"*.

Outline The paper proceeds as follows. Section 2 establishes the theoretical motivation. Section 3 analytically identifies the covariance bias in standard estimations and validates the nowcast-based solution via simulation. Section 4 reports the empirical time-varying estimates of the perceived rule and documents comparison with the forecast-based rules and the actual policy rule. Section 5 develops the SP-MSNK model to analyze general equilibrium implications, focusing on determinacy and historical counterfactuals. Section 6 concludes.

2 Motivating Framework: Perceived vs. Actual Rule

This section uses a simple model to build intuition for why the *perceived* policy rule, rather than the actual rule, governs equilibrium dynamics. Private agents hold a perceived Taylor rule that may differ from the actual rule implemented by the central bank.

The economy is governed by a Fisher equation and an exogenous real rate process r_t, which are structurally common to both the perceived and actual models:

$$i_t = E_t^P [\pi_{t+1}] + r_t \tag{1}$$

$$r_t = \rho_r r_{t-1} + v_t \tag{2}$$

where $E_t^P[\cdot]$ denotes the subjective expectation of the private sector implied by their perceived model. The distinction arises in the monetary policy rule. The private sector perceives the rule as:

$$i_t = \phi_p \pi_t \tag{3}$$

whereas the Fed actually implements:

$$i_t = \phi_f \pi_t \tag{4}$$

If $\phi_p = \phi_f$, the subjective expectation is consistent with the rational expectation ($E_t^P = \mathbb{E}_t$).

Perceived Law of Motion (PLM) The private sector agent is internally rational to her perceived model. Assuming $\phi_p > 1$, the Perceived Law of Motion (PLM) implied by equations (1)–(3) yields:

$$i_{t|t} = \frac{\phi_p}{\phi_p - \rho_r} r_t \qquad \pi_{t|t} = \frac{1}{\phi_p - \rho_r} r_t \tag{5}$$

The subscript $_{t|t}$ denotes the nowcast implied by the agent's subjective model. It represents the anticipated realization of endogenous variables consistent with the agent's subjective model. Although the agent perfectly observes the realized exogenous states (r_t), the misspecification of the policy rule leads to nowcasts that differ from the actual realization. Furthermore, the private sector forms expectations based on these perceived dynamics:

$$E_t^P[i_{t+h}] \equiv i_{t+h|t} = \frac{\rho_r^h \phi_p}{\phi_p - \rho_r} r_t \qquad E_t^P[\pi_{t+h}] \equiv \pi_{t+h|t} = \frac{\rho_r^h}{\phi_p - \rho_r} r_t \tag{6}$$

This equation demonstrates that inflation expectations are inversely related to the perceived policy aggressiveness ϕ_p.

Actual Law of Motion (ALM) Given the subjective expectation (6), combining the actual structural equations (1), (2), and (4) yields the Actual Law of Motion (ALM):

$$i_t = \frac{\phi_p}{\phi_p - \rho_r} r_t \tag{7}$$

$$\pi_t = \frac{\phi_p}{\phi_f} \underbrace{\left[\frac{1}{\phi_p - \rho_r} r_t \right]}_{\text{FIRE}} = \frac{1}{\phi_f} \left[\underbrace{\left(\frac{\rho_r}{\phi_p - \rho_r} r_t \right)}_{=E_t^P[\pi_{t+1}]} + r_t \right] \tag{8}$$

In this simple model, the actual interest rate dynamics (7) coincide with the perceived dynamics (5) because the subjective expectation in the Fisher equation is determined exogenously by the perceived model parameters.[5] However, equation (8) reveals that when $\phi_p \neq \phi_f$, equilibrium inflation dynamics diverge from the agent's perceived law of motion. The term underbraced as FIRE indicates the solution under Full Information Rational Expectations, obtained when the perception is consistent with the actual rule ($\phi_p = \phi_f$). Specifically, if the actual rule falls short of the private sector's perceived rule ($\phi_p > \phi_f$), the impact of structural shocks on inflation is amplified relative to the full-information outcome.

Several insights from this model merit discussion. First, the equilibrium concept employed is within the confines of Temporary Equilibrium (TE, J. M. Grandmont, 1977; J.-M. Grandmont, 1987; Woodford, 2013) where the equilibrium is defined on the basis of the expectations that decision makers happen to entertain at each point in time. It shares resemblance with the other branches of TE, such as the restricted perception equilibrium (Sargent, 1991; Branch, 2006) and the internal rationality (Adam and Marcet, 2011), where agents optimize and form expectations subject to a PLM that differs from the implied ALM.[6] This concept is extended

[5] If there were other endogenous components, such as in a standard IS curve, the interest rate nowcast would also differ from the realization.

[6] RPE typically emerges from under-parameterization, hidden variables, or linear approximations to nonlinear dynamics. While the baseline model shares a resemblance that agents correctly understand the model but

in Section 5 to a general equilibrium environment featuring a regime-switching in the monetary policy rule.

Second, equilibrium determinacy depends on the Taylor principle of the *perceived* rule ($\phi_p > 1$), rather than the actual rule (ϕ_f). Since the model relies on subjective expectations, a unique solution must exist for the subjective model, satisfying the Blanchard and Kahn (1980) condition. Consequently, a unique equilibrium exists even if the actual policy rule is passive ($\phi_f < 1$), provided the perceived Taylor principle is satisfied. The standard FIRE framework is just a special case in which $\phi_p = \phi_f > 1$, explaining why the Taylor principle is conventionally stated in terms of the actual policy rule.

Third, the decomposition in equation (8) reveals how ϕ_p and ϕ_f affect the transmission of structural shocks. The **Perception Channel**, governed by the perceived parameter ϕ_p, shapes macroeconomic dynamics via the private sector's forward-looking expectations ($E_t^P[\pi_{t+1}]$). The **Direct Policy Channel**, governed by the actual parameter ϕ_f, affects inflation by mediating the pass-through of inflation expectation and the contemporaneous effect of the shock. This underscores that the interaction between the contemporaneous policy stance and private sector perceptions about the rule jointly shape macroeconomic dynamics.

Fourth, a natural question is why the discrepancy between perceived and actual rules may persist. While learning seems straightforward in this simple model, inferring the central bank's true reaction function in practice is a challenging task. Agents must infer the precise rule specification as well as the correct inputs, such as whether the Fed reacts to realized outcomes or internal forecasts that are confidential or published with lags. Furthermore, policymakers' preferences are subject to change over time depending on economic environments. This could prevent the perceived rule from perfectly tracking the actual rule.

Lastly, the model clarifies why the conventional identification strategy of using interest rate forecasts appears valid, and where it breaks down. Dividing interest rate expectation by inflation expectation in (6) yields

$$\frac{i_{t+h|t}}{\pi_{t+h|t}} = \phi_p \qquad \forall h \geq 0 \tag{9}$$

Regressing the agent's interest rate forecast on her inflation forecast therefore recovers the perceived rule regardless of the horizon. This justifies the conventional practice of inferring perceived rules from interest rate expectations. However, the practice rests on the assumption of a constant rule. Once relaxed, interest rate forecasts ($h > 0$) no longer identify the perceived rule. Section 3 formalizes this and shows that only interest rate nowcasts ($h = 0$) remain valid.

misperceive the parameter. The key difference is that RPE agents eventually learn and converge to an optimal model within the class of misspecified models, whereas I assume agents' perceptions remain fixed throughout.

3 Identification of the Perceived Rule

3.1 Identification Challenge

For expositional clarity, the motivating framework in Section 2 assumed a constant monetary policy rule. However, a vast empirical literature documents substantial time variation in the monetary policy reaction function (C.-J. Kim and Nelson, 2006; Boivin, 2005; Coibion and Gorodnichenko, 2011), or discrete regime switching (Sims and Zha, 2006; Bianchi, 2013; Bianchi, Lettau, and Ludvigson, 2022). Given this evidence, an identification strategy must account for forward-looking agents who internalize the probability of future policy shifts when forming expectations.

Consider a simple contemporaneous Taylor rule with time-varying parameters:

$$i_t = \phi_{\pi,t}\pi_t + \varepsilon_{M,t} \tag{10}$$

The object of interest is the agent's perception of the policy coefficient $\phi_{\pi,t}$, defined at horizon h as $E_t^P[\phi_{\pi,t+h}]$. At the nowcast horizon ($h = 0$), this represents the rule ($\phi_{\pi,t|t}$) that the agent currently perceives. While I proceed under the simplifying assumption of full observability ($\phi_{\pi,t|t} = \phi_{\pi,t}$), the distinction between perfect and imperfect knowledge at $h = 0$ is immaterial for the derivation and identification of the bias.[7]

Assuming $E_t^P[\varepsilon_{i,t+h}] = 0$, taking the subjective conditional expectation of equation (10) at time t for horizon $t + h$ and decomposing the expectation of the product yields:

$$E_t^P[i_{t+h}] = E_t^P[\phi_{\pi,t+h}]E_t^P[\pi_{t+h}] + \mathrm{Cov}_t^P\left(\phi_{\pi,t+h}, \pi_{t+h}\right). \tag{11}$$

Empirically, the perceived reaction function is typically estimated by projecting expected rates onto expected inflation using OLS or IV (Bauer, Pflueger, and Sunderam, 2024; Jia, Shen, and Zheng, 2023; Bocola et al., 2024). This implies that $\mathrm{Cov}_t^P\left(\phi_{\pi,t+h}, \pi_{t+h}\right)$ will be an omitted variable in the regression and the direction of bias depends on the sign of correlation between $\mathrm{Cov}_t^P\left(\phi_{\pi,t+h}, \pi_{t+h}\right)$ and $E_t^P[\pi_{t+h}]$.

To investigate further, it is useful to define the time-varying coefficient, $\beta_{t,h}$, that describes the linear relationship between expected rates and inflation:

$$E_t^P[i_{t+h}] = \beta_{t,h}E_t^P[\pi_{t+h}], \text{ where } \beta_{t,h} = E_t^P[\phi_{\pi,t+h}] + \frac{\mathrm{Cov}_t^P\left(\phi_{\pi,t+h}, \pi_{t+h}\right)}{E_t^P[\pi_{t+h}]} \tag{12}$$

[7] Under signal extraction, where forecasters hold uncertainty over the current regime s_t, a within-period covariance between ϕ_{π,s_t} and π_t may survive even at the nowcast horizon. Appendix D.1 addresses this concern empirically by re-estimating the perceived rule using only the quarter-end BCEI observation, when forecasters' uncertainty over s_t is supposedly smallest, and shows that the estimates are essentially unchanged.

Equation (12) demonstrates that $\beta_{t,h}$ provides a biased measure of the true perceived policy aggressiveness $E_t^P[\phi_{\pi,t+h}]$. The wedge is driven by the conditional covariance between the policy parameter and inflation, scaled by the level of expected inflation. Thus, the direction of the bias depends on the sign of this covariance relative to expected inflation. In the analysis that follows, the notation $x_{t+h|t} \equiv E_t^P[x_{t+h}]$ is used interchangeably to denote the subjective conditional expectation. In contrast $\mathbb{E}[\cdot]$ without a time subscript denotes the mathematical expectation that represents unconditional expectation taken by the econometrician over the sample.

Under this setup, if the perceived rule is estimated using standard regression techniques, this perceived covariance translates into a bias in the OLS estimator.

Proposition 1. *Assume the data generating process for the forecasted interest rate is given by* $i_{t+h|t} = \beta_{t,h}\pi_{t+h|t}$. *If* $\mathbb{E}[\pi_{t+h|t}] = 0$, *the OLS estimator* β_{OLS} *decomposes as:*

$$\beta_{OLS,h} = \mathbb{E}[\beta_{t,h}] + \frac{\mathbb{E}\left[\left(\beta_{t,h} - \mathbb{E}\left[\beta_{t,h}\right]\right)\pi_{t+h|t}^2\right]}{\text{Var}\left(\pi_{t+h|t}\right)} \tag{13}$$

Proof. See Appendix A.

The first term in the decomposition represents the unconditional mean of the time-varying coefficient $\beta_{t,h}$. The second term reflects variance-adjusted weighting: because OLS seeks to minimize the sum of squared errors, the estimator effectively assigns greater weight to $\beta_{t,h}$ associated with periods of high inflation variance. Numerically, however, this variance-weighting effect proves to be negligible, and for intuition, suppose that $\beta_{OLS,h} \approx \mathbb{E}[\beta_{t,h}]$. Consequently, substituting the definition of $\beta_{t,h}$ from (12), the estimated coefficient approximates:

$$\beta_{OLS,h} \approx \mathbb{E}\left[\phi_{\pi,t+h}\right] + \mathbb{E}\left\{\frac{\text{Cov}_t^P\left[\phi_{\pi,t+h}, \pi_{t+h}\right]}{E_t^P[\pi_{t+h}]}\right\} \tag{14}$$

This decomposition highlights again the structural drivers of the estimation bias: it is the perceived covariance between the policy parameter and inflation scaled by inflation expectation.

To explicitly characterize this covariance, I consider a general setup where the monetary policy rule follows a discrete regime-switching process ($\phi_{\pi,t} \equiv \phi_{\pi,s_t}$) with $s_t \in \{1, \ldots, S\}$. Inflation evolves according to a state-dependent linear system $\pi_t = \mathbf{C}_{\pi,s_t}\mathbf{X}_t$, with state variables following $\mathbf{X}_t = \mathbf{A}_{s_t}\mathbf{X}_{t-1} + \mathbf{B}_{s_t}\boldsymbol{\varepsilon}_t$. Restricting the analysis to a two-state Markov switching process for tractability yields the following proposition. Appendix B.2 shows that the bias discussed here also extends to endogenous regime switching, where transition probabilities depend on the inflation level.

Proposition 2. *Consider a two-state Markov switching process $s_t \in \{1, 2\}$ governing the policy parameter ϕ_{π,s_t} and inflation dynamics. The h-step ahead conditional covariance between the policy parameter and inflation, given the current state $s_t = j$, is:*

$$\text{Cov}^P\left(\phi_{\pi,s_{t+h}}, \pi_{t+h} \mid s_t = j\right) = P_{j1}^{(h)}\left(1 - P_{j1}^{(h)}\right)(\phi_{\pi,1} - \phi_{\pi,2})\left(\pi_{1,t+h}^e - \pi_{2,t+h}^e\right) \tag{15}$$

where $P_{jk}^{(h)} \equiv (\mathbf{P}^h)_{j,k}$ denotes the (j, k)-element of the h-step transition matrix (i.e., the probability of moving from state j to state k in h periods), and $\pi_{k,t+h}^e \equiv E^P[\pi_{t+h} \mid s_t = j, s_{t+h} = k]$ denotes the expected inflation conditional on realizing state k at time $t + h$.

Proof. See Appendix A.

Proposition 2 yields three key insights regarding the sources of identification bias. First, under a constant policy rule ($\phi_{\pi,1} = \phi_{\pi,2}$), the covariance term vanishes, allowing interest rate forecasts to consistently identify the perceived reaction function. Second, even with regime-switching parameters ($\phi_{\pi,1} \neq \phi_{\pi,2}$), the bias disappears if agents perceive the future regime as deterministic (i.e., $P_{j1} \in \{0, 1\}$). For instance, if agents view the current regime as permanent ($P_{jj} = 1$), subsample regressions using forecasts of regime-specific subsamples can successfully recover the perceived reaction function. Third, the bias is eliminated in the absence of general equilibrium feedback. If the interest rate does not feed back into the inflation process, expected inflation becomes invariant to the monetary regime ($\pi_{1,t+h}^e = \pi_{2,t+h}^e$), thereby removing the covariance. In sum, even if the monetary regime switching is exogenous, the bias arises when agents simultaneously internalize i) general equilibrium feedback, *and* ii) the possibility of future regime shifts.

This paper therefore identifies the *perceived* policy rule in a setting where agents are aware of the *general equilibrium* effect of monetary policy, and the fact that the policy rule itself may *evolve over time*. Acknowledging this *joint awareness* is useful because it shows the implicit assumption that justifies the use of interest rate *forecasts* for identification. Some studies allow for perceived time variation in the policy rule but assume a partial equilibrium setting where interest rates do not affect macroeconomic outcomes (Bauer, Pflueger, and Sunderam, 2024).[8] Conversely, other studies work under a full general equilibrium framework but impose a constant policy rule for identification purposes (Hamilton, Pruitt, and Borger, 2011; Bocola et al., 2024). In each case, one of the two conditions in Proposition 2 fails by assumption, and the covariance bias is not relevant for identification.

[8]Bauer, Pflueger, and Sunderam (2024) acknowledge this limitation and discuss the implications for econometric bias in their Appendix. However, the covariance bias is not addressed in their framework.

3.2 Illustration of Bias: A Markov-Switching New Keynesian Model

To further investigate the bias identified in Subsection 3.1, I map the decomposition into a standard three-equation Markov-Switching New Keynesian (MSNK) model (Farmer, Waggoner, and Zha, 2011). The linearized structural equations are given by:

$$x_t = \mathbb{E}_t [x_{t+1}] - \tau (i_t - \mathbb{E}_t [\pi_{t+1}]) + z_{Dt} \tag{16}$$

$$\pi_t = \beta \mathbb{E}_t [\pi_{t+1}] + \kappa x_t + z_{St} \tag{17}$$

$$i_t = \phi_{\pi,s_t} \pi_t + z_{Mt} \tag{18}$$

where x_t denotes the output gap, π_t inflation, and i_t the nominal interest rate. The parameters β, κ, and τ represent the discount factor, the Phillips curve slope, and the intertemporal elasticity of substitution, respectively. The economy is driven by demand (z_{Dt}), supply (z_{St}), and monetary policy (z_{Mt}) shocks, which follow independent AR(1) processes. The regime-dependent coefficient ϕ_{π,s_t} evolves according to a discrete Markov chain.

In this setup, it is assumed that the perceived and actual rules coincide, so that the agent's expectation is fully rational ($E_t^P = \mathbb{E}_t$). This assumption is without loss of generality. The covariance bias is a general property of expectations in regime-switching environments, independent of any misalignment between perceived and actual policies.[9] Put differently, this section warns against the practice of inferring perceived policy rules from interest rate *expectations*, whether or not those expectations are rational. The analysis of bias discussed in this section holds for the full-fledged model with misaligned perceived rule in Section 5.

Let $\mathbf{z}_t = [z_{Dt}, z_{St}, z_{Mt}]'$ be the vector of exogenous shocks with persistence diagonal matrix $\mathbf{R}$. The general result from Proposition 2 maps to the MSNK model as follows.

Corollary 1. *In the MSNK model defined by equations (16)-(18) with a two-state Markov chain $s_t \in \{1, 2\}$, the conditional covariance given current state $s_t = j$ is:*

$$\text{Cov}\left(\phi_{\pi,s_{t+h}}, \pi_{t+h} \mid s_t = j\right) = P_{j1}^{(h)} \left(1 - P_{j1}^{(h)}\right) (\phi_{\pi,1} - \phi_{\pi,2}) (\mathbf{a}_1 - \mathbf{a}_2)^T \mathbf{R}^h \mathbf{z}_t \tag{19}$$

where $\mathbf{a}_k$ is the vector of inflation loadings on $\mathbf{z}_t$ conditional on the future regime being k. The term $(\mathbf{a}_1 - \mathbf{a}_2)^T \mathbf{R}^h \mathbf{z}_t$ represents the expected inflation differential between regimes, defined as $\mathbb{E}_t[\pi_{t+h} \mid s_{t+h} = 1] - \mathbb{E}_t[\pi_{t+h} \mid s_{t+h} = 2]$.

Proof. See Appendix A.

[9] To be precise, the derivation requires only *internal rationality* (Adam and Marcet, 2011), not necessarily Full Information Rational Expectations (FIRE). Internal rationality implies that agents form expectations that are mathematically consistent with their own subjective model, which features interest rate feedback and the possibility of regime shifts. Whether these beliefs coincide with the objective data-generating process is immaterial for the derivation. The bias arises as long as agents internalize the joint distribution of policy parameters and equilibrium outcomes within their perceived model.

To proceed analytically, I simplify the transition probabilities to the i.i.d. case. Furthermore, to clearly distinguish the bias arising from the covariance and to stay consistent with the full model in Section 5.3 where agents perceive the rule as purely systematic, I set the perceived monetary policy shock to zero ($z_{Mt} = 0$) for the following derivation and simulations. This will be discussed in more detail at the end of Section 3.

Corollary 2. *Assume the regime switching process is i.i.d. and let $\phi_{\pi,1} > \phi_{\pi,2}$ (Regime 1 is Hawkish). Under standard parameter restrictions ($\kappa, \tau > 0$) and the absence of monetary policy shocks ($z_{Mt} = 0$), the difference in inflation loadings for demand and supply shocks is strictly negative:*

$$a_{1,D} - a_{2,D} = \frac{-\kappa^2 \tau \left(\phi_{\pi,1} - \phi_{\pi,2}\right)}{\vartheta_D} < 0 \tag{20}$$

$$a_{1,S} - a_{2,S} = \frac{-\left(1 - \rho_S\right) \kappa \tau \left(\phi_{\pi,1} - \phi_{\pi,2}\right)}{\vartheta_S} < 0 \tag{21}$$

where ϑ_D and ϑ_S are positive scalars defined in Appendix A. Consequently, the conditional bias term is strictly negative:

$$\frac{\mathrm{Cov}_t \left[\phi_{\pi,t+h}, \pi_{t+h}\right]}{\mathbb{E}_t[\pi_{t+h}]} < 0 \tag{22}$$

Proof. See Appendix A.

The negative ratio (22) means that the regressor and omitted variable are negatively correlated. To build intuition for this result, suppose there is only supply shock:

$$\mathrm{Cov}\left(\phi_{\pi,s_{t+h}}, \pi_{t+h} \mid s_t = j\right) = \underbrace{p\left(1-p\right)\left(\phi_{\pi,1} - \phi_{\pi,2}\right)\left(a_{1S} - a_{2S}\right)\rho_S^h}_{(-)} z_{St} \tag{23}$$

Since $a_{1S} - a_{2S} < 0$, the underbraced coefficient is negative. Thus, the sign of the covariance is strictly opposite to the sign of the shock z_{St}.

Consider a positive supply shock ($z_{St} > 0$), which generates positive expected inflation ($\mathbb{E}_t[\pi_{t+h}] > 0$). Agents understand that the Hawkish regime ($s_{t+h} = 1$) will produce lower inflation than the dovish regime, implying a negative conditional covariance ($\mathrm{Cov}_t < 0$) and the negative ratio. Conversely, a negative supply shock generates negative expected inflation but a positive covariance (as Hawkish policy will push deflation up), preserving the negative sign of the ratio. This confirms that standard regression estimates will systematically understate the public's perceived responsiveness to inflation.

Table 1: Calibrated Parameters of MSNK model

Parameters	τ	κ	β	ϕ_π	ϕ_x	ρ	ρ_D	ρ_S	σ_D	σ_S
First regime	0.61	0.67	0.99	**2.19**	0.30	0.84	0.76	0.84	0.23	0.62
Second regime	0.61	0.67	0.99	**0.77**	0.17	0.60	0.76	0.84	0.23	0.62

Note: Parameter values are the original ones from Lubik and Schorfheide (2004) and Farmer, Waggoner, and Zha (2011). See Table 3 in Farmer, Waggoner, and Zha (2011).

3.3 Numerical Illustration of Bias

To quantify the theoretical bias derived above, I perform Monte Carlo simulations using the calibrated MSNK model. The structural parameters follow the calibration in Farmer, Waggoner, and Zha (2011) and Lubik and Schorfheide (2004), as summarized in Table 1. The transition probability matrix $\mathbf{P}_{\text{ms}}$ governing the regime switching process is:

$$\mathbf{P}_{\text{ms}} = \begin{bmatrix} p & 1-p \\ 1-p & p \end{bmatrix} \tag{24}$$

where p denotes the persistence of the current regime.

The model is solved and simulated using the method of Farmer, Waggoner, and Zha (2011). I then apply standard OLS estimation to the simulated h-period ahead forecasts. Instrumental variable methods do not resolve this bias, as shown in Appendix B.3.[10]

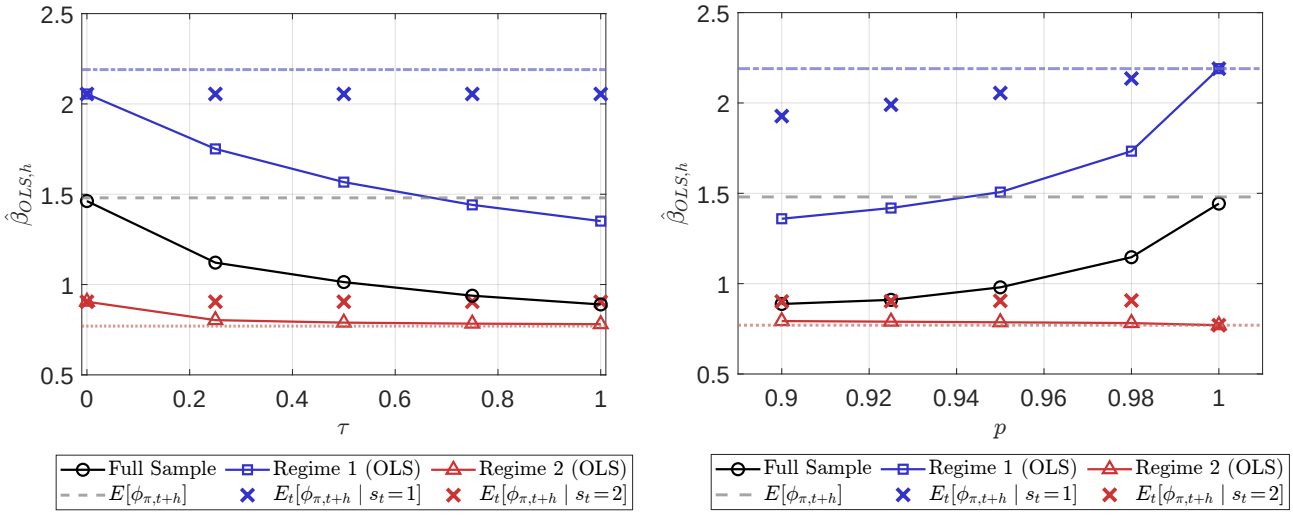
Simple, Univariate Rule I begin with the simplest specification: a univariate Taylor rule without inertia or output gap response. The estimating equation is:

$$\mathbb{E}_t[i_{t+h}] = \beta_{\text{OLS},h}\mathbb{E}_t[\pi_{t+h}] + e_{t,h} \tag{25}$$

where $\beta_{\text{OLS},h}$ is the coefficient obtained by regressing h-period-ahead interest rate expectations on expected inflation. To conserve space, I report results for $h = 2$.

Figure 1 summarizes the simulation results. The horizontal lines in both panels (a) and (b) establish the theoretical benchmarks: the top and bottom lines correspond to the regime-specific coefficients $\phi_{\pi,1}$ and $\phi_{\pi,2}$, while the middle line represents the unconditional ergodic mean, $\mathbb{E}[\phi_{\pi,s_t}]$. The colored markers "x" denote the regime-specific conditional expectations, $\mathbb{E}_t[\phi_{\pi,s_{t+h}}]$. For panel (a), the transition probability is fixed across all cases, making the conditional expectation flat across different cases of τ. For panel (b), $\mathbb{E}_t[\phi_{\pi,s_{t+h}}]$ changes as we vary

[10] In conventional Taylor rule estimation, the error term is an exogenous monetary policy shock, which can be instrumented using lagged variables or other structural shocks. In perceived rule estimation, the "error" is the structural covariance derived in Proposition 1. Since this covariance is an endogenous function of the state variables, no valid instruments exist that satisfy the exclusion restriction.



(a) Sensitivity to Interest Rate Feedback (τ)

(b) Sensitivity to Regime Persistence (p)

Figure 1: OLS Bias in the Perceived Taylor Rule Estimation from the MSNK Model (Univariate Rule)

Note: This figure reports the mean OLS coefficients across simulations for equation (25) at $h = 2$. The left panel varies τ (fixing $p = 0.95$); the right panel varies p (fixing $\tau = 0.61$). The solid black line represents the OLS estimate on the full sample, while the grey dashed line represents the true unconditional expected rule coefficient. Blue and red lines represent OLS estimates on regime-specific subsamples.

p. The solid black line plots the mean OLS coefficient estimated on the full sample, while the blue and red lines display the coefficients obtained from regime-specific subsample regressions. In other words, the blue (red) line is the OLS coefficients estimated on a sample of regime 1 (2) only.

The left panel isolates the role of general equilibrium feedback (τ). Consistent with theoretical predictions, the OLS estimator (the solid black line) is unbiased when monetary policy decouples from the economy ($\tau = 0$). However, as τ increases, the estimator deviates downward from the true belief ($\mathbb{E}[\phi_{\pi,s_{t+h}}] \approx 1.5$), confirming that the bias is driven by the agent's awareness of the endogenous response of inflation to monetary policy.

The right panel illustrates the role of regime uncertainty (p). Under perfect persistence ($p = 1$), beliefs regarding the future regime become deterministic, thereby eliminating the covariance term and yielding unbiased estimates. The solid black line shows that the estimator recovers the ergodic mean ($\mathbb{E}[\phi_{\pi,s_{t+h}}]$) exactly. However, as persistence declines, the downward bias is more pronounced.

Taken together, these highlight important empirical implications. First, under certain degrees of general equilibrium feedback (Panel (a), $\tau = 1$) or regime uncertainty (Panel (b), $p = 0.9$), the econometrician recovers a value below 1, even though the underlying true belief remains near 1.5. Consequently, the econometrician may misleadingly conclude that the

perceived rule violates the Taylor principle.

Furthermore, the panels demonstrate that the regime-specific estimates (blue and red lines) remain biased even when the econometrician conditions on the true regime. In other words, even if the econometrician can perfectly identify the subsamples associated with each regime, regressing the interest rate forecast will not recover the true perception ($\mathbb{E}_t[\phi_{\pi,s_{t+h}}]$, `x` markers). This finding underscores that the bias is not an artifact of mechanically mixing data across regimes, nor is an actual regime switch a precondition for the bias. Instead, it arises because agents structurally internalize the regime uncertainty and the general equilibrium feedback of monetary policy, embedding a non-zero covariance between expected inflation and future policy rule into their interest rate forecasts.

Multivariate, Inertial Rule This bias extends to the generalized inertial Taylor rule:

$$i_t = \Big(1 - \rho_{s_t}\Big)\Big(\phi_{\pi,s_t}\pi_t + \phi_{x,s_t}x_t\Big) + \rho_{s_t}i_{t-1}. \tag{26}$$

Replacing the simple rule (18) with the inertial rule, the corresponding estimation equation using simulated data is:

$$\mathbb{E}_t[i_{t+h}] = \beta_{\pi,h}\mathbb{E}_t[\pi_{t+h}] + \beta_{x,h}\mathbb{E}_t[x_{t+h}] + \beta_{i,h}\mathbb{E}_t[i_{t+h-1}] + e_{t,h} \tag{27}$$

Figure 2 displays the estimated coefficients across forecast horizons h as a function of the transition probability p. The green dashed line represents the benchmark of perfect persistence ($p = 1$). In the absence of regime uncertainty, the estimates are invariant to the horizon and uniformly recover the unconditional expectations (e.g., $\mathbb{E}[(1 - \rho_{s_t})\phi_{\pi,s_t}]$). Conversely, the introduction of regime uncertainty ($p < 1$) induces a significant downward bias in both the inflation and output gap coefficients. Notably, this bias generates a downward term structure in the estimated coefficients as the horizon h extends, qualitatively consistent with the empirical pattern from the actual survey data as documented in Appendix D.2.

3.4 Recovering the Perceived Rule using Interest Rate Nowcasts

Finally, I demonstrate that applying TVP estimation to the nowcast ($h = 0$) enables the consistent recovery of the true perceived reaction function (e.g., ϕ_{π,s_t}). The estimation equation extends the constant-parameter specification (27) to allow for time-variation:

$$\mathbb{E}_t[i_{t+h}] = \beta_{\pi,t,h}\mathbb{E}_t[\pi_{t+h}] + \beta_{x,t,h}\mathbb{E}_t[x_{t+h}] + \beta_{i,t,h}\mathbb{E}_t[i_{t+h-1}] + e_{t,h}. \tag{28}$$

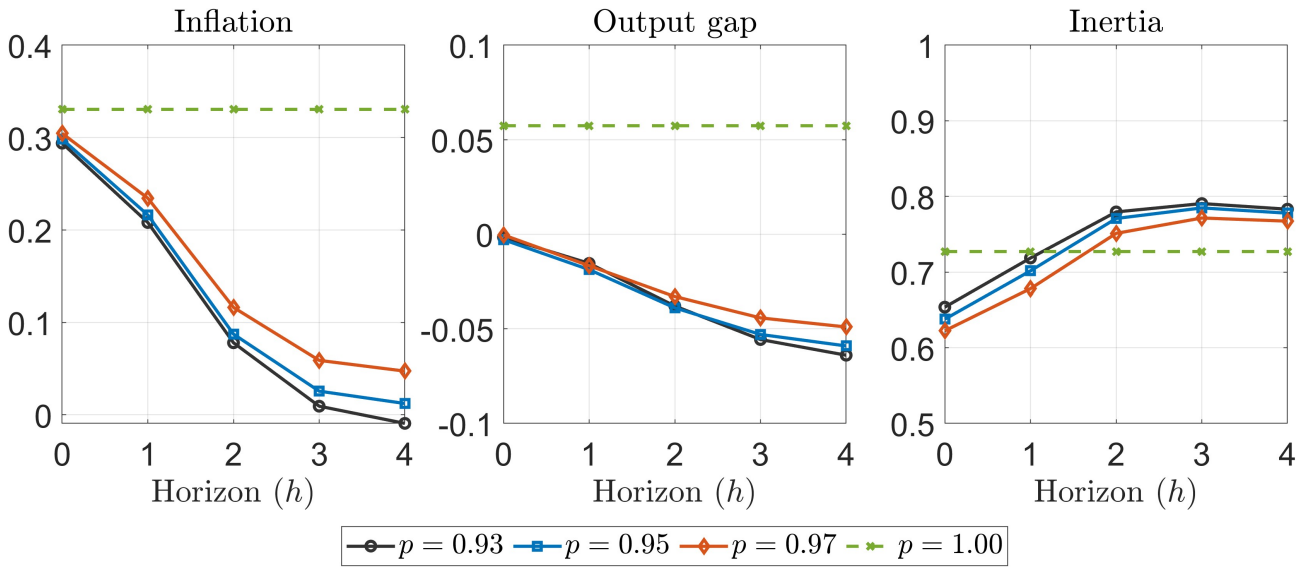


Figure 2: OLS Bias in the MSNK Model (Multivariate Case)

Note: This figure reports the mean OLS coefficients and one standard deviation bands for equation (27) at $h = 2$. The panels show the variation of the OLS coefficients for inflation (left), output gap (middle), and inertia (right) as a function of transition probability (p).

From these reduced-form time-varying coefficients, the implied long-run perceived responses are defined as:

$$\phi_{\pi,t,h} := \frac{\beta_{\pi,t,h}}{1 - \beta_{i,t,h}}, \tag{29}$$

$$\phi_{x,t,h} := \frac{\beta_{x,t,h}}{1 - \beta_{i,t,h}} \tag{30}$$

The estimation is performed on simulated data generated from a deterministic regime path: 100 periods of Regime 2 (Dovish) followed by 100 periods of Regime 1 (Hawkish).[11] The parameters are estimated using the method of Boivin (2005), following the convention in the time-varying Taylor rule literature (Coibion and Gorodnichenko, 2011).

Figure 3 presents the results for the baseline regime-switching economy ($p = 0.85$). The dashed lines represent the target conditional expectations (e.g., $\mathbb{E}_t[\phi_{\pi,s_{t+h}}]$), while the solid lines plot the estimated long-run coefficients. The forecast-based estimates ($h = 2$, red solid line) remain structurally biased relative to the true conditional expectation (red dashed line). In contrast, the nowcast-based estimates ($h = 0$, blue solid line) correctly track the true long-run parameters $\phi_{\pi,t}$. This result implies that while survey forecasts ($h > 0$) are contaminated by the perceived covariance bias, the survey nowcast ($h = 0$) provides a clean identification of the

[11] This regime frequency is consistent with the ergodic distribution of the Markov process $\mathbf{P}_{ms}$.

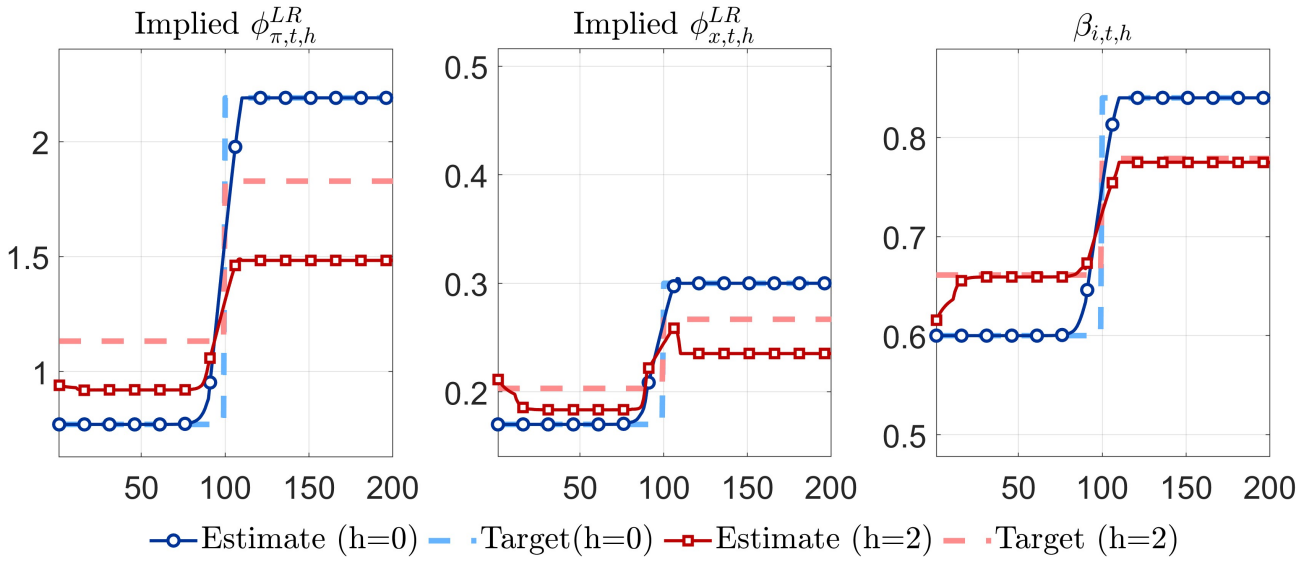


Figure 3: TVP Estimation with Regime Switching ($p = 0.85$)

Note: This figure compares the estimation results of equation (28) for $h = 0$ (blue) and $h = 2$ (red) under $p = 0.85$. The dashed lines indicate the target conditional expectations (e.g., $\mathbb{E}_t[\phi_{\pi,s_{t+h}}]$), generated based on the fixed regime path (100 periods of $s_t = 2$, followed by 100 periods of $s_t = 1$). The solid lines with markers indicate the estimated long-run coefficients.

private sector's real-time perceived policy rule. In the Appendix, Figure B.1 demonstrates that the covariance bias vanishes under regime certainty, where interest rate forecasts successfully recover the perceived rule without covariance bias, analogous to the OLS case.

Two main identification assumptions are worth highlighting. First is that the nowcasts reported in the survey data are consistent with the agents' subjective models. This assumption is standard in the literature, where the interest rate nowcasts are actively used (Bauer, Pflueger, and Sunderam, 2024; Crump, Eusepi, and Moench, 2024). One related concern regarding interest rate nowcast is that it could be a mechanical read-off from already realized data. Nevertheless, it remains a valuable instrument for identifying the perceived monetary policy rule. Even if the nowcast lies close to the realization, identification is valid as long as it is a reflection of the subjective model. Moreover, as shown in Appendix D.4, the nowcast errors are not trivial in either size or persistence, implying that they are not a mechanical restatement of the observed rate.

Second, the derivations and simulations have abstracted from the monetary policy shocks. The underlying assumption is that the perceived monetary policy shocks do not materially confound the estimation of the perceived Taylor rule. This is consistent with the consensus in the literature on conventional Taylor-rule estimation (Coibion and Gorodnichenko, 2011; Carvalho, Nechio, and Tristao, 2021; Bauer, Pflueger, and Sunderam, 2024). In Appendix

D.3, I show that this consensus also applies to the perceived Taylor rule by following the empirical test of Carvalho, Nechio, and Tristao (2021). Furthermore, this consensus holds even in environments characterized by large monetary policy shocks, such as the rapid tightening of the late 1970s under Chairman Volcker (Coibion and Gorodnichenko, 2011; Coibion, 2012).

Relatedly, it is worth clarifying that the non-zero covariance environment discussed in this section is not a prerequisite for using interest rate nowcasts. The covariance bias explains why interest rate forecasts may fail to provide an unbiased estimate of the perceived policy rule. But it is not an additional assumption required to justify the nowcast approach. As shown in equation (9) in Section 2, interest rate nowcasts identify the perceived rule correctly even in a constant rule environment as long as the aforementioned two assumptions hold. In other words, the nowcast approach remains robust under the standard assumptions of existing works and requires no new assumptions.

Additionally, although time-varying parameter (TVP) estimation applied to nowcasts corrects for this covariance bias, constant-parameter methods such as OLS or IV remain biased even when nowcasts are used. When the econometrician's sample spans multiple regimes, the OLS estimator places disproportionate weight on regimes associated with greater regressor variance. In this case, the estimator fails to recover the unconditional mean of the perceived policy coefficients across regimes.

4 Empirical Results

4.1 Data

To identify the perceived monetary policy rule, I utilize expectations data from the Blue Chip Economic Indicators (BCEI) and the Survey of Professional Forecasters (SPF). Specifically, I use the consensus now/forecasts of the short-term interest rate (Three-Month Treasury Bill), inflation (GDP deflator), and the unemployment rate.[12]

One concern is that the consensus nowcast may be subject to aggregation bias and that the three-month Treasury bill nowcasts may contain a risk premium. Appendix D.5 addresses both concerns by re-estimating the perceived rule using BCFF microdata, forecaster by forecaster, with each respondent's own federal funds rate, inflation, and output gap forecasts constructed from real output growth forecasts. The cross-sectional median of the individual estimates closely tracks the consensus-based estimate, suggesting that neither aggregation nor the choice of short rate materially distorts the estimated perceived rule. BCFF microdata are only used

[12]Quarterly inflation forecasts are transformed into year-over-year rates to align with the actual practice of the monetary policy. This transformation utilizes the Real-Time Data Set for Macroeconomists (RTDSM) maintained by the Federal Reserve Bank of Philadelphia, following Bauer, Pflueger, and Sunderam (2024). Robustness checks confirm that the estimation results are similar if quarterly inflation series are used directly.

as a robustness check rather than the baseline for two reasons. First, constructing individual output gap forecasts requires an additional assumption of a common estimate of potential GDP across forecasters. Second, BCEI's unemployment forecast is more closely aligned with the Fed's mandate of maximum employment. BCFF's constructed output gap forecast is valid via Okun's law, but this relationship is found to be unstable post-2020.

To estimate the actual monetary policy reaction function, I rely on the Greenbook/Tealbook forecasts prepared by Federal Reserve staff prior to each FOMC meeting.[13] This choice is motivated by the substantial literature arguing that the central bank sets policy based on its own internal forecasts available in real-time (Orphanides, 2004; Boivin, 2005; Coibion and Gorodnichenko, 2011). To ensure comparability with the perceived rule estimation, I utilize the Greenbook forecasts for the GDP deflator and the unemployment rate, aligning the forecast horizons with those in the private sector surveys. The policy instrument is defined as the target federal funds rate established at each corresponding FOMC meeting.

4.2 TVP Estimation of the Perceived Rule

To precisely identify the time-varying perceived reaction function at a quarterly frequency, I pool information from multiple surveys available within each quarter. Specifically, I stack the monthly Blue Chip Economic Indicators (BCEI) nowcasts and forecasts alongside the quarterly Survey of Professional Forecasters (SPF) nowcasts and forecasts. This approach increases the number of observations per period (quarter), improving the precision of the quarterly parameter estimates while maintaining the real-time nature of the identification.[14]

Let t index the survey period and h denote the forecast horizon, both measured in quarters. I formulate the model as a system where the observables are vectors containing all available survey observations within that quarter for a specific horizon h. The measurement equation is given by:

$$\begin{aligned}\mathbf{i}^{BC}_{t+h|t} &= \beta_{\pi,t,h}\boldsymbol{\pi}^{BC}_{t+h+j|t} + \beta_{x,t,h}\left(-2\cdot\mathbf{u}^{BC}_{t+h+j|t}\right) + \beta_{i1,t,h}\mathbf{i}^{BC}_{t+h-1|t} + \beta_{i2,t,h}\mathbf{i}^{BC}_{t+h-2|t} + c_{t,h}\mathbf{1} + \boldsymbol{\varepsilon}_{t,h} \\ &= \mathbf{X}_{t,h,j}\boldsymbol{\Theta}_{t,h} + \boldsymbol{\varepsilon}_{t,h} \end{aligned} \tag{31}$$

where bold variables $(\mathbf{i}, \boldsymbol{\pi}, \mathbf{u})$ denote vectors of stacked survey expectations (BCEI and SPF) for quarter t, and $\boldsymbol{\Theta}_{t,h} = [\beta_{\pi,t,h}, \beta_{x,t,h}, \beta_{i1,t,h}, \beta_{i2,t,h}, c_{t,h}]'$ represents the vector of time-varying coefficients. The case $h = 0$ corresponds to the nowcast specification, while $h > 0$ corresponds to forecasts. The term $c_{t,h}$ denotes the time-varying constant, which controls for time variation

[13] An alternative would be the Summary of Economic Projections. Its usefulness here is limited by a short sample that begins only in 2007, a quarterly rather than FOMC-meeting frequency, and calendar-year rather than fixed-horizon forecasts.

[14] In Appendix D, the model is re-estimated at a monthly frequency using only the BCEI monthly consensus forecasts. The resulting monthly time series closely tracks the quarterly baseline,

in long-run target values as in Coibion and Gorodnichenko (2011). To conserve space, I report the result based on a forward-looking horizon of $j = 1$. The specification with forward-looking ($j = 1$) rule with AR(2) inertia is selected by information criteria, consistent with findings in the conventional Taylor rule literature (Boivin, 2005; Coibion and Gorodnichenko, 2011). In Appendix D, I assess the sensitivity of the empirical results to alternative horizons j.

The time-varying parameters are assumed to be common across the stacked observations and evolve according to a random walk:

$$\boldsymbol{\Theta}_{t,h} = \boldsymbol{\Theta}_{t-1,h} + \boldsymbol{\nu}_{t,h}, \quad \boldsymbol{\nu}_{t,h} \sim \mathcal{N}(0, \boldsymbol{\Sigma}_{\nu,h}) \tag{32}$$

I account for heteroskedasticity by allowing the variance of the observation vector to follow a stochastic volatility process:

$$\boldsymbol{\varepsilon}_{t,h} \sim \mathcal{N}(\mathbf{0}, \sigma_{t,h}^2 \mathbf{I}), \quad \sigma_{t,h} \equiv e^{\eta_{t,h}}, \quad \eta_{t,h} = \eta_{t-1,h} + \xi_{t,h}, \quad \xi_{t,h} \sim \mathcal{N}(0, \sigma_{\xi,h}^2) \tag{33}$$

For estimation, I follow the established method in time-varying Taylor rule literature of Boivin (2005) and Coibion and Gorodnichenko (2011). The method allows for structural breaks in the variance-covariance matrix of innovations to the time-varying parameters. One innovation in this specification is the inclusion of stochastic volatility in equation (33), which follows the methodology by S. Kim, Shephard, and Chib (1998). As argued by Sims (2001), omitting stochastic volatility can lead to spurious time variation in coefficients, as the model may misattribute changes in shock volatility to changes in the time-varying parameters. A detailed description of the estimation procedure is provided in Appendix C.

4.2.1 The Real-Time Perceived Rule using Interest Rate Nowcasts

Building on Section 3.4, I use the estimated coefficients from equation (31) at interest rate nowcast ($h = 0$) and define the total inertia and the implied long-run *perceived* responses as:

$$\rho_{t|t} := \beta_{i1,t,0} + \beta_{i2,t,0} \quad \phi_{\pi,t|t} := \frac{\beta_{\pi,t,0}}{1 - \rho_{t|t}} \quad \phi_{x,t|t} := \frac{\beta_{x,t,0}}{1 - \rho_{t|t}} \tag{34}$$

Figure 4 presents the evolution of the perceived policy coefficients and the stochastic volatility. The plots display posterior medians alongside 68% and 90% credible intervals.[15] The estimates exhibit substantial time variation, underscoring the inadequacy of constant-parameter approximations for capturing private sector beliefs.

First, the pre-ZLB era provides compelling support for the "Good Perceived Policy" hy-

[15] Appendix D.6 re-runs the whole procedure on bootstrap samples and discusses how much the bands widen to account for hyperparameter uncertainty.

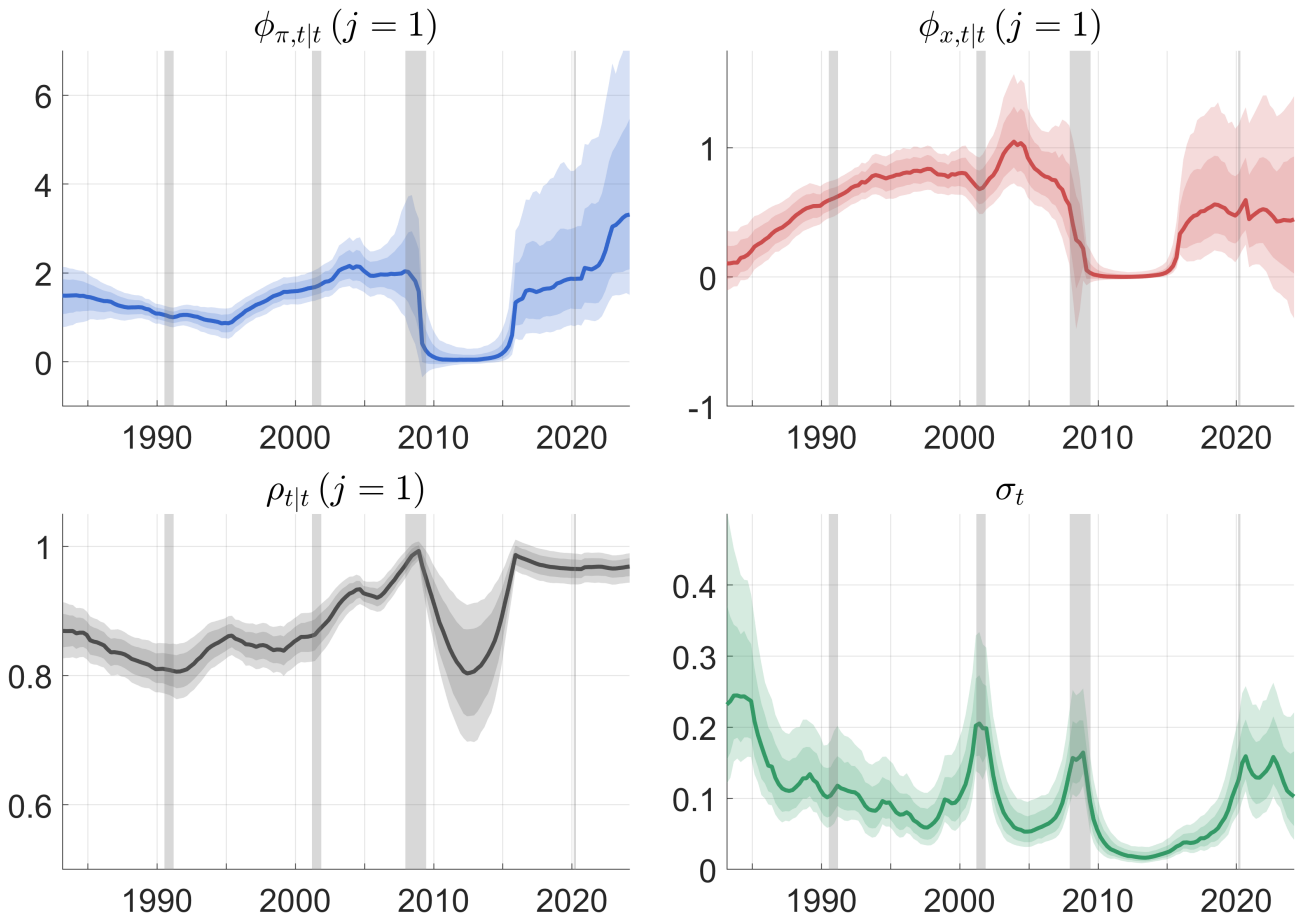


Figure 4: Time-Varying Estimates of the Perceived Policy Rule (Nowcast Specification)

Note: The figure plots the posterior median (solid line) along with the 68% and 90% bands (shaded regions) for the long-run response coefficients to inflation and the output gap, the total inertia parameter, and the stochastic volatility of the policy shock.

pothesis regarding the Great Moderation. Even after controlling for stochastic volatility, the perceived inflation coefficient $\phi_{\pi,t|t}$ consistently satisfies the Taylor principle throughout the period. This suggests that the private sector's anchored belief in the central bank's active stabilization stance was important in achieving stability during the Great Moderation.[16]

Second, the steady rise in the output gap coefficient $\phi_{x,t|t}$ throughout the pre-crisis period suggests a structural shift in private beliefs toward a dual mandate framework.[17] Third, the dynamics of the inertia parameter $\rho_{t|t}$ during the ZLB episode reflect evolving expectations re-

[16]The third panel of Figure 5 corroborates this by applying the Blanchard–Kahn determinacy condition draw-by-draw to the calibrated 3-equation NK model. The posterior probability of a determinate equilibrium under the nowcast-based perceived rule remains close to one over the sample period except for ZLB period (2009-2015).

[17]Menon (2025) uses a structural model where the private sector learns about the central bank's objective. The paper documents that the perception of the Fed's weight on unemployment-gap stabilization has been increasing over time.

garding lift-off. The initial sharp decline suggests forecasters viewed the effective lower bound as a temporary constraint at the onset, whereas the subsequent rebound around 2011 is consistent with agents learning that the constraint would be long-lasting. Finally, in the post-ZLB period, perceived inertia remains entrenched at a level structurally higher than its pre-crisis average, likely reflecting the private sector's internalization of the central bank's active reliance on forward guidance (Bauer, Pflueger, and Sunderam, 2024).

A natural concern is that the zero lower bound (ZLB) could bias the estimated perceived rule, since the interest rate nowcast is censored during these periods. In practice, this concern is minimal for this paper's analysis. Following González-Astudillo and Tanvir (2023) and Carriero et al. (2025), I re-estimate the model replacing the nowcast inside the ZLB windows with a latent shadow rate. Appendix D.7 reports this result. Except for the first ZLB period (2009-2015), the estimates are statistically indistinguishable from the baseline, and the second ZLB episode (2020-2022) does not show much change from the baseline estimates. The shadow-rate estimation moves the long-run coefficients meaningfully only during and around the first ZLB, a window not relevant for the analysis in Section 5. This finding parallels the corresponding check for the *actual* Taylor rule using the Wu and Xia (2016) shadow rate, documented in Appendix D.8.

4.2.2 Comparison of Horizons

The nowcast estimates above rely on the identification argument that nowcasts purge the perceived covariance bias affecting forecast-based regressions. A direct way to verify this is to re-estimate equation (31) at a longer LHS horizon and compare. If agents view the current regime as permanent or perceives a constant rule, the estimated time-varying coefficients should be invariant across interest rate forecast horizons. In contrast, under the bias mechanism in Section 3, forecast-based estimates ($h > 0$) should be systematically attenuated relative to nowcast-based estimates ($h = 0$) due to the latent beliefs about possible regime shifts.

Figure 5 reports the comparison.[18] The first two panels show that the implied long-run responses to inflation and to the output gap are systematically attenuated under the forecast specification ($h = 2$, red dashed) relative to the nowcast specification ($h = 0$, blue solid). Notably, the long-run inflation response under $h = 2$ falls below unity for a large portion of the sample, suggesting a violation of the Taylor principle. The missing Taylor principle in survey data is also consistent with the evidence in Bauer, Pflueger, and Sunderam (2024). The third panel assesses the determinacy implication of these coefficients. Using a textbook New Keynesian model with calibration matched to Section 5 and Taylor rule specification mirroring 31, plugging in parameter draws delivers a posterior probability of determinacy in

[18]The results are qualitatively similar across alternative forecast horizons. Appendix D.9 reports the implied long-run rules estimated at $h \in \{0, 1, 2, 3\}$.

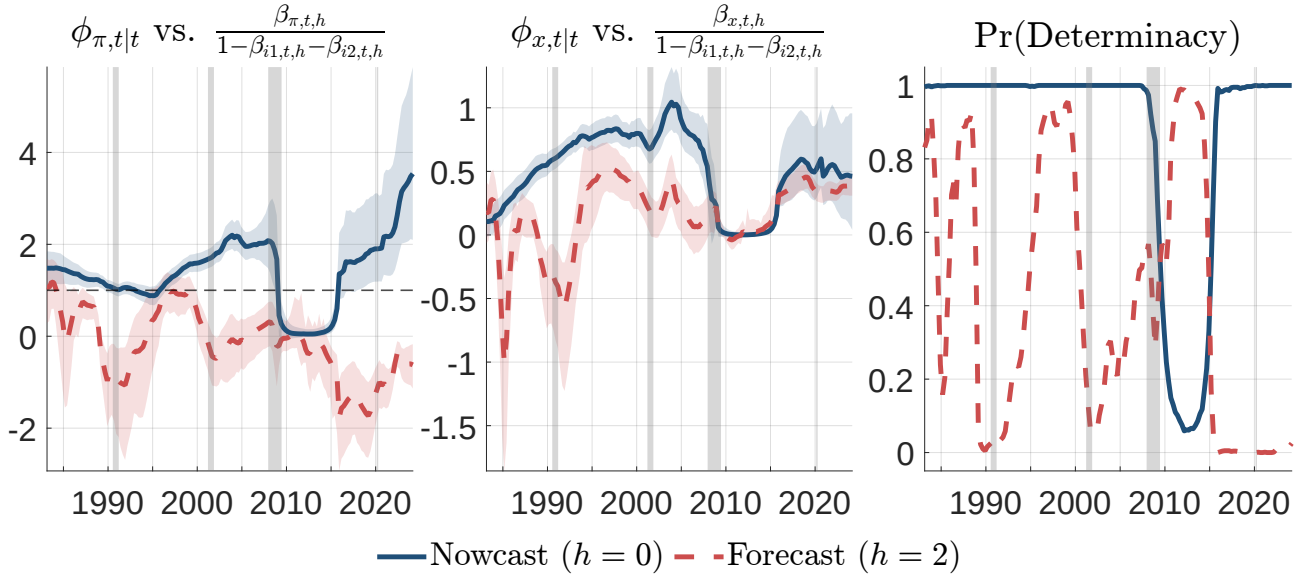


Figure 5: Comparison of perceived long-run rule and probability of determinacy

Note: The figure compares estimates from equation (31) using nowcasts ($h = 0$, blue solid) against 2-quarter ahead forecasts ($h = 2$, red dashed). The first two panels report the implied long-run responses to inflation and to the output gap. Shaded areas are 16-84th percentile bands. The third panel reports the posterior probability that a three-equation New Keynesian model is determinate, evaluated draw by draw at each date.

each time period. While nowcast-based coefficients imply that the perception is consistent with determinacy, with the only exception of the ZLB window, under the forecast specification it oscillates well below one for much of the sample, implying indeterminacy. This implies that conducting inference about the perceived Taylor principle from forecast-based regressions can produce misleading conclusions.

There may be a concern that the attenuation at longer h is mechanical. First, longer-horizon expectations of inflation and the output gap (unemployment rate) may vary less as they converge to their long-run targets, so the regressors carry less identifying variation. Second, the hyperparameters of time-varying parameters are separately tuned at each horizon, which could generate cross-horizon differences on their own. Third, the interest rate forecast and the two inertia regressors are three points on the same smooth forecasted rate path and are nearly collinear, so the lag terms could absorb the variation and leave the inflation response identified off residual noise. Appendices D.10–D.12 address these concerns in turn and show that the attenuation of perceived inflation rule is robust, indicating that it is not a property of the estimation.

4.3 Actual vs. Perceived Rule

To compare the private sector's perception against the central bank's actual conduct, I estimate the actual monetary policy rule using Greenbook forecasts, following the established literature (Orphanides, 2001; Boivin, 2005; Coibion and Gorodnichenko, 2011; Coibion, 2012). The specification mirrors the quarterly stacked model used for the perceived rule but utilizes Greenbook data at the FOMC meeting frequency (denoted by superscript GB):

$$i_t = (1 - \rho_{1,t} - \rho_{2,t}) \left[\phi_{\pi,t} \pi^{GB}_{t+j|t} + \phi_{x,t} \left(-2 \cdot u^{GB}_{t+j|t} \right) \right] + \rho_{1,t} i_{t-1} + \rho_{2,t} i_{t-2} + c_t + \varepsilon_t \tag{35}$$

Here, i_t corresponds to the target federal funds rate, as opposed to the interest rate nowcast $i_{t|t}$ used for perceived rule estimation. This estimation yields the time path of the actual policy coefficients $\phi_{\pi,t}$ and $\phi_{x,t}$, which I then contrast with the perceived coefficients $\phi_{\pi,t|t}$ and $\phi_{x,t|t}$ recovered from the survey data. I maintain $j = 1$ as the baseline horizon to ensure strict comparability with the perceived rule estimates. The sample period runs from 1969 to 2019 due to publication lag of Greenbook data. In Appendix D.8, I also re-estimate the actual policy rule with the shadow rate of Wu and Xia (2016) for possible concern related to ZLB, but the resulting time series of coefficients are very similar to this baseline.

Figure 6 superimposes the estimated time-varying coefficients for the actual and perceived rules. While the two series generally comove closely, some periods of divergence are evident. To rigorously assess these discrepancies, the bottom panel of the figure plots the difference in these coefficients along with the 68% and 90% credible intervals.

A prominent episode of divergence emerges during the "preemptive tightening" of the mid-1990s. Under Chairman Greenspan, the actual response to the mean inflation forecast became statistically insignificant compared to the perceived rule. This discrepancy arose from the Federal Reserve's adoption of a "risk management" approach to monetary policy, in which the target rate was adjusted to mitigate potential tail risks despite stable baseline macroeconomic expectations (Greenspan, 2004; Bernanke, 2022). This has led to a weaker statistical relationship between the target rate and the Greenbook forecasts. However, while this discretionary strategy manifested as a systematic deviation in the actual reaction function, the private sector's perceived rule remained structurally intact.

The distinct divergence episode is also informative about the nature of the perceived rule itself. If forecasters were merely reading the realized federal funds rate off recent history, the nowcast-based perceived coefficients would mechanically track the Greenbook-based actual coefficients period by period. The fact that the two series visibly come apart in the mid-1990s, and again at points during the post-Volcker sample, indicates that interest rate nowcast embeds a genuine belief about the anticipated realization of the forecasters.

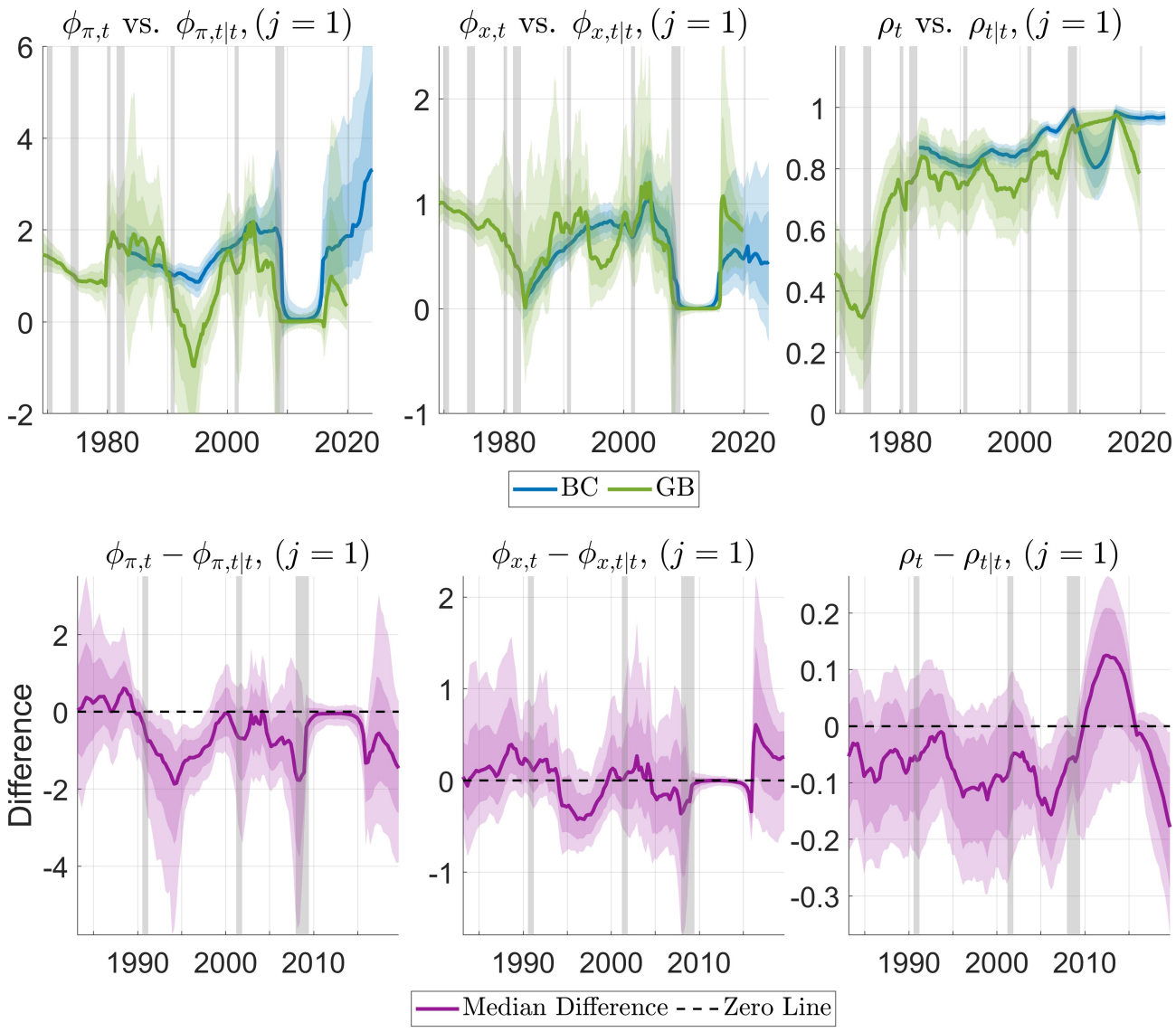


Figure 6: Actual vs. Perceived Monetary Policy Rules

Note: **Top panel:** The figure superimposes the estimated long-run implied long-run inertia coefficients from the actual rule (green) from the Greenbook forecasts and the perceived rule (blue) from Blue Chip and SPF survey data. The confidence bands are 68% and 90% respectively. **Bottom panel:** The figure plots the difference $\phi_{\pi,t} - \phi_{\pi,t|t}$ with 68% and 90% confidence bands. Positive values indicate the Fed was more aggressive than perceived. The differences are calculated for estimates observed at March, June, September, and December. The inertia coefficient estimated from Greenbook data has been adjusted from the FOMC to the quarterly frequency following Coibion and Gorodnichenko (2011).

5 Theory of Perceived vs. Actual Rule

5.1 General Framework

5.1.1 Setup

Consider an economy described by a vector of endogenous variables $\mathbf{y}_t \in \mathbb{R}^n$ and a vector of actual structural shocks $\boldsymbol{\varepsilon}_t \in \mathbb{R}^m$. The economy is subject to two distinct regime-switching processes. First, the *perceived regime*, denoted by $s_t^p \in \mathcal{S}^p = \{1, 2, \ldots, S_p\}$, represents the state of the economy perceived by the private sector. Agents believe s_t^p evolves according to a subjective transition probability matrix $\widehat{\mathbf{P}}$ with elements $\widehat{P}_{jk} = \Pr\left(s_{t+1}^p = k \mid s_t^p = j\right)$. Second, the *central bank's actual regime*, denoted by $s_t^f \in \mathcal{S}^f = \{1, 2, \ldots, S_f\}$, characterizes the true policy stance implemented by the monetary authority. Consequently, the *actual* state of the economy is the tuple $s_t = \left(s_t^p, s_t^f\right)$ within the joint state space $\mathcal{S} = \mathcal{S}^p \times \mathcal{S}^f$, where the total number of joint states is $S = S_p \times S_f$. For notational convenience, I index the joint state by a single integer $\ell \in \{1, 2, \ldots, S\}$. The economy evolves according to the actual transition probability matrix $\mathbf{P}$ with elements $P_{\ell\ell'} = \Pr\left(s_{t+1} = \ell' \mid s_t = \ell\right)$.[19]

I distinguish between the full vector of structural shocks hitting the economy, $\boldsymbol{\varepsilon}_t$, and the subset of shocks observed by the private sector, denoted by $\boldsymbol{\varepsilon}_t^p$. Let $\mathbf{Q}$ be a selection matrix such that:

$$\boldsymbol{\varepsilon}_t^p = \mathbf{Q}\boldsymbol{\varepsilon}_t \tag{36}$$

For instance, in the specific model developed later, I assume agents perceive the Taylor rule as purely systematic and cannot observe the true monetary policy shock. This restriction reflects the information constraint that agents cannot empirically distinguish whether the discrepancy between their interest rate nowcast and the realized rate arises from a systematic deviation in the reaction function or a temporary monetary policy shock. Given this signal extraction problem, the agent cannot tell the source of their forecast error, having no incentive to detect and change their misspecified model. In this case, the perceived shock vector $\boldsymbol{\varepsilon}_t^p$ contains only non-policy shocks (e.g., demand and supply), whereas the actual shock vector $\boldsymbol{\varepsilon}_t$ includes the monetary policy shock.

[19] The dimension of $\mathbf{P}$ is $S \times S$, whereas the dimension of $\widehat{\mathbf{P}}$ is $S_p \times S_p$.

5.1.2 Perceived Law of Motion

Conditional on the perceived state $s_t^p = j$, the private sector assumes the economy evolves according to the linear structural system:

$$\mathbf{\Gamma}_{0,j}^{\mathrm{P}} E_t^{\mathrm{P}}\left[\mathbf{y}_{t+1}\right] + \mathbf{\Gamma}_{1,j}^{\mathrm{P}} \mathbf{y}_t + \mathbf{\Gamma}_{2,j}^{\mathrm{P}} \mathbf{y}_{t-1} + \mathbf{\Gamma}_{3,j}^{\mathrm{P}} \boldsymbol{\varepsilon}_t^p = \mathbf{0} \tag{37}$$

where $E_t^{\mathrm{P}}[\cdot]$ denotes the expectation operator conditional on the agents' information set and the subjective transition probabilities $\widehat{\mathbf{P}}$.

Assumption 1 (MSV Solution). I assume that for the system defined by $\left\{\mathbf{\Gamma}_{i,j}^{\mathrm{P}}\right\}$ and $\widehat{\mathbf{P}}$, there exists a mean-square stable Minimum State Variable (MSV) solution. This implies the Perceived Law of Motion (PLM) takes the form of a Markov-Switching VAR driven by the perceived shocks:

$$\mathbf{y}_{t|t} = \widehat{\mathbf{A}}_{s_t^p} \mathbf{y}_{t-1} + \widehat{\mathbf{B}}_{s_t^p} \boldsymbol{\varepsilon}_t^p \tag{38}$$

The matrices $\widehat{\mathbf{A}}_j$ and $\widehat{\mathbf{B}}_j$ are structurally fundamental to the agent's belief system and are obtained via standard MS-DSGE solution algorithms (e.g., Maih, 2015). Note that $\mathbf{y}_{t|t}$ denotes the private sector's *nowcast* of the endogenous variables given the state variables $\mathbf{y}_{t-1}$ and the perceived shocks $\boldsymbol{\varepsilon}_t^p$. This nowcast is consistent with the perceived law of motion but may differ from the actual realization $\mathbf{y}_t$ observed in the economy, which is generated by the combination of agent's subjective expectation, actual model of the economy and full vector of actual shocks $\boldsymbol{\varepsilon}_t$.

Agents form expectations using their subjective model. The one-period ahead forecast given the current perceived state j is:

$$E_t^{\mathrm{P}}\left[\mathbf{y}_{t+1}\right] = \sum_{k=1}^{S_p} \widehat{P}_{jk} E_t^{\mathrm{P}}\left[\mathbf{y}_{t+1} \mid s_{t+1}^p = k\right] \tag{39}$$

$$= \tilde{\mathbf{A}}_j \left(\widehat{\mathbf{A}}_j \mathbf{y}_{t-1} + \widehat{\mathbf{B}}_j \boldsymbol{\varepsilon}_t^p\right) \tag{40}$$

where $\tilde{\mathbf{A}}_j \equiv \sum_{k=1}^{S_p} \widehat{P}_{jk} \widehat{\mathbf{A}}_k$ is the expected transition matrix. I define the sensitivity of expectations to history and perceived shocks as $\mathbf{\Gamma}_{y,j} \equiv \tilde{\mathbf{A}}_j \widehat{\mathbf{A}}_j$ and $\mathbf{\Gamma}_{\varepsilon,j} \equiv \tilde{\mathbf{A}}_j \widehat{\mathbf{B}}_j$, respectively. Thus, the behavioral equation for the agent's forecast is given by $E_t^{\mathrm{P}}\left[\mathbf{y}_{t+1}\right] = \mathbf{\Gamma}_{y,j} \mathbf{y}_{t-1} + \mathbf{\Gamma}_{\varepsilon,j} \boldsymbol{\varepsilon}_t^p$.

5.1.3 Actual Law of Motion

The equilibrium is characterized as a Temporary Equilibrium (J. M. Grandmont, 1977; J.-M. Grandmont, 1987; Woodford, 2013) extended to a Markov-switching environment. In each period, equilibrium is defined by the expectations agents form based on their perceived law

of motion (PLM), which in turn determines the actual law of motion (ALM) for aggregate variables.

The actual evolution of the economy is governed by the joint state s_t, indexed by ℓ, and the actual shocks $\boldsymbol{\varepsilon}_t$. The private sector's subjective expectation $E_t^{\mathrm{P}}[\cdot]$ is formed under the perceived model. The wedge between the perceived and the actual economy is the regime-dependent, structural parameters summarized by $\left\{\boldsymbol{\Gamma}_{i,\ell}^{\mathrm{A}}\right\}$ loadings:[20]

$$\boldsymbol{\Gamma}_{0,\ell}^{\mathrm{A}} E_t^{\mathrm{P}}\left[\mathbf{y}_{t+1}\right] + \boldsymbol{\Gamma}_{1,\ell}^{\mathrm{A}} \mathbf{y}_t + \boldsymbol{\Gamma}_{2,\ell}^{\mathrm{A}} \mathbf{y}_{t-1} + \boldsymbol{\Gamma}_{3,\ell}^{\mathrm{A}} \boldsymbol{\varepsilon}_t = \mathbf{0} \tag{41}$$

The perceived and actual systems follow the same index convention, with $\boldsymbol{\Gamma}_{i,j}^{\mathrm{P}}$ in (37) indexed by the perceived state j and $\boldsymbol{\Gamma}_{i,\ell}^{\mathrm{A}}$ by the joint state ℓ.

Substituting the expectation formation mechanism derived in (40) into the actual structural system and utilizing the mapping $\boldsymbol{\varepsilon}_t^p = \mathbf{Q}\boldsymbol{\varepsilon}_t$ yields:

$$\boldsymbol{\Gamma}_{0,\ell}^{\mathrm{A}} \left(\boldsymbol{\Gamma}_{y,j} \mathbf{y}_{t-1} + \boldsymbol{\Gamma}_{\varepsilon,j} \mathbf{Q} \boldsymbol{\varepsilon}_t\right) + \boldsymbol{\Gamma}_{1,\ell}^{\mathrm{A}} \mathbf{y}_t + \boldsymbol{\Gamma}_{2,\ell}^{\mathrm{A}} \mathbf{y}_{t-1} + \boldsymbol{\Gamma}_{3,\ell}^{\mathrm{A}} \boldsymbol{\varepsilon}_t = \mathbf{0} \tag{42}$$

Rearranging the system leaves a purely backward-looking system:

$$\mathbf{H}_\ell \mathbf{y}_t + \mathbf{F}_\ell \mathbf{y}_{t-1} + \mathbf{C}_\ell \boldsymbol{\varepsilon}_t = \mathbf{0} \tag{43}$$

where $\mathbf{H}_\ell = \boldsymbol{\Gamma}_{1,\ell}^{\mathrm{A}}$, $\mathbf{F}_\ell = \boldsymbol{\Gamma}_{0,\ell}^{\mathrm{A}} \boldsymbol{\Gamma}_{y,j} + \boldsymbol{\Gamma}_{2,\ell}^{\mathrm{A}}$, and $\mathbf{C}_\ell = \boldsymbol{\Gamma}_{0,\ell}^{\mathrm{A}} \boldsymbol{\Gamma}_{\varepsilon,j} \mathbf{Q} + \boldsymbol{\Gamma}_{3,\ell}^{\mathrm{A}}$ aggregate terms from both the perception channel and the direct structural impact. Provided that $\mathbf{H}_\ell$ is non-singular, the Actual Law of Motion (ALM) follows in a closed form, time-varying VAR:

$$\mathbf{y}_t = \underbrace{-\mathbf{H}_\ell^{-1} \mathbf{F}_\ell}_{\mathbf{A}_\ell^{\mathrm{ALM}}} \mathbf{y}_{t-1} + \underbrace{-\mathbf{H}_\ell^{-1} \mathbf{C}_\ell}_{\mathbf{B}_\ell^{\mathrm{ALM}}} \boldsymbol{\varepsilon}_t \tag{44}$$

This system characterizes the dynamics of an economy where agents possess a persistent misperception of the true model of the economy and nests the full information markov-switching rational expectation model as a case.

Nothing in this framework is specific to monetary policy yet. The wedge between perceived and actual models can be placed anywhere in the structural system. So any mismatch between the subjective and actual models in a regime-switching environment can be represented this way. For example, a perceived Phillips curve with the wrong slope, a perceived Euler equation with the wrong intertemporal elasticity, or misperceived shock persistence would each be a different restriction on the same objects. The framework can also accommodate cases in which

[20] The framework can be extended to make the private sector and the central bank to hold different expectation operators. For example, the framework can allow the central bank to hold rational expectation with the knowledge of the private sector's misperception. Appendix B.4 lays out this case in the general form.

the subjective model is a simple New Keynesian model, while the actual economy is governed by a larger linear system. In this paper, however, I restrict attention only to the misspecification in monetary policy rule parameters to focus on the paper's main question.

5.2 Perceived Taylor Principle and Determinacy

A central question in monetary economics is how policy should be conducted to prevent self-fulfilling equilibria. Under Full Information Rational Expectations (FIRE), determinacy requires the central bank to satisfy the Taylor principle. This subsection shows that once perceived and actual rules are separated, what matters is that the principle holds in perception.

I work in the constant-parameter limit ($S_p = S_f = 1$) nested within the SP-MSNK model. The private sector's perceived model is

$$x_t = E_t^{\mathrm{P}}\left[x_{t+1}\right] - \tau\left(i_t - E_t^{\mathrm{P}}\left[\pi_{t+1}\right] - r_t^n\right) \tag{45}$$

$$\pi_t = \beta E_t^{\mathrm{P}}\left[\pi_{t+1}\right] + \kappa x_t \tag{46}$$

$$i_t = r_t^n + \phi_\pi^{\mathrm{P}} \pi_t \tag{47}$$

while the central bank implements

$$i_t = r_t^n + \phi_\pi^{\mathrm{F}} \pi_t. \tag{48}$$

Both rules track the natural rate and differ only in the inflation response ($\phi_\pi^{\mathrm{F}}, \phi_\pi^{\mathrm{P}}$).

Proposition 3 (Perceived Taylor principle, determinacy, and divine coincidence)**.** *Consider the economy* (45)–(48) *with* $\tau, \kappa > 0$ *and* ϕ_π^P, $\phi_\pi^F \geq 0$. *The economy admits a unique bounded equilibrium* if and only if *the perceived Taylor principle holds,*

$$\phi_\pi^P > 1. \tag{49}$$

And $i_t = r_t^n$ *supports divine coincidence* ($\pi_t = x_t = 0$) *in equilibrium for all* t.

Proof. See Appendix A.4.

The natural-rate peg $i_t = r_t^n$ is a textbook example of optimal monetary policy that is consistent with divine coincidence but suffers from indeterminacy (Galí, 2015). Under this policy, equilibrium outcomes may unexpectedly depart from divine coincidence. Consequently, the policy prescription was to adopt a rule that satisfies the Taylor principle such that the divine coincidence is achieved as a unique outcome. This prescription has become a central element of monetary policy design that guides the central banks globally.

Proposition 3 reorients this view. When the perceived Taylor principle is satisfied, the natural-rate peg $i_t = r_t^n$ does not suffer from indeterminacy and can achieve divine coincidence. This has important empirical and policy implications. The empirical monetary literature has disproportionately focused on whether the actual conduct of monetary policy is consistent with the Taylor principle. But this result suggests that the relevant object is whether the perceived policy rule is in line with the Taylor principle. From the policymaker's point of view, it is therefore not sufficient to just implement the Taylor principle. It should also ensure that the public believes the policy rule satisfies the Taylor principle.

5.3 Subjective Perception Markov Switching New Keynesian Model

To quantitatively analyze the macroeconomic effects of the perceived monetary policy rules, I introduce Subjective Perception Markov-Switching New Keynesian (SP-MSNK) model. This specification fits the general framework and adheres to a standard three-equation New Keynesian structure.

5.3.1 Setup

Assume the economy is characterized by S_p perceived states of monetary policy, $s_t^p \in \mathcal{S}^p$. The representative agent's perceived law of motion (PLM) is given by the following system:

$$x_t = E_t^{\mathrm{P}}\left[x_{t+1}\right] - \tau\left(i_t - E_t^{\mathrm{P}}\left[\pi_{t+1}\right]\right) + z_{Dt} \tag{50}$$

$$\pi_t = \beta E_t^{\mathrm{P}}\left[\pi_{t+1}\right] + \kappa x_t + z_{St} \tag{51}$$

$$i_t = \left(1 - \rho_{s_t^p}^{\mathrm{P}}\right)\left[\phi_{\pi,s_t^p}^{\mathrm{P}} E_t^{\mathrm{P}}\left[\pi_{t+1}\right] + \phi_{x,s_t^p}^{\mathrm{P}} E_t^{\mathrm{P}}\left[x_{t+1}\right]\right] + \rho_{1,s_t^p}^{\mathrm{P}} i_{t-1} + \rho_{2,s_t^p}^{\mathrm{P}} i_{t-2} \tag{52}$$

$$z_{Dt} = \rho_D z_{Dt-1} + \varepsilon_{D,t} \tag{53}$$

$$z_{St} = \rho_S z_{St-1} + \varepsilon_{S,t} \tag{54}$$

The perceived state evolves according to the transition probability matrix $\widehat{\mathbf{P}}$, where $\widehat{P}_{jk} = \Pr(s_{t+1}^p = k \mid s_t^p = j)$. The total perceived inertia is defined as $\rho_{s_t^p}^{\mathrm{P}} \equiv \rho_{1,s_t^p}^{\mathrm{P}} + \rho_{2,s_t^p}^{\mathrm{P}}$.[21]

Defining the vector of state variables as $\mathbf{y}_t = [x_t, \pi_t, i_t, i_{t-1}, z_{Dt}, z_{St}]'$, the perceived system yields a state-space representation $\mathbf{y}_{t|t} = \widehat{\mathbf{A}}_{s_t^p}\mathbf{y}_{t-1} + \widehat{\mathbf{B}}_{s_t^p}\boldsymbol{\varepsilon}_t^p$. Notably, the perceived model does not contain a structural monetary policy shock. From the agent's perspective, deviations of the realized interest rate i_t from the rule-implied rate could be due to systematic rule or monetary policy shocks. The AR(2) smoothing structure is included to maintain consistency with the

[21] The model is subject to the forward guidance puzzle. In Appendix B.6, I introduce the parameter $\theta \in (0, 1]$ that governs the degree of cognitive discounting in the IS curve Following (Gabaix, 2020; Pfäuti and Seyrich, 2022).Setting $\theta < 1$ dampens the quantitative magnitude of the perception channel but does not alter the qualitative conclusions.

empirical specification presented in Section 4.

The Federal Reserve conducts monetary policy according to an *actual* Taylor rule. This rule is governed by the state $s_t^f \in \mathcal{S}^f$ and may differ from the perceived rule:

$$i_t = \left(1 - \rho^{\mathrm{F}}_{s_t^f}\right)\left[\phi^{\mathrm{F}}_{\pi,s_t^f} E_t^{\mathrm{P}}\left[\pi_{t+1}\right] + \phi^{\mathrm{F}}_{x,s_t^f} E_t^{\mathrm{P}}\left[x_{t+1}\right]\right] + \rho^{\mathrm{F}}_{1,s_t^f} i_{t-1} + \rho^{\mathrm{F}}_{2,s_t^f} i_{t-2} + \varepsilon_{Mt} \tag{55}$$

where $\rho^{\mathrm{F}}_{s_t^f} \equiv \rho^{\mathrm{F}}_{1,s_t^f} + \rho^{\mathrm{F}}_{2,s_t^f}$. The actual state of the economy is the tuple $s_t = \left(s_t^p, s_t^f\right)$ indexed by ℓ.[22] I assume that the Fed observes and responds to private sector expectations. This assumption is introduced to mainly isolate the effect of disagreement about the policy rule without confounding the expectational disagreement. In Appendix B.5, I relax this assumption by allowing the Fed to use a forward-looking Taylor rule with rational expectations, which yields quantitatively similar results.

Solving for the expectation coefficients $\tilde{\mathbf{A}}_j$ (where j corresponds to the perceived component of ℓ) and substituting them into the structural equations yields the Actual Law of Motion (ALM):

$$\mathbf{y}_t = \mathbf{A}_\ell^{\mathrm{ALM}} \mathbf{y}_{t-1} + \mathbf{B}_\ell^{\mathrm{ALM}} \boldsymbol{\varepsilon}_t \tag{56}$$

where the matrices $\mathbf{A}_\ell^{\mathrm{ALM}}$ and $\mathbf{B}_\ell^{\mathrm{ALM}}$ are defined and derived explicitly in Appendix A.5. This system serves as the laboratory for analyzing the implications of the perceived rule and empirical findings in Section 4.

Calibration The model calibration is standard and follows Pfäuti (2023). The discount factor is $\beta = 0.99$, the intertemporal elasticity of substitution is $\tau = 1.00$, and the Phillips curve slope is $\kappa = 0.057$, implying firms reoptimize prices approximately every four quarters. The shock processes assume persistence $\rho_D = \rho_S = 0.80$.

The regime-dependent policy parameters are calibrated by discretizing the estimated time-varying perceived Taylor rule from Section 4 using a Hidden Markov Model (HMM). Specifically, I fit a univariate Gaussian HMM with $K = 5$ states to the inflation response coefficient $\phi_{\pi,t|t}$, with the number of states selected by the Bayesian Information Criterion. The rationale for clustering on $\phi_{\pi,t|t}$ alone is that the hawkish–dovish distinction is primarily defined by the inflation response. Allowing all four TVPs to drive the classification yields economically implausible groupings. Given the estimated regime sequence from this univariate HMM, the remaining coefficients are recovered as conditional, within-regime averages of the TVP estimates.[23] The resulting five perceived regimes capture distinct monetary policy stances observed in the TVP

[22] For example, in a six-regime case with $S_p = S_f = 6$, the joint state space has $S = 36$ elements, indexed by $\ell = (s_t^p - 1) \times 6 + s_t^f$.

[23] Matching additional moments in addition to the mean yields similar results.

Table 2: Regime-Specific Perceived Policy Parameters

	1: Eagle	**2: Str. Hawk**	**3: Hawk**	**4: Weak Hawk**	**5: ZLB**	**6: Dove**
$\phi_{\pi,s_t^{\mathrm{p,f}}}$	3.182	2.003	1.539	1.094	0.126	0.770
$\phi_{x,s_t^{\mathrm{p,f}}}$	0.471	0.686	0.504	0.658	0.019	0.680
$\rho_{1,s_t^{\mathrm{p,f}}}$	1.482	1.405	1.219	1.155	1.141	0.600
$\rho_{2,s_t^{\mathrm{p,f}}}$	−0.515	−0.461	−0.333	−0.323	−0.279	0.000
$\rho_{s_t^{\mathrm{p,f}}}$	0.967	0.945	0.886	0.832	0.862	0.600
$\widehat{P}_{kk}$	0.785	0.859	0.840	0.867	0.839	0.987

Note: Regimes 1–5 are identified by fitting a univariate Hidden Markov Model to the TVP-estimates of perceived inflation response ($\phi_{\pi,t|t}$). The other coefficients are recovered as within-regime averages of the TVP estimates. $\phi_{x,s_t^{\mathrm{p,f}}}$ is reported at annualized values. It is divided by 4 when input to the quarterly model. $\rho_{s_t^{\mathrm{p,f}}} \equiv \rho_{1,s_t^{\mathrm{p,f}}} + \rho_{2,s_t^{\mathrm{p,f}}}$ denotes total inertia. Regime 6 (Dove) is calibrated separately based on pre-Volcker era estimates from Lubik and Schorfheide (2004) and Farmer, Waggoner, and Zha (2011), with $\phi_{\pi,s_t^{\mathrm{p,f}}} = 0.77$, $\phi_{x,s_t^{\mathrm{p,f}}} = 0.68$ (annualized), and $\rho = 0.60$. This regime is unreachable from the other regimes ($\widehat{P}_{k6} = 0$ for $k = 1, \ldots, 5$) and is introduced only for the purpose of counterfactual exercises.

estimates and are labeled in the order of their hawkishness toward inflation: Eagle, Strong Hawk, Hawk, Weak Hawk, and ZLB. These five regimes are then augmented with a sixth Dovish regime, described in detail below, that is reserved for counterfactual exercises. Table 2 reports the estimated parameters and transition probabilities (diagonal elements) for all six regimes.

Finally, I augment the five estimated regimes with a sixth regime calibrated to capture a Dovish policy stance based on pre-Volcker era estimates from Lubik and Schorfheide (2004) and Farmer, Waggoner, and Zha (2011). The Dove regime is mainly characterized by a passive inflation response ($\phi_{\pi,6} = 0.77$). The purpose of its inclusion is to enrich the analysis by providing a concrete Dovish anchor without distorting the baseline PLM regimes 1–5. This regime is assumed to be unreachable from the other five baseline percevied regimes: $\widehat{P}_{k6} = 0$ for $k = 1, \ldots, 5$.

Figure 7 displays the resulting five-regime HMM approximation of the estimated perceived time-varying coefficients. The discretization captures the key episodes in the perceived rule: the Eagle regime emerges during the post-2020 tightening cycle, the Hawk regime characterizes much of the 1990s and 2010s, and the ZLB regime spans the period following the Global Financial Crisis.

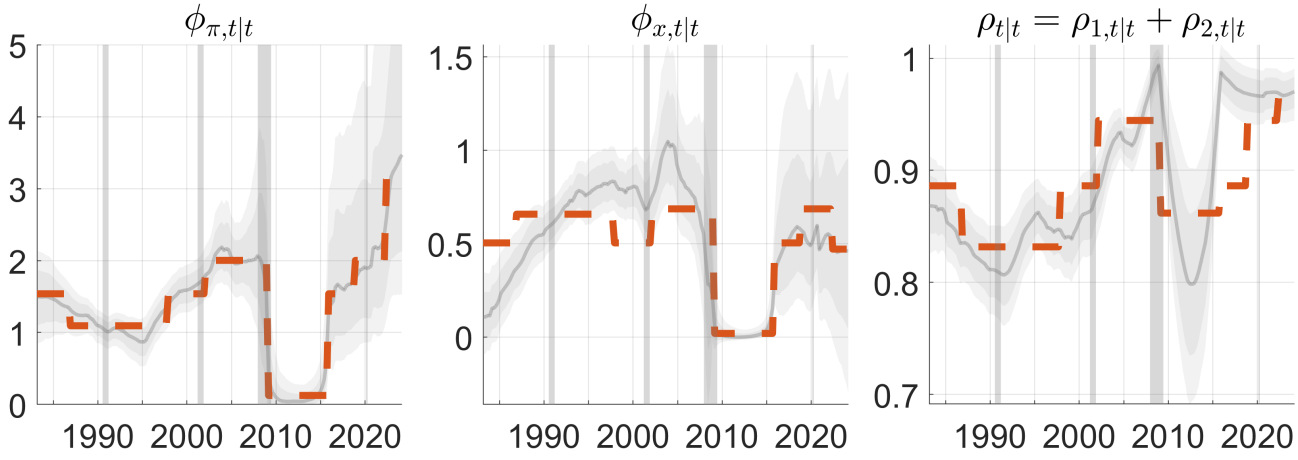


Figure 7: HMM Discretization of the Perceived Taylor Rule (5 regimes)

Note: The figure presents the result of discrete approximation of TVP-estimates of perceived rule.

5.4 Counterfactual Analysis

This section quantifies the historical implications and importance of perceived monetary policy rules through two counterfactual exercises. The first examines the role of monetary policy perceptions during the Great Moderation (1986–1999) by decomposing their contribution into a perception channel and a direct policy channel. The second examines the consequences of mismeasuring perceptions using interest rate forecasts, focusing on the post-pandemic tightening cycle (2021–2023).

5.4.1 Perception vs. Direct Policy Channel: Great Moderation

A large literature has focused on the role of monetary policy in the Great Moderation (Clarida, Gali, and Gertler, 2000; Coibion and Gorodnichenko, 2011; Bianchi, 2013). The main question was whether the decline in macroeconomic volatility is attributable to a shift in monetary policy regime. This section revisits this question from the perspective of perceived monetary policy rules. Specifically, I ask whether the decline in macroeconomic volatility would have been as pronounced in the absence of an accompanying perceived regime shift.

More specifically, I compare two channels through which monetary policy can affect macroeconomic stability. One is the *perception channel*, which operates through private-sector expectations, and the other is the *direct policy channel*, which operates through the actual policy rule. This distinction allows me to quantify the relative contribution of changes in perceptions and changes in the actual conduct of monetary policy. I exploit the Dove regime (regime 6) as a concrete counterfactual anchor and consider two independent counterfactual scenarios:

1. **Dovish Perception:** The perceived regime over the Great Moderation (1986–1999) is replaced with that of the Dove regime ($s^p = 6$), while the actual policy parameters remain

at their baseline values. This isolates the perception channel by asking: What if the private sector had perceived the Fed as dovish, while the actual rule remained hawkish?

2. **Dovish Actual Rule:** The actual policy rule parameters over the Great Moderation (1986–1999) are replaced with those of the Dove regime ($s^f = 6$), while the perceived rule remains at its baseline. This isolates the direct policy channel by asking: What if the Fed were dovish, while the perception remained hawkish?

Comparing the outcomes of the two scenarios will be informative of which channel brings about larger macroeconomic consequences.

Methodology The counterfactual simulation proceeds in three steps. First, I recover the sequence of structural shocks $\boldsymbol{\varepsilon}_t = [\varepsilon_{D,t}, \varepsilon_{S,t}, \varepsilon_{M,t}]'$ from the observed data. I treat the realized output gap, inflation, and policy rate as observables and use the state-space representation of the actual law of motion (56), where the sequence of regime-specific matrices $\mathbf{A}_\ell^{\text{ALM}}$ and $\mathbf{B}_\ell^{\text{ALM}}$ follow the HMM regime sequence in Figure 7. In the baseline, I assume that the perceived and actual regimes coincide at each point in time, $s_t^p = s_t^f$. Second, I construct a counterfactual ALM by selectively altering the ALM matrices with the ones associated with the counterfactual perceived or actual policy parameters. Third, I feed the same structural shocks through the counterfactual ALM to generate the counterfactual trajectories. The observable data are as follows. The output gap is the percentage deviation of real GDP from the CBO potential real GDP estimate. The inflation is the quarterly inflation measured by the GDP deflator. Interest rate is the federal funds rate but I employ the shadow rates of Wu and Xia (2016) during the ZLB periods (2009Q1–2015Q4, 2020Q2–2022Q1).

Results Figure 8 presents the counterfactual trajectories for the Great Moderation (1986–1999). In each panel, the solid black line depicts the baseline historical data, the dashed blue line plots the perception counterfactual (Dovish Perception, Hawkish Actual), and the dotted orange line (Hawkish Perception, Dovish Actual) plots the actual rule counterfactual.

The perception counterfactual reveals that had the public perceived the Fed as Dovish throughout the Great Moderation, inflation would have been substantially more volatile, despite the hawkish actual rule. The belief in a passive policy unanchors private sector expectations $E_t^P[\pi_{t+1}]$ and $E_t^P[x_{t+1}]$, which feed directly into realized outcomes through the IS and Phillips curves.[24] In contrast, the actual-rule counterfactual generates a comparatively modest increase in volatility, operating through the direct policy channel. The quantitative dominance of the perception channel is summarized in panel (d). It shows that under the dovish perception, the

[24]Appendix B.7 develops this mechanism further in a welfare exercise. It shows that a Dovish perception raises inflation volatility regardless of the actual stance, and that the resulting welfare cost depends on how persistent agents believe the regime to be.

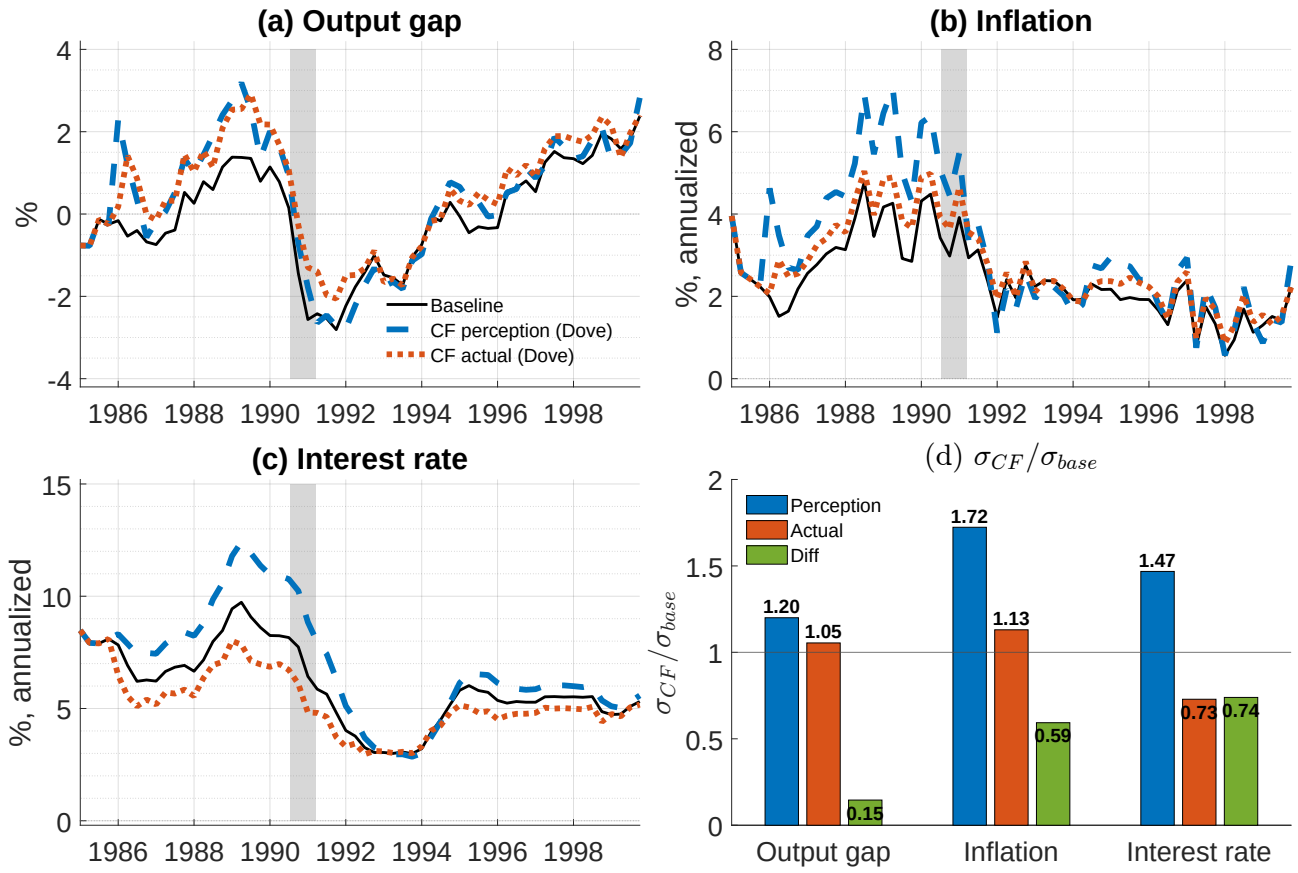


Figure 8: Counterfactual Dynamics and Volatility: Great Moderation

Note: (a)–(c) Solid black: baseline; dashed blue: CF perception (Dove PLM); dotted orange: CF actual (Dove rule). (d) $\sigma_{CF}/\sigma_{base}$ over target-regime quarters; values above 1 indicate higher volatility.

volatility increases by a factor of 1.72, whereas in dovish actual case, it only increases mildly (1.13 times). This suggests that the stability of the Great Moderation would not have been possible without the successful anchoring of perceived rule.

5.4.2 Misreading the Perceived Rule: The Post-Pandemic Lift-off (2021–2023)

During the post-pandemic inflation surge, the Federal Reserve decided to raise interest rates in 2022Q1. The timing of the lift-off decision was at the center of the policy discussion. I revisit this question from the perspective of perception and illustrate the importance of measuring perception precisely.

Setup I compare two central banks that have different views about the public's perceived inflation rule. The *nowcast-based* Fed carries the baseline perceived rule of Table 2. In the *forecast-based* Fed, the perceived inflation response of the Eagle regime is attenuated from $\phi_\pi^{\mathrm{P}} = 3.182$ to 0.248 from 2021Q1 onward, that is, from comfortably active to decisively passive. This attenuated coefficient is calibrated from the forecast-based (h = 1–3 mean) estimate of

the long-run inflation response, over 2021Q1–2023Q4. The perceived output-gap response, the perceived inertia, the actual rule, and the regime path, which is held at the Eagle regime throughout the window, are common to the two economies at every date. The differences in counteractual outcomes only come from the differences in the calibration of the perceived inflation rule.

Methodology The two central banks solve the model separately, taking the realized path of inflation, output gap, and interest rates as given under their respective perception calibration. This gives the two economies distinct PLM and ALM matrices, and different time series of structural shocks.[25] Next, over the time window 2021Q1–2023Q4, each central bank experiments counterfactual lift-off timing. The policy rate is set to the realized rate shifted q quarters earlier, implemented by changing the monetary policy shock $\varepsilon_{M,t}$ alone, with the demand and supply shocks held at their recovered historical values. The menu consists of $q \in \{-4, \ldots, 4\}$, with $q = 0$ corresponding to the actual lift-off date (2022Q1).

Given a lift-off scenario at $q \neq 0$, each central bank scores the counterfactual path of output gap and inflation over the window 2021-2023 with a conventional loss function:

$$\mathcal{L} = \mathbb{E}\left[(\pi_t^{\text{yoy}} - \pi^*)^2 + \lambda x_t^2\right]. \tag{57}$$

The inflation target is set at $\pi^* = 2\%$ in year-over-year terms and the output gap stabilization weight is set at 0.007 following Pfäuti (2023) and Adam and Billi (2006).[26] The central banks evaluate each lift-off scenario for $q \in \{-4, \cdots, 4\}$.

Results Figure 9 shows the results. The first column shows a common interest rate path assessed under the two calibrations. The second column presents the counterfactual inflation path under the two calibrations. The third column shows the value of loss function (57). In the upper row, tthe tightening takes place four quarters earlier at 2021Q1, instead of 2022Q1. In the bottom row, the tightening takes place at 2021Q3. The two dates are optimal lift-off dates based on nowcast- and forecast-based calibrations, respectively.

In the first scenario, the early lift-off at 2021Q1 is viewed as optimal from the nowcast-calibrated Fed. It reads the earlier tightening as smoothly bringing average year-over-year inflation back to level around the 2% target. In contrast, the forecast-calibrated central bank associates the same rate path with a much larger disinflation, and even deflation. Under the nowcast calibration, agents believe the Fed responds strongly to inflation. They expect that the tightening will reverse once inflation is contained. Thus, their expectations remain anchored,

[25] The observed interest rate is replaced by the shadow rate of Wu and Xia (2016) inside the ZLB windows.

[26] The results discussed in the main text are not sensitive to λ. Appendix B.8 reports the realized paths under each central bank's own choice.

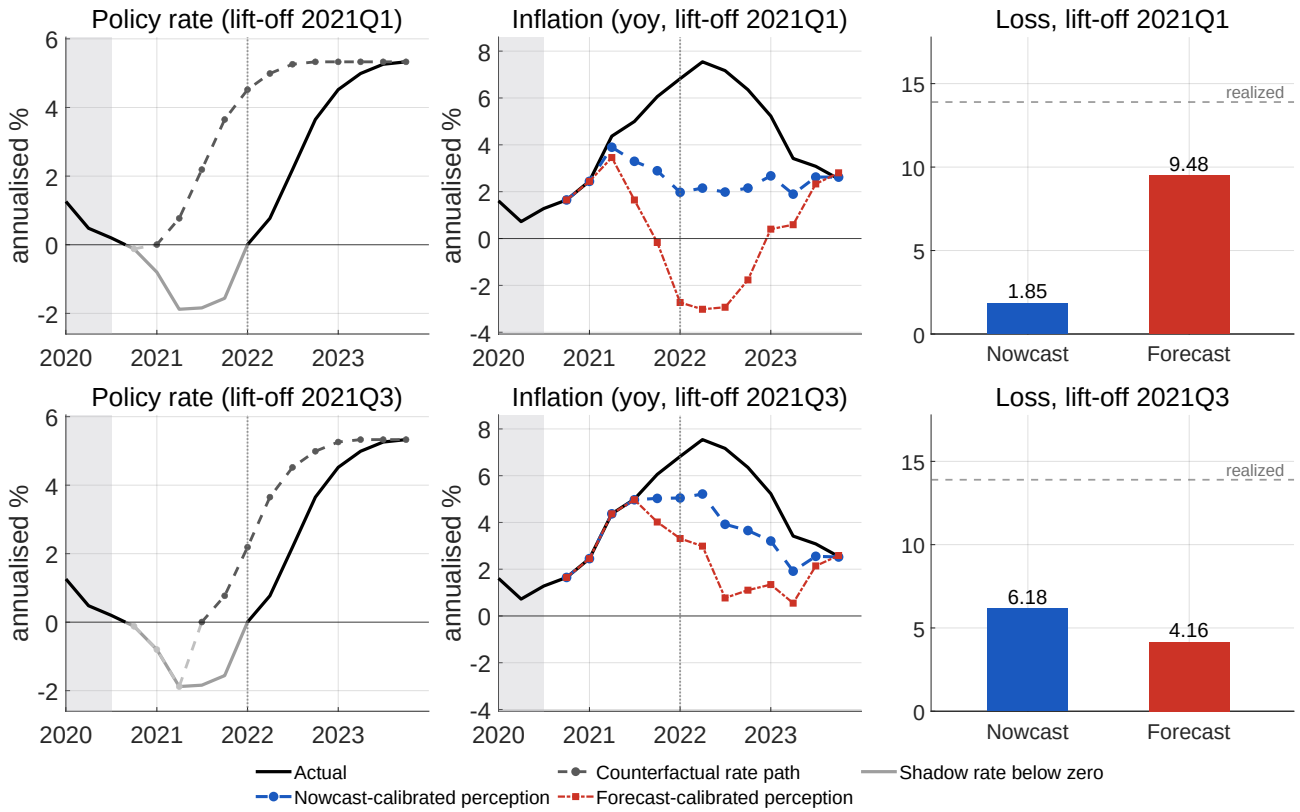


Figure 9: Early Lift-off Counterfactuals: Nowcast vs. Forecast-based Calibration

Note: The solid line denotes actual data. The dashed lines mark the counterfactual outcomes under an imposed lift-off in 2021Q1. Panel (b) scores each candidate lift-off date in the true economy (blue) and as the forecast-calibrated central bank sees it (red).

and the tightening passes through to inflation in a contained way. Under the forecast calibration, however, the perceived inflation response is close to zero. The agent expects that the current tightening will persist longer due to inertia, even if inflation reverses its course. Thus, the expectations become less anchored, and the same tightening creates much larger disinflationary effects. As a consequence, as the third panel shows, the loss assessed by the forecast-calibrated Fed is much higher than that of the nowcast-based Fed (9.48 vs. 1.85).

The second row shows the optimal lift-off scenario under the forecast-based calibration. Inflation comes down quicker to the target level (red) than the nowcast case (blue). As aforementioned, the forecast-calibrated agent perceives the same interest rate path as more contractionary and this brings down inflation more quickly than the nowcast-based model. As a consequence, the loss evaluated in this scenario is lower for the forecast-calibrated central bank (4.16 vs. 6.18). However, from the point of view of nowcast-based calibration, this path incurs a substantially more loss than the previous case (1.85 vs. 6.18).

This divergence matters because it changes the optimal policy that would be selected. It shows that if the central bank uses the forecast-based calibration, it may misjudge the cost

of a given policy. Then, the optimal policy under nowcast-based perception will be viewed as suboptimal and counterproductive in an alternative, forecast-based perception.

One qualification is that this exercise holds the public's beliefs fixed even if the policy rate changes. In practice, an earlier lift-off would itself convey information about the rule and could shift perceptions. The quantitative outcomes of this exercise should not be interpreted as a definite scenario of what would have happened under the each early lift-off case, or as a prescription for what the Fed should have done. However, the exercise nevertheless highlights a broader policy implication. If a central bank reads the perceived rule as more attenuated than it actually is, it may systematically misjudge the timing of policy intervention and may overinvest in communication to correct a perception that is already anchored.

Taken together, the two counterfactuals make complementary points. The Great Moderation exercise establishes that anchored perceptions, rather than the actual rule alone, were essential in achieving macroeconomic stability. The post-pandemic exercise establishes that the precise measurement of those perceptions is itself of first-order policy relevance. A central bank that reads the perceived rule off interest rate forecasts may misjudge the transmission of its own policy path and make mistakes on the timing of intervention.

6 Conclusion

Full-information rational expectation has long been a leading framework in macroeconomics, which effectively equates the policy rule perceived by the public with the rule actually implemented. As a result, the question of how policy is understood has received less attention than what the actual policy rule is. This paper reorients this focus and shows that the monetary policy reaction function *perceived* by the public can be a more important driver of macroeconomic dynamics than the *actual* rule.

Estimating the perceived rule from interest rate *forecasts* produces a downward-biased estimate of the true perceived inflation rule. Agents may anticipate changes in policymakers' preferences over the forecast horizon and this uncertainty generates an omitted variable bias through the perceived covariance between the future policy rule and inflation. However, interest rate *nowcasts* avoid this bias because the current regime is already revealed. Applying this insight to survey data, I show that nowcast-based estimates recover a perceived rule consistent with the Taylor principle that is spuriously rejected by the forecast-based estimates.

Correctly identifying the perceived rule matters because it governs equilibrium dynamics. In a Subjective Perception Markov-Switching New Keynesian model, I show that equilibrium determinacy depends on whether the public believes in the Taylor principle, not on whether the central bank actually adheres to it. Stabilizing the economy therefore requires not just implementing the Taylor principle but also anchoring public perceptions to it. This perspective also

yields important implications for both interpretation of historical episodes and policy analysis. The Great Moderation required not just a shift toward a more hawkish regime but a corresponding shift in perceptions. Using the post-pandemic tightening cycle, I show that relying on forecast-based estimates can mislead policy analysis, potentially leading to suboptimal policy decisions and mistakes.

The analysis clarifies its scope and points to an important direction for future research. First, I abstract from learning and take perceptions as given, allowing the analysis to isolate the consequences of changes in perceptions rather than explain how they arise. Second, the empirical analysis relies on professional forecasters, who are likely to track monetary policy rules more closely and therefore provide an upper bound on how well perceptions are anchored. If households and firms are more inattentive and hold looser views of the policy rule, the marginal effects of anchoring perception may be larger. Thus, measuring and comparing perceptions across different agents remains an important direction for future research.

References

Adam, Klaus and Roberto M Billi (2006). "Optimal monetary policy under commitment with a zero bound on nominal interest rates". In: *Journal of Money, credit and Banking*, pp. 1877–1905.

Adam, Klaus and Albert Marcet (2011). "Internal rationality, imperfect market knowledge and asset prices". In: *Journal of Economic Theory* 146.3, pp. 1224–1252.

Barthélemy, Jean and Magali Marx (2017). "Solving endogenous regime switching models". In: *Journal of Economic Dynamics and Control* 77, pp. 1–25.

Bauer, Michael D, Carolin E Pflueger, and Adi Sunderam (2024). "Perceptions about monetary policy". In: *The Quarterly Journal of Economics* 139.4, pp. 2227–2278.

Bernanke, Ben S (2022). *21st century monetary policy: The Federal Reserve from the great inflation to COVID-19*. WW Norton & Company.

Bianchi, Francesco (2013). "Regime switches, agents' beliefs, and post-World War II US macroeconomic dynamics". In: *Review of Economic studies* 80.2, pp. 463–490.

Bianchi, Francesco, Martin Lettau, and Sydney C Ludvigson (2022). "Monetary policy and asset valuation". In: *The Journal of Finance* 77.2, pp. 967–1017.

Blanchard, Olivier Jean and Charles M Kahn (1980). "The solution of linear difference models under rational expectations". In: *Econometrica: Journal of the Econometric Society*, pp. 1305–1311.

Bocola, Luigi et al. (2024). *Bond market views of the fed*. Tech. rep. National Bureau of Economic Research.

Bocola, Luigi et al. (2025). *Monetary Policy without an Anchor*. Tech. rep. National Bureau of Economic Research.

Boivin, Jean (2005). *Has US monetary policy changed? Evidence from drifting coefficients and real-time data*.

Branch, William A (2006). "Restricted perceptions equilibria and learning in macroeconomics". In: *Post Walrasian Macroeconomics: Beyond the Dynamic Stochastic General Equilibrium Model. Cambridge University Press, New York*, pp. 135–160.

Bullard, James and Kaushik Mitra (2002). "Learning about monetary policy rules". In: *Journal of monetary economics* 49.6, pp. 1105–1129.

Bundick, Brent (2015). "Estimating the Monetary Policy Rule Perceived by Forecasters." In: *Economic Review (01612387)* 100.4.

Carriero, Andrea et al. (2025). "Forecasting with shadow rate VARs". In: *Quantitative Economics* 16.3, pp. 795–822.

Carter, Chris K and Robert Kohn (1994). "On Gibbs sampling for state space models". In: *Biometrika* 81.3, pp. 541–553.

Carvalho, Carlos, Fernanda Nechio, and Tiago Tristao (2021). "Taylor rule estimation by OLS". In: *Journal of Monetary Economics* 124, pp. 140–154.

Clarida, Richard, Jordi Gali, and Mark Gertler (2000). "Monetary policy rules and macroeconomic stability: evidence and some theory". In: *The Quarterly journal of economics* 115.1, pp. 147–180.

Cogley, Timothy, Christian Matthes, and Argia M Sbordone (2015). "Optimized Taylor rules for disinflation when agents are learning". In: *Journal of Monetary Economics* 72, pp. 131–147.

Coibion, Olivier (2012). "Are the effects of monetary policy shocks big or small?" In: *American Economic Journal: Macroeconomics* 4.2, pp. 1–32.

Coibion, Olivier and Yuriy Gorodnichenko (2011). "Monetary policy, trend inflation, and the great moderation: An alternative interpretation". In: *American Economic Review* 101.1, pp. 341–370.

Crump, Richard K, Stefano Eusepi, and Emanuel Moench (2024). "Is There Hope for the Expectations Hypothesis?" In: *FRB of New York Staff Report* 1098.

Davig, Troy and Eric M Leeper (2007). "Generalizing the Taylor principle". In: *American Economic Review* 97.3, pp. 607–635.

Diercks, Anthony M, Joon Kim, and Eric T Swanson (2026). "Underestimating the Fed". In: *Available at SSRN 7118460*.

Evans, George W and Seppo Honkapohja (2003a). "Adaptive learning and monetary policy design". In: *Journal of Money, Credit and Banking*, pp. 1045–1072.

Evans, George W and Seppo Honkapohja (2003b). "Expectations and the stability problem for optimal monetary policies". In: *The Review of Economic Studies* 70.4, pp. 807–824.

Farmer, Roger EA, Daniel F Waggoner, and Tao Zha (2011). "Minimal state variable solutions to Markov-switching rational expectations models". In: *Journal of Economic Dynamics and Control* 35.12, pp. 2150–2166.

Gabaix, Xavier (2020). "A behavioral New Keynesian model". In: *American Economic Review* 110.8, pp. 2271–2327.

Galí, Jordi (2015). *Monetary policy, inflation, and the business cycle: an introduction to the new Keynesian framework and its applications*. Princeton University Press.

González-Astudillo, Manuel and Rakeen Tanvir (2023). "Hawkish or Dovish Fed? Estimating a Time-Varying Reaction Function of the Federal Open Market Committee's Median Participant". In.

Grandmont, J-M (1987). "Temporary equilibrium". In: *The New Palgrave Dictionary of Economics*. Springer, pp. 1–6.

Grandmont, Jean Michel (1977). "Temporary general equilibrium theory". In: *Econometrica: Journal of the Econometric Society*, pp. 535–572.

Greenspan, Alan (2004). "Risk and uncertainty in monetary policy". In: *American Economic Review* 94.2, pp. 33–40.

Hamilton, James D, Seth Pruitt, and Scott Borger (2011). "Estimating the market-perceived monetary policy rule". In: *American Economic Journal: Macroeconomics* 3.3, pp. 1–28.

Jia, Pengfei, Haopeng Shen, and Shikun Zheng (2023). "Monetary policy rules and opinionated markets". In: *Economics Letters* 223, p. 110995.

Kim, Chang-Jin and Charles R Nelson (2006). "Estimation of a forward-looking monetary policy rule: A time-varying parameter model using ex post data". In: *Journal of Monetary Economics* 53.8, pp. 1949–1966.

Kim, Sangjoon, Neil Shephard, and Siddhartha Chib (1998). "Stochastic volatility: likelihood inference and comparison with ARCH models". In: *The Review of Economic Studies* 65.3, pp. 361–393.

Lubik, Thomas A and Frank Schorfheide (2004). "Testing for indeterminacy: An application to US monetary policy". In: *American Economic Review* 94.1, pp. 190–217.

Maih, Junior (2015). *Efficient perturbation methods for solving regime-switching DSGE models*. 01/2015. Working paper.

Menon, Shreeyesh (2025). "Keeping up with the Curve: Learning, Evolving Beliefs and the Anchoring of Expectations". PhD thesis. University of Texas at Austin.

Nakamura, Emi, Venance Riblier, and Jón Steinsson (2025). *Beyond the taylor rule*. Tech. rep. National Bureau of Economic Research.

Orphanides, Athanasios (2001). "Monetary policy rules based on real-time data". In: *American Economic Review* 91.4, pp. 964–985.

— (2004). "Monetary policy rules, macroeconomic stability, and inflation: A view from the trenches". In: *Journal of Money, Credit and Banking*, pp. 151–175.

Pfäuti, Oliver (2023). "The inflation attention threshold and inflation surges". In: *arXiv preprint arXiv:2308.09480*, p. 2.

Pfäuti, Oliver and Fabian Seyrich (2022). "A behavioral heterogeneous agent new keynesian model". In.

Pierdzioch, Christian, Jan-Christoph Rülke, and Georg Stadtmann (2012). "Who believes in the Taylor principle? Evidence from the Livingston survey". In: *Economics Letters* 117.1, pp. 96–98.

Sargent, Thomas J (1991). "Equilibrium with signal extraction from endogenous variables". In: *Journal of Economic Dynamics and Control* 15.2, pp. 245–273.

Schorfheide, Frank (2005). "Learning and monetary policy shifts". In: *Review of Economic dynamics* 8.2, pp. 392–419.

Sims, Christopher A (2001). "Comment on Sargent and Cogley's 'Evolving US postwar inflation dynamics'". In: *NBER Macroeconomics Annual* 16, pp. 373–379.

Sims, Christopher A and Tao Zha (2006). "Were there regime switches in US monetary policy?" In: *American Economic Review* 96.1, pp. 54–81.

Stock, James H and Mark W Watson (1998). "Median unbiased estimation of coefficient variance in a time-varying parameter model". In: *Journal of the American Statistical Association* 93.441, pp. 349–358.

Woodford, Michael (2013). "Macroeconomic analysis without the rational expectations hypothesis". In: *Annu. Rev. Econ.* 5.1, pp. 303–346.

Wu, Jing Cynthia and Fan Dora Xia (2016). "Measuring the macroeconomic impact of monetary policy at the zero lower bound". In: *Journal of Money, Credit and Banking* 48.2-3, pp. 253–291.

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A Proofs and Derivations

A.1 Proof of Proposition 1

Let the data generating process be $i_{t+h|t} = \beta_{t,h}\pi_{t+h|t}$, where $\pi_{t+h|t} \equiv E_t^P[\pi_{t+h}]$. The OLS estimator is defined as:

$$\beta_{OLS} = \frac{\text{Cov}\left(\beta_{t,h}\pi_{t+h|t}, \pi_{t+h|t}\right)}{\text{Var}\left(\pi_{t+h|t}\right)} \tag{A.1}$$

Expanding the covariance term in the numerator:

$$\begin{aligned}\text{Cov}\left(\beta_{t,h}\pi_{t+h|t}, \pi_{t+h|t}\right) &= \mathbb{E}\left[\beta_{t,h}\right]\text{Cov}\left(\pi_{t+h|t}, \pi_{t+h|t}\right) \\ &+ \mathbb{E}\left[\pi_{t+h|t}\right]\text{Cov}\left(\beta_{t,h}, \pi_{t+h|t}\right) + \mathbb{E}\left[\left(\beta_{t,h} - \mathbb{E}\left[\beta_{t,h}\right]\right)\left(\pi_{t+h|t} - \mathbb{E}\left[\pi_{t+h|t}\right]\right)^2\right]\end{aligned} \tag{A.2}$$

Assuming $\mathbb{E}\left[\pi_{t+h|t}\right] = 0$, the middle term vanishes, and the expression simplifies to:

$$\beta_{OLS} = \frac{\mathbb{E}\left[\beta_{t,h}\right]\text{Var}(\pi_{t+h|t}) + \mathbb{E}\left[\left(\beta_{t,h} - \mathbb{E}\left[\beta_{t,h}\right]\right)\pi_{t+h|t}^2\right]}{\text{Var}\left(\pi_{t+h|t}\right)} \tag{A.3}$$

$$= \mathbb{E}\left[\beta_{t,h}\right] + \frac{\mathbb{E}\left[\left(\beta_{t,h} - \mathbb{E}\left[\beta_{t,h}\right]\right)\pi_{t+h|t}^2\right]}{\text{Var}\left(\pi_{t+h|t}\right)} \tag{A.4}$$

This concludes the proof. ■

A.2 Proof of Proposition 2

Let the state space system be defined by $\pi_t = \mathbf{C}_{\pi,s_t}\mathbf{X}_t$ and $\mathbf{X}_t = \mathbf{A}_{s_t}\mathbf{X}_{t-1} + \mathbf{B}_{s_t}\boldsymbol{\varepsilon}_t$. The objective is to compute the subjective conditional covariance $\text{Cov}^P\left[\phi_{\pi,s_{t+h}}, \pi_{t+h} \mid s_t = j\right]$. First, the conditional expectation of inflation given the future state k is defined as follows:

$$\pi_{k,t+h}^e \equiv E^P\left[\pi_{t+h} \mid s_t = j, s_{t+h} = k\right] \tag{A.5}$$

$$= \mathbf{C}_{\pi,k}E^P\left[\mathbf{X}_{t+h} \mid s_t = j, s_{t+h} = k\right] \tag{A.6}$$

$$= \mathbf{C}_{\pi,k}\mathbf{M}_h^{(j\to k)}\mathbf{X}_t \tag{A.7}$$

where $\mathbf{M}_h^{(j\to k)}$ is the expected propagation matrix conditional on starting in state j and ending in state k:

$$\mathbf{M}_h^{(j\to k)} = \frac{1}{P_{jk}^{(h)}} \sum_{m_1,\cdots,m_{h-1}} \left(\prod_{\ell=0}^{h-1} P_{m_\ell, m_{\ell+1}} \right) \mathbf{A}_{m_h} \mathbf{A}_{m_{h-1}} \cdots \mathbf{A}_{m_1} \tag{A.8}$$

with $m_0 = j$ and $m_h = k$. Using the definition of covariance:

$$\mathrm{Cov}^P \left[\phi_{\pi,s_{t+h}}, \pi_{t+h} \mid s_t = j \right] = E^P \left[\phi_{\pi,s_{t+h}} \pi_{t+h} \mid s_t = j \right] - E^P \left[\phi_{\pi,s_{t+h}} \mid s_t = j \right] E^P \left[\pi_{t+h} \mid s_t = j \right] \tag{A.9}$$

Expanding the terms using the law of iterated expectations over the future state $k \in \{1, 2\}$:

$$E^P \left[\phi_{\pi,s_{t+h}} \pi_{t+h} \mid s_t = j \right] = \sum_k P_{jk}^{(h)} \phi_{\pi,k} \pi_{k,t+h}^e \tag{A.10}$$

$$E^P \left[\pi_{t+h} \mid s_t = j \right] = \sum_k P_{jk}^{(h)} \pi_{k,t+h}^e \tag{A.11}$$

$$E^P \left[\phi_{\pi,s_{t+h}} \mid s_t = j \right] = \sum_k P_{jk}^{(h)} \phi_{\pi,k} \tag{A.12}$$

Substituting these back into the covariance expression:

$$\mathrm{Cov}_t^P = \sum_k P_{jk}^{(h)} \phi_{\pi,k} \pi_{k,t+h}^e - \left(\sum_k P_{jk}^{(h)} \pi_{k,t+h}^e \right) \left(\sum_k P_{jk}^{(h)} \phi_{\pi,k} \right) \tag{A.13}$$

$$\begin{aligned} &= P_{j1}^{(h)} \phi_{\pi,1} \pi_{1,t+h}^e + P_{j2}^{(h)} \phi_{\pi,2} \pi_{2,t+h}^e \\ &\quad - \left(P_{j1}^{(h)} \pi_{1,t+h}^e + P_{j2}^{(h)} \pi_{2,t+h}^e \right) \left(P_{j1}^{(h)} \phi_{\pi,1} + P_{j2}^{(h)} \phi_{\pi,2} \right) \end{aligned} \tag{A.14}$$

Expanding and collecting terms, noting that $P_{j2}^{(h)} = 1 - P_{j1}^{(h)}$ in the two-state case:

$$\begin{aligned} \mathrm{Cov}_t^P &= P_{j1}^{(h)} \pi_{1,t+h}^e \left[\phi_{\pi,1} - \left(P_{j1}^{(h)} \phi_{\pi,1} + P_{j2}^{(h)} \phi_{\pi,2} \right) \right] \\ &\quad + P_{j2}^{(h)} \pi_{2,t+h}^e \left[\phi_{\pi,2} - \left(P_{j1}^{(h)} \phi_{\pi,1} + P_{j2}^{(h)} \phi_{\pi,2} \right) \right] \end{aligned} \tag{A.15}$$

$$= P_{j1}^{(h)} \pi_{1,t+h}^e \left[P_{j2}^{(h)} (\phi_{\pi,1} - \phi_{\pi,2}) \right] + P_{j2}^{(h)} \pi_{2,t+h}^e \left[P_{j1}^{(h)} (\phi_{\pi,2} - \phi_{\pi,1}) \right] \tag{A.16}$$

$$= P_{j1}^{(h)} P_{j2}^{(h)} (\phi_{\pi,1} - \phi_{\pi,2}) \pi_{1,t+h}^e - P_{j1}^{(h)} P_{j2}^{(h)} (\phi_{\pi,1} - \phi_{\pi,2}) \pi_{2,t+h}^e \tag{A.17}$$

$$= P_{j1}^{(h)} (1 - P_{j1}^{(h)}) (\phi_{\pi,1} - \phi_{\pi,2}) (\pi_{1,t+h}^e - \pi_{2,t+h}^e) \tag{A.18}$$

This concludes the proof. ■

A.3 Derivation of Corollary 2 (i.i.d. Case)

Using the method of undetermined coefficients, the inflation and output gap loadings on the exogenous shocks are solved under the assumption of i.i.d. regime switching. It is conjectured that the equilibrium laws of motion for inflation and the output gap are linear in the exogenous state vector $\mathbf{z}_t = [z_{Dt}, z_{St}, z_{Mt}]'$ with regime-dependent coefficients:

$$\pi_t = \mathbf{a}_{s_t}\mathbf{z}_t \tag{A.19}$$

$$x_t = \mathbf{b}_{s_t}\mathbf{z}_t \tag{A.20}$$

where $\mathbf{a}_{s_t}$ and $\mathbf{b}_{s_t}$ are 1×2 vectors corresponding to the regime $s_t \in \{1, 2\}$. Specifically, for the demand and supply components, the scalars $a_{s_t,k}$ and $b_{s_t,k}$ for $k \in \{D, S\}$ are solved. The resulting coefficients for the demand shock (z_D) are:

$$\begin{bmatrix} a_{1,D} \\ a_{2,D} \\ b_{1,D} \\ b_{2,D} \end{bmatrix} = \kappa\vartheta_D^{-1} \begin{bmatrix} 1 + \kappa\tau\phi_{\pi,2} \\ 1 + \kappa\tau\phi_{\pi,1} \\ (1 + \kappa\tau\phi_{\pi,2}) - \beta\rho_D[p(1 + \kappa\tau\phi_{\pi,2}) + (1-p)(1 + \kappa\tau\phi_{\pi,1})] \\ (1 + \kappa\tau\phi_{\pi,1}) - \beta\rho_D[p(1 + \kappa\tau\phi_{\pi,2}) + (1-p)(1 + \kappa\tau\phi_{\pi,1})] \end{bmatrix} \tag{A.21}$$

where the scalar denominator ϑ_D is defined as:

$$\vartheta_D = (1 + \kappa\tau\phi_{\pi,1})(1 + \kappa\tau\phi_{\pi,2}) + \rho_D(-\kappa\tau - \zeta)[(1-p)(1 + \kappa\tau\phi_{\pi,1}) + p(1 + \kappa\tau\phi_{\pi,2})] \tag{A.22}$$

Similarly, for the supply shock (z_S), the coefficients are given by:

$$\begin{bmatrix} a_{1,S} \\ a_{2,S} \\ b_{1,S} \\ b_{2,S} \end{bmatrix} = \begin{bmatrix} \frac{(1-\rho_S)(1+\kappa\tau\phi_{\pi,2})}{\vartheta_S} \\ \frac{(1-\rho_S)(1+\kappa\tau\phi_{\pi,1})}{\vartheta_S} \\ \frac{(1-\rho_S)[(1-\beta p\rho_S)(1+\kappa\tau\phi_{\pi,2}) - \beta(1-p)\rho_S(1+\kappa\tau\phi_{\pi,1})] - \vartheta_S}{\kappa\vartheta_S} \\ \frac{(1-\rho_S)[-\beta p\rho_S(1+\kappa\tau\phi_{\pi,2}) + (1-\beta(1-p)\rho_S)(1+\kappa\tau\phi_{\pi,1})] - \vartheta_S}{\kappa\vartheta_S} \end{bmatrix} \tag{A.23}$$

where the denominator ϑ_S is defined as:

$$\vartheta_S = (1 + \kappa\tau\phi_{\pi,1})(1 + \kappa\tau\phi_{\pi,2}) + \rho_S\zeta_S[p(1 + \kappa\tau\phi_{\pi,2}) + (1-p)(1 + \kappa\tau\phi_{\pi,1})] \tag{A.24}$$

Substituting these coefficients into the expressions for $a_{1,D} - a_{2,D}$ and $a_{1,S} - a_{2,S}$ yields the results presented in Corollary 2. ■

A.4 Proof of Proposition 3

Step 1: the perceived block. The subjective model is a standard three equation New Keynesian model. It is determinate if and only if $\phi_\pi^{\mathrm{P}} > 1$, the proof being standard as noted in Galí (2015). The subjective solutions are $\pi_{t|t} = x_{t|t} = 0$, hence

$$E_t^{\mathrm{P}}\left[\pi_{t+1}\right] = E_t^{\mathrm{P}}\left[x_{t+1}\right] = 0 \qquad \text{for all } t. \tag{A.25}$$

Step 2: the actual block. Imposing (A.25) on the actual IS and Phillips curves, transmission collapses to the contemporaneous real-rate gap,

$$x_t = -\tau\left(i_t - r_t^n\right), \qquad \pi_t = \kappa x_t. \tag{A.26}$$

The actual rule (48) gives $i_t - r_t^n = \phi_\pi^{\mathrm{F}}\pi_t$, so (A.26) reduces to $\left(1 + \tau\kappa\phi_\pi^{\mathrm{F}}\right)x_t = 0$. Since $\tau, \kappa > 0$ and $\phi_\pi^{\mathrm{F}} \geq 0$ the bracket is bounded below by one, so the unique solution is $x_t = \pi_t = 0$ and therefore $i_t = r_t^n$. ■

A.5 Derivation of the ALM for the SP-MSNK Model

In this appendix, the structural matrices for the actual law of motion described in Section 5.3 are derived. Since the actual Taylor rule is also based on the private sector's expectation, there is no rational expectation component in the state-space. This means that after substituting the subjective expectations, we are left with

$$\mathbf{H}_\ell \mathbf{y}_t + \mathbf{F}_\ell \mathbf{y}_{t-1} + \mathbf{C}_\ell \boldsymbol{\varepsilon}_t = \mathbf{0} \tag{A.27}$$

Provided that $\mathbf{H}_\ell$ is non-singular, the Actual Law of Motion (ALM) is simply given by:

$$\mathbf{y}_t = \underbrace{-\mathbf{H}_\ell^{-1}\mathbf{F}_\ell}_{\mathbf{A}_\ell^{\mathrm{ALM}}}\mathbf{y}_{t-1} + \underbrace{-\mathbf{H}_\ell^{-1}\mathbf{C}_\ell}_{\mathbf{B}_\ell^{\mathrm{ALM}}}\boldsymbol{\varepsilon}_t \tag{A.28}$$

Hence, it boils down to specifying the three loading matrices.

Let the actual joint state be $\ell \in \mathcal{S}$, corresponding to the pair $(s_t^p = j, s_t^f)$. The private sector forms expectations based on the perceived state j, whereas the parameters of the actual rule depend only on s_t^f. The combined matrices $\mathbf{H}_\ell$, $\mathbf{F}_\ell$ and $\mathbf{C}_\ell$ therefore carry the joint index ℓ, since each mixes the two sides. Using the derived behavioral equation $E_t^{\mathrm{P}}\left[\mathbf{y}_{t+1}\right] = \boldsymbol{\Gamma}_{y,j}\mathbf{y}_{t-1} + \boldsymbol{\Gamma}_{\varepsilon,j}\boldsymbol{\varepsilon}_t$, I decompose the inflation and output gap expectations. Let e_π and e_x be selection vectors such

that $\pi = e_\pi \mathbf{y}$ and $x = e_x \mathbf{y}$. The expectations are:

$$E_t^{\mathrm{P}}\left[\pi_{t+1}\right] = e_\pi\left(\mathbf{\Gamma}_{y,j}\mathbf{y}_{t-1} + \mathbf{\Gamma}_{\varepsilon,j}\boldsymbol{\varepsilon}_t\right) \tag{A.29}$$

$$E_t^{\mathrm{P}}\left[x_{t+1}\right] = e_x\left(\mathbf{\Gamma}_{y,j}\mathbf{y}_{t-1} + \mathbf{\Gamma}_{\varepsilon,j}\boldsymbol{\varepsilon}_t\right) \tag{A.30}$$

I define the aggregate expectation coefficients for the structural equations as:

$$\tilde{\mathbf{\Gamma}}_A = \theta e_x \mathbf{\Gamma}_{y,j} + \tau e_\pi \mathbf{\Gamma}_{y,j} \qquad \tilde{\mathbf{\Gamma}}_B = \theta e_x \mathbf{\Gamma}_{\varepsilon,j} + \tau e_\pi \mathbf{\Gamma}_{\varepsilon,j} \tag{A.31}$$

$$\tilde{\mathbf{\Gamma}}_C = \beta e_\pi \mathbf{\Gamma}_{y,j} \qquad \tilde{\mathbf{\Gamma}}_D = \beta e_\pi \mathbf{\Gamma}_{\varepsilon,j} \tag{A.32}$$

and for the Taylor Rule (with smoothing parameter $\Lambda_{s_t^f} = 1 - \rho^{\mathrm{F}}_{1,s_t^f} - \rho^{\mathrm{F}}_{2,s_t^f}$):

$$\tilde{\mathbf{\Gamma}}_E = \Lambda_{s_t^f}\left(\phi^{\mathrm{F}}_{\pi,s_t^f} e_\pi \mathbf{\Gamma}_{y,j} + \phi^{\mathrm{F}}_{x,s_t^f} e_x \mathbf{\Gamma}_{y,j}\right) \qquad \tilde{\mathbf{\Gamma}}_F = \Lambda_{s_t^f}\left(\phi^{\mathrm{F}}_{\pi,s_t^f} e_\pi \mathbf{\Gamma}_{\varepsilon,j} + \phi^{\mathrm{F}}_{x,s_t^f} e_x \mathbf{\Gamma}_{\varepsilon,j}\right) \tag{A.33}$$

These expressions are substituted into the actual structural equations.

1. IS Curve:

$$\begin{aligned} x_t &= \theta E_t^{\mathrm{P}}\left[x_{t+1}\right] - \tau\left(i_t - E_t^{\mathrm{P}}\left[\pi_{t+1}\right]\right) + z_{Dt} \\ 0 &= -x_t - \tau i_t + z_{Dt} + \tilde{\mathbf{\Gamma}}_A \mathbf{y}_{t-1} + \tilde{\mathbf{\Gamma}}_B \boldsymbol{\varepsilon}_t \end{aligned} \tag{A.34}$$

2. Phillips Curve:

$$\begin{aligned} \pi_t &= \beta E_t^{\mathrm{P}}\left[\pi_{t+1}\right] + \kappa x_t + z_{St} \\ 0 &= \kappa x_t - \pi_t + z_{St} + \tilde{\mathbf{\Gamma}}_C \mathbf{y}_{t-1} + \tilde{\mathbf{\Gamma}}_D \boldsymbol{\varepsilon}_t \end{aligned} \tag{A.35}$$

3. Actual Taylor Rule: The actual rule includes the explicit monetary policy shock ε_{Mt}, which is absent from $\boldsymbol{\varepsilon}_t^P$.

$$\begin{aligned} i_t &= \Lambda_{s_t^f}\left[\phi^{\mathrm{F}}_{\pi,s_t^f} E_t^{\mathrm{P}}\left[\pi_{t+1}\right] + \phi^{\mathrm{F}}_{x,s_t^f} E_t^{\mathrm{P}}\left[x_{t+1}\right]\right] + \rho^{\mathrm{F}}_{1,s_t^f} i_{t-1} + \rho^{\mathrm{F}}_{2,s_t^f} i_{t-2} + \varepsilon_{Mt} \\ 0 &= -i_t + \tilde{\mathbf{\Gamma}}_E \mathbf{y}_{t-1} + \tilde{\mathbf{\Gamma}}_F \boldsymbol{\varepsilon}_t + \rho^{\mathrm{F}}_{1,s_t^f} i_{t-1} + \rho^{\mathrm{F}}_{2,s_t^f} i_{t-2} + \varepsilon_{Mt} \end{aligned} \tag{A.36}$$

This system is written in the matrix form $\mathbf{0} = \mathbf{H}_\ell \mathbf{y}_t + \mathbf{F}_\ell \mathbf{y}_{t-1} + \mathbf{C}_\ell \boldsymbol{\varepsilon}_t$. The matrix $\mathbf{H}_\ell$ is:

$$\mathbf{H}_\ell = \begin{bmatrix} -1 & 0 & -\tau & 0 & 1 & 0 \\ \kappa & -1 & 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 \end{bmatrix} \tag{A.37}$$

The matrix $\mathbf{F}_\ell$ incorporates the expectation effects plus the structural persistence:

$$\mathbf{F}_\ell = \begin{bmatrix} \tilde{\mathbf{\Gamma}}_A \\ \tilde{\mathbf{\Gamma}}_C \\ \tilde{\mathbf{\Gamma}}_E + \mathbf{R}_{\text{lag}} \\ \mathbf{I}_{i_{t-1}} \\ \mathbf{I}_D \\ \mathbf{I}_S \end{bmatrix} \tag{A.38}$$

where $\mathbf{R}_{\text{lag}}$ adds $\rho^{\text{F}}_{1,s_t^f}$ to the i_{t-1} column and $\rho^{\text{F}}_{2,s_t^f}$ to the i_{t-2} column. The lower blocks $\mathbf{I}_{(\cdot)}$ represent the AR processes for i_{t-1}, z_D, and z_S (e.g., $i_{t-1} = i_{t-1}$, $z_{Dt} = \rho_D z_{Dt-1} + \varepsilon_{Dt}$).

Finally, the shock matrix $\mathbf{C}_\ell$ is:

$$\mathbf{C}_\ell = \begin{bmatrix} \tilde{\mathbf{\Gamma}}_{B,D} & \tilde{\mathbf{\Gamma}}_{B,S} & 0 \\ \tilde{\mathbf{\Gamma}}_{D,D} & \tilde{\mathbf{\Gamma}}_{D,S} & 0 \\ \tilde{\mathbf{\Gamma}}_{F,D} & \tilde{\mathbf{\Gamma}}_{F,S} & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \tag{A.39}$$

where $\tilde{\mathbf{\Gamma}}_{B,D}$ denotes the column of $\tilde{\mathbf{\Gamma}}_B$ corresponding to the demand shock. Note that the $(3,3)$ element is exactly 1, representing the actual monetary policy shock, while the perceived impact of this shock via $\tilde{\mathbf{\Gamma}}_F$ is zero. ■

B Additional Simulation Results

B.1 TVP estimation with no regime uncertainty

Figure B1 serves as a consistency check for the case of permanent regimes ($p = 1.00$). In the absence of regime uncertainty, both the nowcast and forecast specifications recover the true

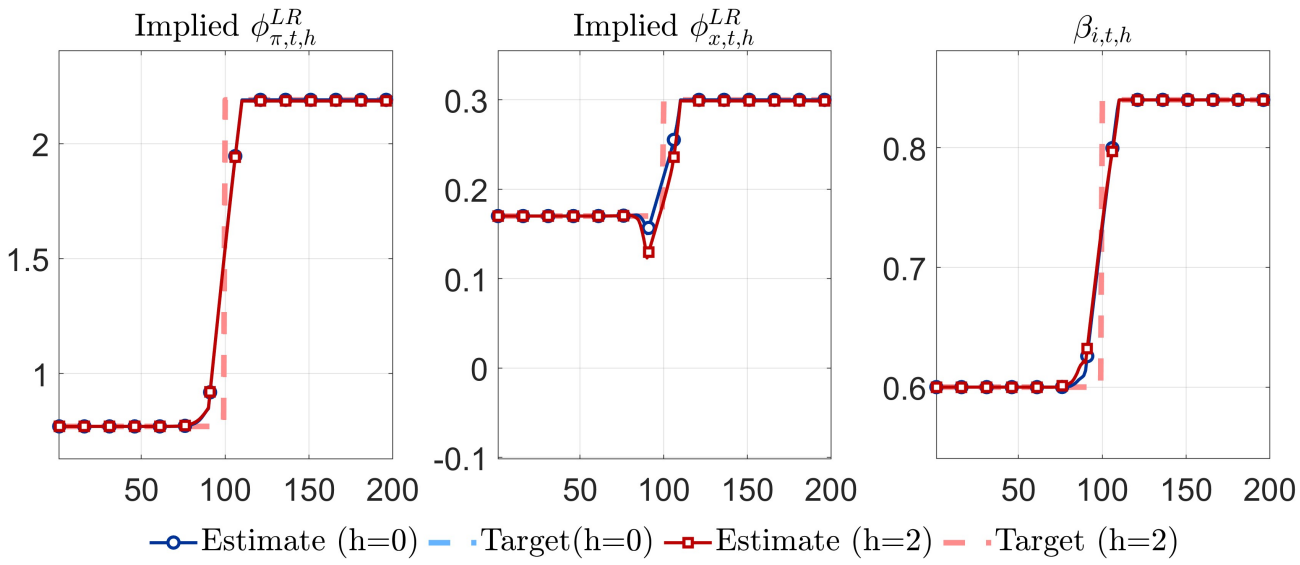


Figure B1: TVP Estimation with No Switching ($p = 1.00$)

Note: This figure compares the estimation results of equation (28) for $h = 0$ (blue) and $h = 2$ (red) under $p = 1.00$. The dashed lines indicate the target conditional expectations $\mathbb{E}_t[\phi_{\pi,s_{t+h}}]$. The solid lines with markers indicate the estimated long-run coefficients.

parameters. However, as documented in Section 4, the empirical data strongly reject the null hypothesis of constant parameters.

B.2 Endogenous Regime Switching

This section explores the robustness of the identification strategy to endogenous regime switching, where the probability of a policy regime shift depends on the state of the economy. Specifically, I consider a specification where the transition probabilities are functions of the current inflation level.

Simulation Setup The simulation follows the baseline Subjective Perception MSNK model with a state-dependent transition matrix $\mathbf{P}(\pi_t)$. The diagonal elements $p_{jj}(\pi_t)$, representing regime persistence, are modeled using a logistic function centered around the inflation target. Let $L(\pi_t)$ be the logistic component:

$$L(\pi_t) = \frac{1}{1 + \exp\left(-\frac{g}{\sigma}(\pi_t - c_\pi)\right)} \tag{B.1}$$

where g governs the steepness and σ is a scaling parameter. The regime-specific persistence probabilities are defined as:

$$p_{11}(\pi_t) = \text{clip}\left(\bar{p} + \delta_p \left(L(\pi_t) - 0.5\right)\right) \tag{B.2}$$

$$p_{22}(\pi_t) = \text{clip}\left(\bar{p} - \delta_p \left(L(\pi_t) - 0.5\right)\right) \tag{B.3}$$

where the function $\text{clip}(x) = \min(\max(x, 0.001), 0.999)$ ensures probabilities remain within valid bounds. $\bar{p}$ is the steady-state persistence (0.95), and δ_p controls the maximum amplitude of the probability deviation. Under this specification, higher inflation ($\pi_t > c_\pi$) increases the persistence of the Hawkish regime (p_{11}) and decreases the persistence of the Dovish regime (p_{22}), making a switch to (or retention of) active policy more likely.

Results The TVP estimation results on the simulated data confirm the findings from the exogenous switching case:

1. **Nowcast ($h = 0$):** The nowcast-based estimator successfully recovers the true time-varying perceived policy coefficient $\phi_{\pi,t}$ without systematic bias.

2. **Forecast ($h = 2$):** The forecast-based estimates exhibit a significant bias, confirming that the "perceived covariance" bias persists when agents internalize endogenous regime dynamics.

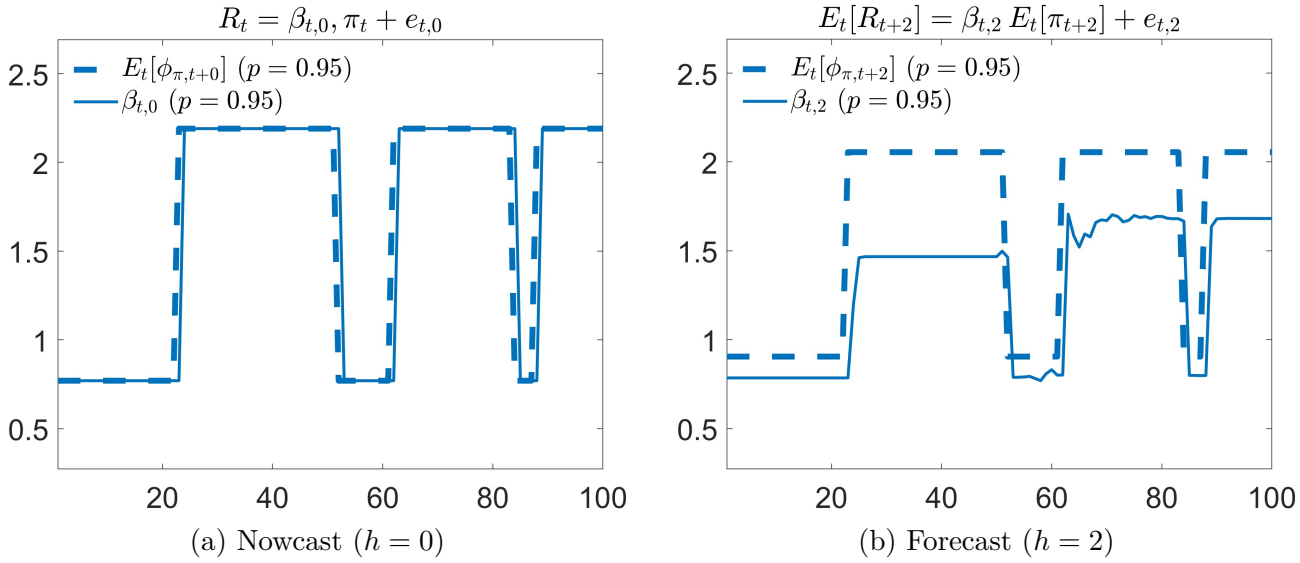
B.3 Instrumental Variable Estimation

A potential concern with the OLS results presented in Section 3 is that the bias might stem from standard simultaneity issues—specifically, the correlation between inflation and monetary policy shocks—rather than the structural "perceived covariance" channel. To address this, I investigate whether standard Instrumental Variable (IV) strategies can recover the true policy parameter in the presence of regime switching.

Methodology I estimate the policy rule on data simulated from the MSNK model using the Generalized Method of Moments (GMM), following the standard approach in the literature (Clarida, Gali, and Gertler, 2000). The estimating equation is specified in terms of h-period ahead expectations, consistent with the identification of the perceived rule:

$$\mathbb{E}_t[i_{t+h}] = \beta_{IV} \mathbb{E}_t[\pi_{t+h}] + u_{t,h} \tag{B.4}$$

I consider two distinct sets of instruments $\mathbf{Z}_t \in \mathcal{I}_t$:



(a) Nowcast $(h = 0)$ (b) Forecast $(h = 2)$

Figure B2: TVP Estimation under Endogenous Regime Switching

Note: This figure displays the TVP estimation results on data simulated from the endogenous regime switching model described in Appendix B.2. The left panel shows the estimates using nowcasts $(h = 0)$, and the right panel shows the estimates using 2-period ahead forecasts $(h = 2)$. The solid colored lines represent the posterior median of the estimated time-varying coefficient $\beta_{t,h}$, while the dashed lines represent the true model-implied conditional expectation $E_t[\phi_{\pi,t+h}]$. Shaded regions indicate 68% confidence intervals.

1. **Lagged Variables:** $\mathbf{Z}_t = [\pi_{t-1}, \pi_{t-2}]'$. This follows the standard New Keynesian identification strategy assuming that lagged variables are orthogonal to the current monetary policy shock (Clarida, Gali, and Gertler, 2000).

2. **Structural Shocks:** $\mathbf{Z}_t = [z_{D,t}, z_{S,t}]'$. This set uses the realized non-monetary structural shocks (demand and supply) as instruments. This approach mirrors the identification logic of Carvalho, Nechio, and Tristao (2021), who use identified exogenous shocks to instrument for inflation.

Results To conserve space, result for $h = 2$ is reported. Figure B3 compares the OLS estimates with the two GMM-IV estimators across varying degrees of regime persistence p. The results demonstrate that the IV estimates (dashed lines) are virtually indistinguishable from the biased OLS estimates (solid blue line). As regime uncertainty increases (p decreases), all estimators systematically diverge from the true unconditional mean (black dashed line).

This failure of IV estimation confirms that the bias is not driven by standard simultaneity between $\mathbb{E}_t[\pi_{t+h}]$ and an exogenous policy shock. Instead, as derived in Proposition 1, the effective "error term" in this regression contains the conditional covariance between the policy parameter and inflation, $\text{Cov}_t(\phi_{\pi,t+h}, \pi_{t+h})$. Because this covariance is an endogenous object that depends on the state of the economy, it is correlated with both the regressors (expected

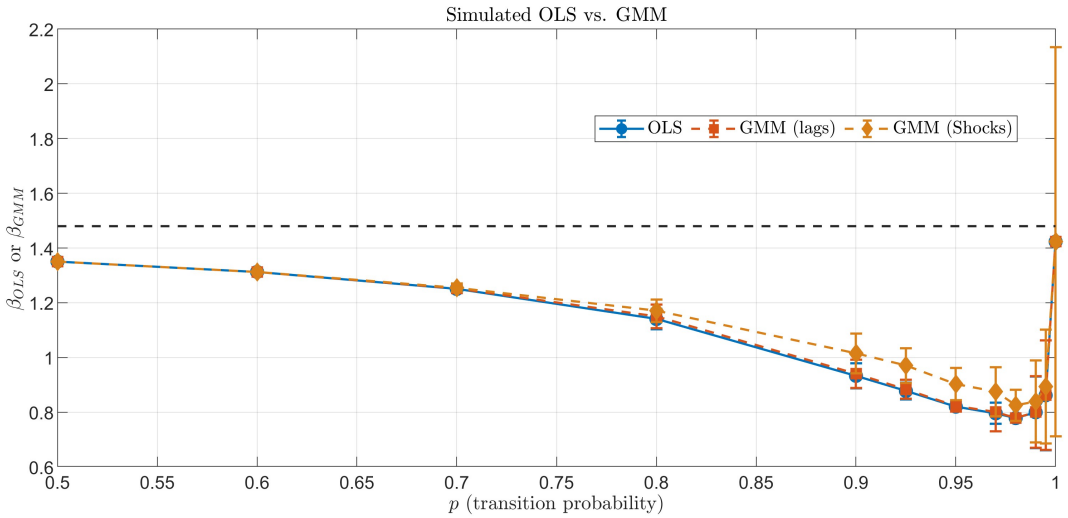


Figure B3: Comparison of OLS and IV Estimates under Regime Uncertainty

Note: The figure displays Monte Carlo estimates of the policy coefficient β across varying degrees of regime persistence p for the forecast horizon $h = 2$. The blue line represents OLS estimates. The red and yellow dashed lines represent GMM estimates using lagged variables and structural shocks as instruments, respectively. The black dashed line indicates the true unconditional mean of the policy parameter.

inflation) *and* the valid instruments (lags or supply shocks). Consequently, the exclusion restriction is structurally violated, rendering standard IV methods ineffective for correcting the perceived covariance bias.

B.4 General Framework with Heterogeneous Expectations Operators

Section 5.1 derives the actual law of motion under the assumption that every expectation entering the actual system is the private sector's subjective expectation. This appendix records the more general case, in which the central bank and the private sector hold different expectations operators. The private sector's side of the model is untouched, so the perceived law of motion (38) and the behavioral equation (40) carry over verbatim.

Let the central bank form rational expectations $\mathbb{E}_t[\cdot]$ under the actual transition matrix $\mathbf{P}$ and the true joint state s_t, so that its information set weakly nests the private sector's. The actual structural system then carries both operators:

$$\boldsymbol{\Gamma}^{\mathrm{A}}_{\mathrm{RE},\ell}\mathbb{E}_t\left[\mathbf{y}_{t+1}\right] + \boldsymbol{\Gamma}^{\mathrm{A}}_{0,\ell}E^{\mathrm{P}}_t\left[\mathbf{y}_{t+1}\right] + \boldsymbol{\Gamma}^{\mathrm{A}}_{1,\ell}\mathbf{y}_t + \boldsymbol{\Gamma}^{\mathrm{A}}_{2,\ell}\mathbf{y}_{t-1} + \boldsymbol{\Gamma}^{\mathrm{A}}_{3,\ell}\boldsymbol{\varepsilon}_t = \mathbf{0} \tag{B.5}$$

where the numbered matrices are exactly those of (41) and $\boldsymbol{\Gamma}^{\mathrm{A}}_{\mathrm{RE},\ell} = \mathbf{0}$ recovers the baseline. Substituting (40) together with the mapping $\boldsymbol{\varepsilon}^p_t = \mathbf{Q}\boldsymbol{\varepsilon}_t$ eliminates the subjective expectation

but leaves the rational expectation term intact:

$$\mathbf{\Gamma}^{\mathrm{A}}_{\mathrm{RE},\ell}\mathbb{E}_t\left[\mathbf{y}_{t+1}\right] + \mathbf{H}_\ell \mathbf{y}_t + \mathbf{F}_\ell \mathbf{y}_{t-1} + \mathbf{C}_\ell \boldsymbol{\varepsilon}_t = \mathbf{0} \tag{B.6}$$

with $\mathbf{H}_\ell$, $\mathbf{F}_\ell$ and $\mathbf{C}_\ell$ defined exactly as in Section 5.1. The contrast with the baseline is that this system is no longer backward-looking, so it cannot be inverted state by state. It must be solved jointly across all $S = S_p \times S_f$ joint states with a Markov-switching rational expectations algorithm (Maih, 2015), which delivers an ALM of the same form as (44),

$$\mathbf{y}_t = \mathbf{A}_\ell^{\mathrm{ALM,RE}}\mathbf{y}_{t-1} + \mathbf{B}_\ell^{\mathrm{ALM,RE}}\boldsymbol{\varepsilon}_t \tag{B.7}$$

though with loading matrices that now depend on the entire joint transition matrix $\mathbf{P}$ rather than on the state ℓ alone. Appendix B.5 implements this case in the three-equation model of Section 5.3 and reports the quantitative comparison, which is close to indistinguishable from the baseline.

B.5 SP-MSNK Model with Rational Expectation of the Fed

In the baseline model presented in Section 5, the Fed is assumed to respond to the private sector's expectations ($E_t^{\mathrm{P}}[\cdot]$). This assumption simplifies the solution of the actual law of motion to a recursive, backward-looking system. In this section, I relax this assumption and consider a central bank that operates under Full Information Rational Expectations (FIRE). Specifically, the Fed knows the true structure of the economy, including the agents' misperceived law of motion, and responds to the mathematical rational expectation of future variables ($\mathbb{E}_t[\cdot]$). This case is meaningful as the only departure from the standard FIRE 3-eq NK model would be the misspecified monetary policy rule. The change in determinacy conditions will be solely attributed to the separation of perceived and actual rules.

Model Setup The private sector's behavior remains identical to the baseline. Agents form expectations using the Perceived Law of Motion (PLM) defined by the matrices $\widehat{\mathbf{A}}_{s_t^p}$ and $\widehat{\mathbf{B}}_{s_t^p}$, and the subjective transition matrix $\widehat{\mathbf{P}}$. Consequently, the behavioral equations for private expectations are predetermined linear functions of the state vector:

$$E_t^{\mathrm{P}}\left[\mathbf{y}_{t+1}\right] = \mathbf{\Gamma}_{y,s_t^p}\mathbf{y}_{t-1} + \mathbf{\Gamma}_{\varepsilon,s_t^p}\boldsymbol{\varepsilon}_t \tag{B.8}$$

where $\boldsymbol{\varepsilon}_t$ is the vector of actual structural shocks.

The actual economy is governed by the joint state $s_t = (s_t^p, s_t^f)$. The IS and Phillips curves

are structurally identical to the baseline, as they describe private sector decision-making rules:

$$x_t = \theta E_t^{\mathrm{P}}\left[x_{t+1}\right] - \tau\left(i_t - E_t^{\mathrm{P}}\left[\pi_{t+1}\right]\right) + z_{Dt} \tag{B.9}$$

$$\pi_t = \beta E_t^{\mathrm{P}}\left[\pi_{t+1}\right] + \kappa x_t + z_{St} \tag{B.10}$$

However, the Federal Reserve's reaction function is now specified as:

$$i_t = \left(1 - \rho^{\mathrm{F}}_{s_t^f}\right)\left[\phi^{\mathrm{F}}_{\pi,s_t^f}\mathbb{E}_t\left[\pi_{t+1}\right] + \phi^{\mathrm{F}}_{x,s_t^f}\mathbb{E}_t\left[x_{t+1}\right]\right] + \rho^{\mathrm{F}}_{1,s_t^f} i_{t-1} + \rho^{\mathrm{F}}_{2,s_t^f} i_{t-2} + \varepsilon_{Mt} \tag{B.11}$$

where $\mathbb{E}_t[\cdot]$ denotes the mathematical expectation operator conditional on the actual joint regime s_t and the actual transition matrix $\mathbf{P}$. I assume that the Fed's state s_t^f transitions with the same probability as $\widehat{\mathbf{P}}$, which gives an implied actual transition probability matrix $\mathbf{P}$. However, the results are robust to alternative characterization of the actual transition probability matrix $\mathbf{P}$ since the perceived transition probability is what eventually matters for expectation.

Derivation and Solution Substituting the private sector's predetermined expectations into the IS and Phillips curves allow us to express equations (B.9) and (B.10) as:

$$0 = -x_t - \tau i_t + z_{Dt} + \tilde{\mathbf{\Gamma}}_A \mathbf{y}_{t-1} + \tilde{\mathbf{\Gamma}}_B \boldsymbol{\varepsilon}_t \tag{B.12}$$

$$0 = \kappa x_t - \pi_t + z_{St} + \tilde{\mathbf{\Gamma}}_C \mathbf{y}_{t-1} + \tilde{\mathbf{\Gamma}}_D \boldsymbol{\varepsilon}_t \tag{B.13}$$

where ℓ indexes the joint state s_t. The aggregate coefficient matrices $\tilde{\mathbf{\Gamma}}$ are exactly those of Appendix A.5, so that $\tilde{\mathbf{\Gamma}}_A = \theta e_x \mathbf{\Gamma}_{y,j} + \tau e_\pi \mathbf{\Gamma}_{y,j}$ evaluated at the perceived component j of ℓ, and similarly for the remaining blocks.

Combining these with the RE Taylor rule (B.11) and collecting terms places the system in the general form of Appendix B.4:

$$\mathbf{0} = \mathbf{\Gamma}^{\mathrm{A}}_{\mathrm{RE},\ell}\mathbb{E}_t\left[\mathbf{y}_{t+1}\right] + \mathbf{H}_\ell \mathbf{y}_t + \mathbf{F}_\ell \mathbf{y}_{t-1} + \mathbf{C}_\ell \boldsymbol{\varepsilon}_t \tag{B.14}$$

The matrices $\mathbf{H}_\ell$, $\mathbf{F}_\ell$ and $\mathbf{C}_\ell$ are the ones derived in Appendix A.5, so the only new object is $\mathbf{\Gamma}^{\mathrm{A}}_{\mathrm{RE},\ell}$. It is zero in the IS and Phillips curve rows, and in the row of the Taylor rule it collects the Fed's response to the rational expectation of inflation and the output gap,

$$\left(1 - \rho^{\mathrm{F}}_{s_t^f}\right)\left(\phi^{\mathrm{F}}_{\pi,s_t^f} e_\pi + \phi^{\mathrm{F}}_{x,s_t^f} e_x\right) \tag{B.15}$$

Unlike the baseline model, where the actual law of motion reduces to a purely backward-looking system, this system retains forward-looking elements due to the Fed's reaction function. I solve

it with the algorithm of Maih (2015), which returns the loadings $\mathbf{A}_\ell^{\text{ALM,RE}}$ and $\mathbf{B}_\ell^{\text{ALM,RE}}$ of Appendix B.4.

Two-State Calibration The exercise in this appendix is implemented under a two-state alternative calibration rather than the six-state specification used in the main text. The reason is computational: under the RE-Fed specification, the actual law of motion is no longer backward-looking, and the Markov-Switching Rational Expectations system must be solved jointly across all $S = S_p \times S_f$ regimes. With the baseline six perceived regimes and six actual regimes, this entails solving a 36-regime MS-RE system, which is prohibitively costly given the iterative nature of the Maih (2015) algorithm. A two-state Hawk–Dove specification ($S_p = S_f = 2$, hence $S = 4$) preserves the qualitative comparison between the PE-Fed and RE-Fed models at a fraction of the computational cost.

The calibration follows the earlier version of this working paper, in which the TVP-estimated perceived rule was approximated by a two-state (Hawk–Dove) Markov chain rather than the six-regime HMM discretization used in the current main text. Structural parameters are $\beta = 0.99$, $\tau = 1.00$, $\kappa = 0.057$, and $\theta = 0.85$, with the cognitive discounting factor $\theta < 1$ included to attenuate the forward-guidance puzzle in the RE solution (Gabaix, 2020; Pfäuti and Seyrich, 2022). Shock persistence is set to $\rho_D = \rho_S = 0.85$. The transition probabilities are calibrated by maximum likelihood on the historical regime frequency over 1983–2019, yielding $\widehat{p}_{11} = 0.978$ (Hawk) and $\widehat{p}_{22} = 0.964$ (Dove). The regime-dependent policy parameters are obtained by matching the first and second moments of the time-varying perceived rule estimates: the Hawk regime has $\phi_{\pi,1} = 1.723$, $\phi_{x,1} = 0.699$ (annualized), $\rho_{1,1} = 1.287$, $\rho_{2,1} = -0.391$; the Dove regime has $\phi_{\pi,2} = 0.451$, $\phi_{x,2} = 0.203$ (annualized), $\rho_{1,2} = 1.130$, $\rho_{2,2} = -0.282$. Because this exercise is intended only to verify that the Fed's expectations operator does not materially affect the conclusions, the qualitative comparison between PE-Fed and RE-Fed specifications carries over to the six-state baseline.

Figure B4 and B5 compare the counterfactual simulation results of the baseline model (Fed responds to E^P) against the RE-Fed model (Fed responds to $\mathbb{E}$). PE Fed refers to the Fed responding to the private sector's subjective expectation (dashed blue or red). The dashed green line represents the result under the rational expectation of the actual rule. The dynamics are virtually indistinguishable. The marginal impact of the Fed adopting Rational Expectations and, hence, the disagreement in expectation is negligible compared to the impact of private sector misperceptions about the rule.

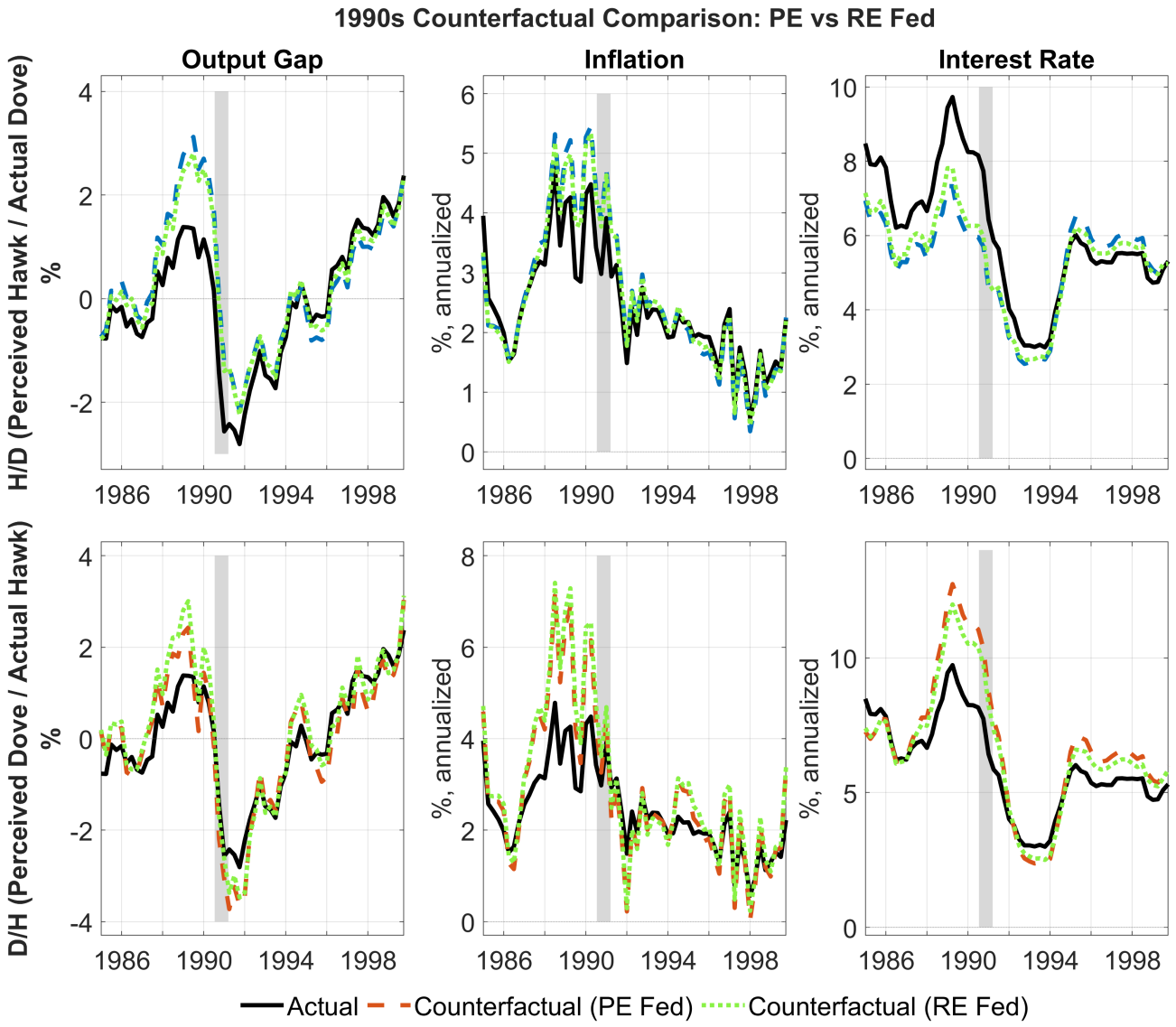


Figure B4: Counterfactual simulations: Fed Responding to E^P vs. $\mathbb{E}$

Note: The figure compares impulse responses to a demand shock under the baseline specification (blue solid line), where the Fed responds to private sector expectations E_t^P, and the alternative specification (red dashed line), where the Fed responds to rational expectations $\mathbb{E}_t$.

B.6 Alternative Calibration ($\theta = 0.85$)

The baseline calibration of the SP-MSNK model in Section 5 sets the cognitive discounting parameter to $\theta = 1.00$, which corresponds to the standard New Keynesian IS curve without behavioral discounting. While this maintains comparability with the textbook benchmark, the resulting model can amplify the response of current outcomes to expected future variables—a manifestation of the well-known forward-guidance puzzle (Gabaix, 2020; Pfäuti and Seyrich, 2022). Since the perception channel in the SP-MSNK model operates precisely through these forward-looking expectations, it is natural to ask whether the conclusions are sensitive to at-

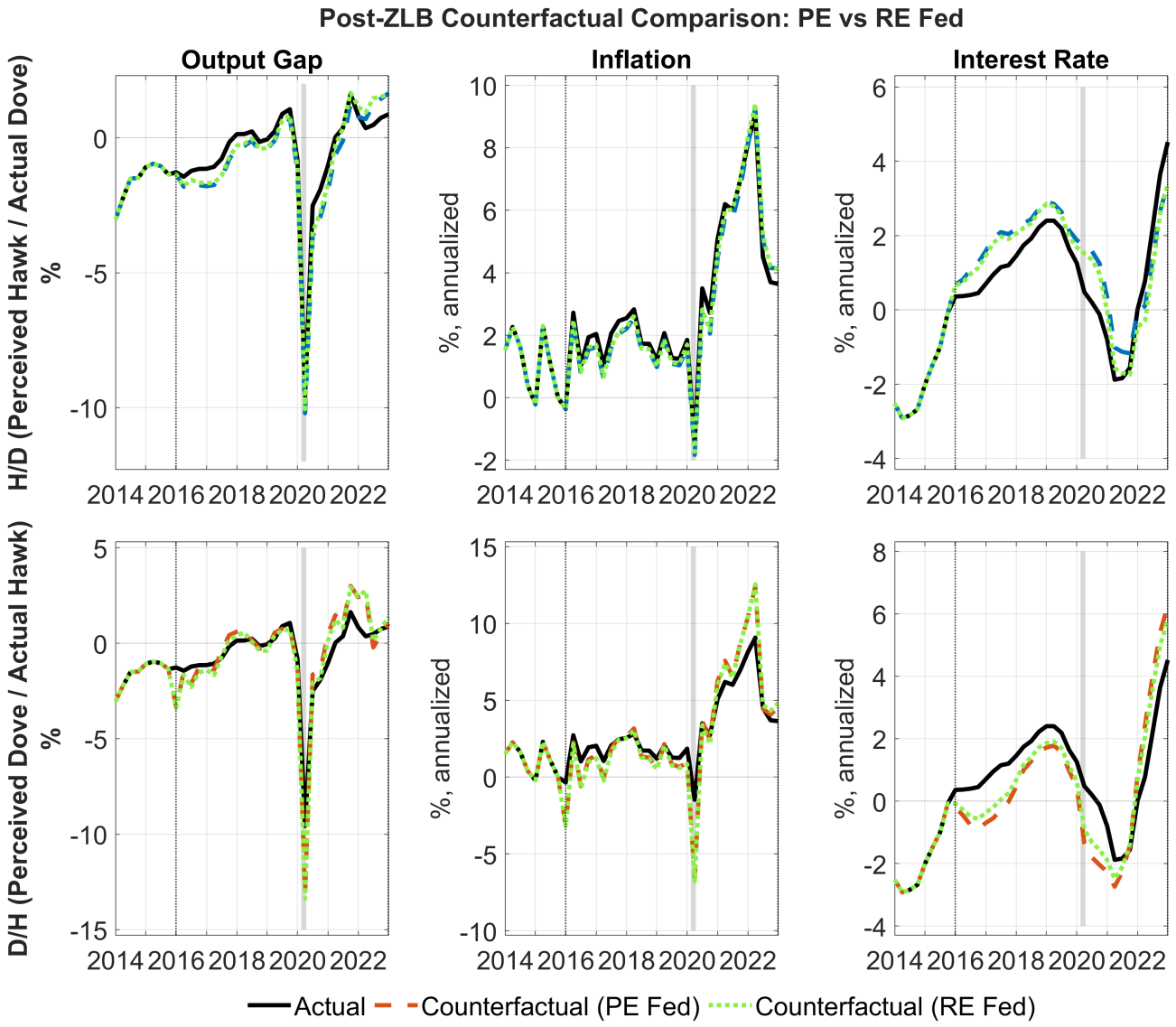


Figure B5: Counterfactual simulations: Fed Responding to E^P vs. $\mathbb{E}$

Note: The figure replicates the Perceived Dovish counterfactual (Counterfactual 2) from Section 5.4. The red dashed line represents the baseline counterfactual (Fed responds to E^P), and the green dotted line represents the case where the Fed responds to $\mathbb{E}_t$.

tenuating this channel.

To address this, I re-run the Great Moderation counterfactual of Section 5.4.1 under $\theta = 0.85$, following the cognitive discounting calibration of Gabaix (2020) and Pfäuti and Seyrich (2022). All other structural parameters, the regime-specific policy coefficients in Table 2, and the perceived transition matrix $\widehat{\mathbf{P}}$ are held at their baseline values. Reducing θ below unity attenuates the forward-guidance effect by lowering the weight that agents place on expected future output gaps in the IS curve, which in turn dampens the equilibrium pass-through of perceived policy parameters into current dynamics. The perception-bias counterfactual of Section 5.4.2 is conducted at the baseline value $\theta = 1.00$ throughout.

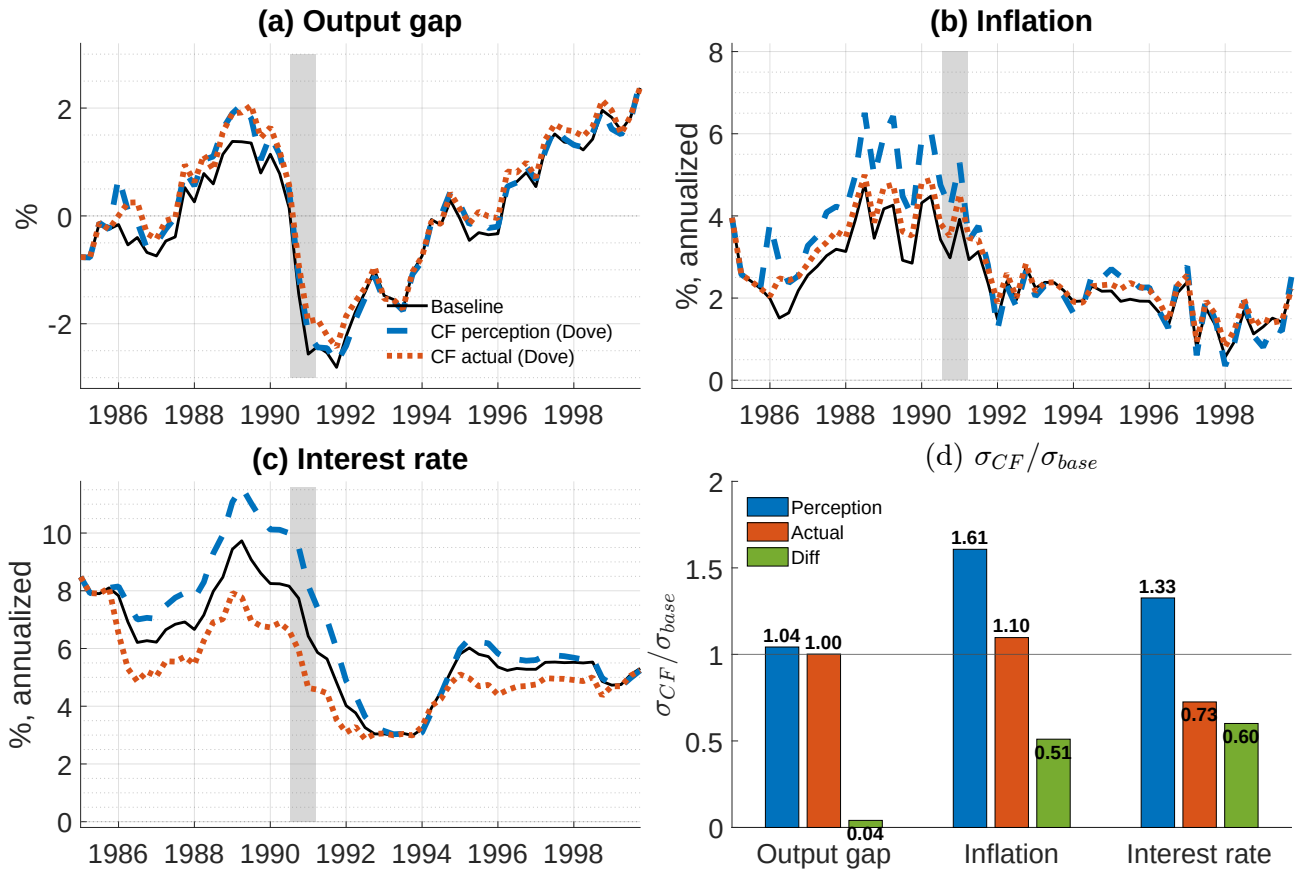


Figure B6: Great Moderation Counterfactual under $\theta = 0.85$

Note: Replicates the Great Moderation counterfactual of Figure 8 under the alternative calibration $\theta = 0.85$. (a)–(c) Solid black: baseline; dashed blue: CF perception (Dove PLM); dotted orange: CF actual (Dove rule). (d) $\sigma_{CF}/\sigma_{base}$ over target-regime quarters.

Figure B6 replicates the Great Moderation counterfactual under $\theta = 0.85$. The qualitative conclusions of the main text are preserved: the perception counterfactual (Dovish PLM, Hawkish actual rule) continues to generate substantially larger inflation volatility than the actual-rule counterfactual (Hawkish PLM, Dovish actual rule), confirming the dominance of the perception channel over the direct policy channel.

Quantitatively, the magnitudes are attenuated relative to the $\theta = 1.00$ baseline, as expected. With cognitive discounting, agents place less weight on the longer end of the expectations chain, which mechanically reduces the influence of perceived future policy on current outcomes. Nevertheless, the volatility ratios in panel (d) remain economically large, indicating that the perception channel identified in the main text is not an artifact of the standard New Keynesian IS specification. The robustness of this result to the inclusion of behavioral discounting suggests that the main conclusions hold under the more conservative information-processing structure that mitigates the forward-guidance puzzle.

B.7 Welfare Analysis: The Cost of Misperception and Its Persistence

This appendix develops the welfare implications of misperception, the mechanism behind the perception channel documented in Section 5.4.1. It asks how costly a wrong perception is, and how that cost depends on how long agents expect the wrong regime to last. The exercise uses a two-regime calibration drawn from the six-regime model rather than the full regime set of Table 2, because the trade-off is clearest when a single pair of regimes is varied at a time.

This section evaluates the model's implications for monetary policy by analyzing welfare under four combined regimes drawn from the six-regime model. I focus on two regimes that span a wide range of the policy spectrum: the Hawk regime ($s = 3$) and the Dove regime ($s = 6$). The four focal scenarios correspond to the possible joint state realizations $s_t = (s_t^p, s_t^f)$ formed from these two regimes:

1. **HH**: Agents perceive a Hawk regime, and the Fed implements a Hawk rule. This is the aligned-rule regime under an active policy.
2. **DD**: Agents perceive a Dove regime, and the Fed implements a Dove rule. This is the aligned-rule regime under a passive policy.
3. **HD**: Agents perceive a Hawk regime, but the Fed actually implements a Dove rule. This captures a case of misalignment where the central bank's stance is overestimated by the public.
4. **DH**: Agents perceive a Dove regime, but the Fed actually implements a Hawk rule. This is a case of misalignment where the central bank's stance is underestimated by the public.

It is important to clarify the distinction between the *realized* regime path and the *perceived* regime dynamics in this exercise. While the actual law of motion (ALM) is simulated conditional on the fixed joint state s_t, the private sector forms expectations based on the belief that the perceived state is s_t^p, evolving stochastically according to the full six-regime transition matrix $\widehat{\mathbf{P}}$. The PLM is solved globally over all six perceived regimes, so even when the realized state is fixed at the Hawk or Dove regime, agents' expectations incorporate the possibility of transitioning to any of the other regimes. For this exercise, I vary only the perceived persistence of the two focal regimes by setting $\widehat{P}_{33(HH)} = \widehat{P}_{66(DD)} = \widehat{p}$ and $\widehat{P}_{36(HD)} = \widehat{P}_{63(DH)} = 1 - \widehat{p}$, while keeping all other rows of $\widehat{\mathbf{P}}$ at their baseline values.

Methodology Conditional on the specific policy regime ℓ, and the associated the state-space matrices $\mathbf{A}_\ell^{\text{ALM}}$ and $\mathbf{B}_\ell^{\text{ALM}}$, I simulate the economy and assess the welfare loss using a standard

quadratic loss function:

$$\mathcal{L} = \frac{1}{2} E_0 \sum_{t=0}^{\infty} \beta^t \left[\pi_t^2 + \lambda_x x_t^2 \right] \tag{B.16}$$

where the relative weight on output gap stabilization is set to $\lambda_x = 0.1$.

Results Figure B7 illustrates the unconditional variance trade-offs between inflation and the output gap across the four regimes. The regimes are denoted by a two-letter code where the first letter represents the perceived regime and the second represents the actual rule; circle markers identify the aligned-rule benchmarks (HH and DD), while cross markers denote the misperception regimes (HD and DH).

The left panel, calibrated to a high perceived persistence ($\widehat{p} = 0.98$), establishes the baseline stability properties of misperception. Regimes characterized by a Dove perception (DD, DH) generate strictly higher volatility in inflation compared to their Hawk-perception counterparts (HD, HH). Consequently, the Dove-perception equilibria cluster to the north of the Hawk-perception equilibria regardless of the actual policy rule in place. This separation underscores the primacy of the Perception Channel. Since private agents form expectations ($E_t^P[\cdot]$) based on the perceived Taylor rule, a belief in passive policy unanchors expectations in the IS and Phillips curves. Consequently, even if the Federal Reserve actually adheres to the Taylor principle (Regime DH), it cannot fully offset the destabilizing feedback loop generated by the private sector's belief in accommodation. This confirms that the *perceived* reaction function is the primary determinant of equilibrium dynamics beyond the actual policy rate.

The right panel ($\widehat{p} = 0.50$) illustrates how these trade-offs evolve when perceived persistence declines. The reduction in perceived persistence triggers a visible convergence in terms of inflation and welfare cost. The welfare outcomes in the misperception regimes (HD and DH) and their respective aligned-rule benchmarks (HH and DD) come together. The intuition is that when agents view the current regime as less persistent, their forward-looking expectations are more heavily weighted toward the alternative regime. Consequently, the destabilizing (stabilizing) effects of a perceived Dove (Hawk) regime are attenuated (strengthened) if the private sector anticipates a faster reversion. This finding highlights an important dimension of monetary communication: effective stabilization requires managing not only the perceived *stance* of the current rule (e.g., ϕ_{π,s_t}) but also its perceived *duration* ($\widehat{p}$), ensuring that necessary accommodative deviations are viewed as strictly transitory.

B.8 The Perception-Bias Counterfactual: Additional Results

This appendix reports a further result from the perception-bias counterfactual of Section 5.4.2.

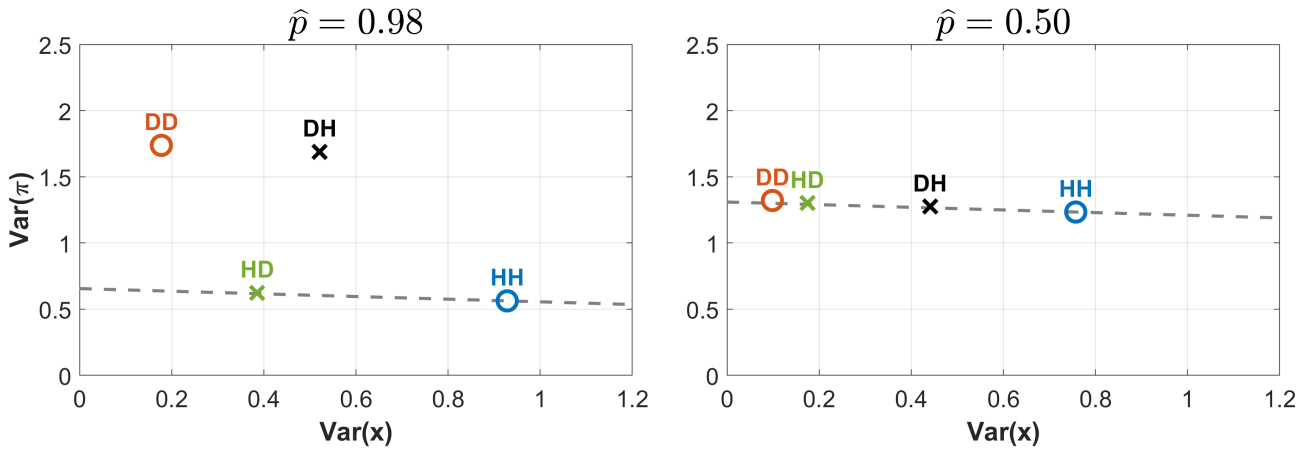


Figure B7: Policy Trade-offs and Perceived Regime Persistence ($\widehat{p}$)

Note: The figure shows the welfare costs for four regimes (Hawk × Dove). Circles: aligned-rule benchmarks; crosses: misperception. Dashed line: minimum iso-loss curve.

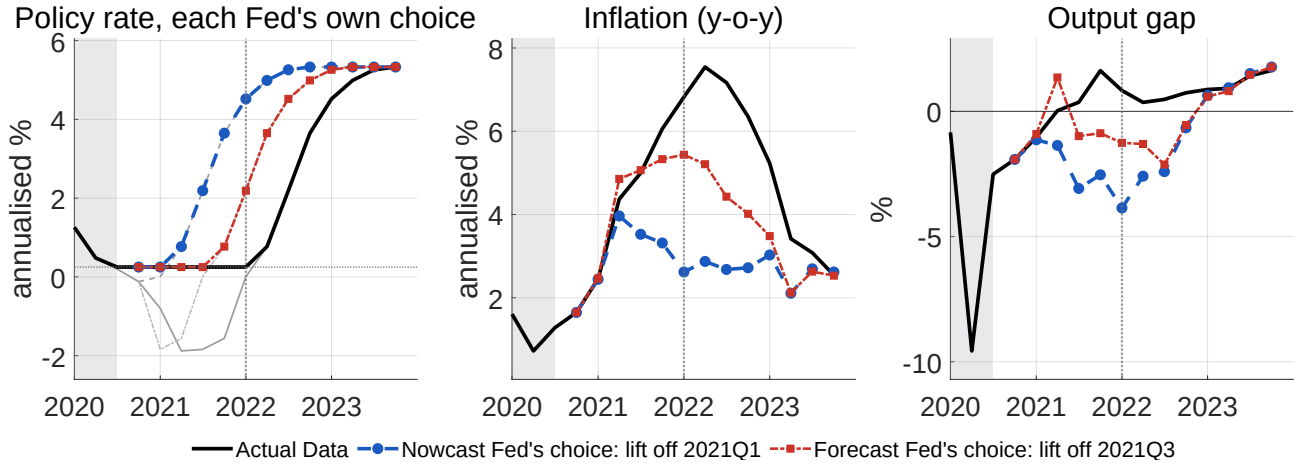


Figure B8: Realized Paths under Each Central Bank's Own Choice of Lift-off Date

Note: The solid line denotes actual data. The dashed lines mark the counterfactual outcomes under the lift-off date each central bank selects for itself, rather than the common path of Figure 9. Both are evaluated in the true economy.

Figure B8 plots the realized paths when each central bank implements its own preferred lift-off date rather than the common path of Figure 9. Both paths are evaluated in the true economy, so the comparison isolates the consequence of the choice itself. The forecast-calibrated central bank's two-quarter delay leaves year-over-year inflation above 3% throughout 2022, whereas the earlier lift-off chosen under the nowcast calibration holds it below 3% from 2022 onward, at the cost of a deeper but temporary output gap that closes by 2023.

C Estimation Procedure

The perceived monetary policy rule is estimated using a Time-Varying Parameter (TVP) model with Stochastic Volatility (SV). I employ a "stacked" measurement equation pooling monthly and quarterly survey data within each period to improve identification. The estimation proceeds in three steps:

1. **State-Space Representation:** Let $t = 1, \ldots, T$ index quarters and N be the number of survey observations in quarter t. The measurement equation stacks these observations:

$$\mathbf{i}_{t+h|t} = \mathbf{X}_{t,h}\boldsymbol{\Theta}_t + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \mathcal{N}\left(\mathbf{0}, e^{h_t}\mathbf{I}_N\right) \tag{C.1}$$

   where $\boldsymbol{\Theta}_t$ follows a random walk with regime-dependent covariance $\boldsymbol{\Sigma}_{\nu,s}$, and the log-volatility h_t follows a geometric random walk with variance σ_h^2.

2. **Prior Calibration:** Priors are calibrated using a training sample of the first 48 observations, following Boivin (2005):

   - **Initial States ($\boldsymbol{\Theta}_0, h_0$):** Set using OLS point estimates and residual variances from the initial 48 months of observations.
   - **Coefficient Covariance ($\boldsymbol{\Sigma}_{\nu,s}$):** The matrix is calibrated via the Stock and Watson (1998) median-unbiased estimator based on the Quandt Likelihood Ratio (QLR) statistic following Boivin (2005). Subscript s stands for possible breaks in the variance of innovation to the TVPs. I give structural break points around ZLB. For the actual rule using Greenbook. I give additional break around the inauguration of Paul Volcker similar to Coibion and Gorodnichenko (2011).

3. **Gibbs Sampling Algorithm:** The posterior is approximated using a Gibbs sampler with $M = 10,000$ iterations (5,000 burn-in)[27]:

   (a) **Coefficients ($\boldsymbol{\beta}_{1:T}$):** Drawn using the multimove simulation smoother of Carter and Kohn (1994). The stacked observations are handled by setting the measurement error covariance to $R_t = e^{h_t}\mathbf{I}_N$.

   (b) **Stochastic Volatility ($h_{1:T}$):** Drawn using the auxiliary mixture sampler of S. Kim, Shephard, and Chib (1998). I construct a quarterly volatility proxy from the stacked residuals to isolate the aggregate signal.

[27] The results are robust to a higher number of simulations. Increasing the number of simulations only adds to the computation time.

D Robustness and Additional Empirical Results

This appendix provides a set of robustness checks and additional empirical results that supplement the main empirical findings.

D.1 Within-Period Uncertainty: Quarter-End BCEI Only

The covariance-bias derivation in Section 3 treats the operative perceived coefficient $\phi_{\pi,t|t}$ as measurable with respect to the agent's time-t information set. If forecasters instead face a signal-extraction problem and hold a non-degenerate posterior over the current regime s_t, a within-period covariance between the rule (ϕ_{π,s_t}) and macro aggregate ($\pi_{t+j|t}$) may affect the estimation even at the nowcast horizon.

To check if this concern is large, I re-estimate the perceived rule using only the third-month BCEI observation of each quarter. This timing is deliberate: the quarter-end BCEI survey is collected after forecasters have observed nearly all current-quarter macroeconomic data and FOMC actions, so their uncertainty over the current regime (s_t) should be at its lowest within the quarter. Under this setup, the within-period covariance should be minimal; any divergence from the baseline result would indicate that within-period signal extraction materially biases the baseline estimates.

Figure D1 compares the two specifications. The two posterior median paths track each other closely across all four parameters and over the full sample, with the third-month only specification's credible bands wider as expected from the smaller number of observations used in the estimation. This suggests that within-period signal-extraction uncertainty over the current regime is empirically negligible for the recovered perceived rule and most likely relevant for interest rate forecast-based estimates.

D.2 OLS Estimation of perceived rule

In this subsection, I present results obtained via constant-parameter Ordinary Least Squares (OLS). This exercise mirrors the simulation analysis in Section 3, establishing a baseline to assess the empirical consequences of neglecting time-variation. As demonstrated below, the resulting estimates exhibit a downward-sloping term structure for the inflation and output gap coefficients that qualitatively replicates the simulation findings, thereby providing empirical evidence of the covariance bias.

Using the Blue Chip consensus data, I estimate the following constant-parameter reaction function for forecast horizons $h \geq 0$:

$$i^{BC}_{t+h|t} = \beta_{\pi,h}\pi^{BC}_{t+h+j|t} + \beta_{x,h}\left(-2 \cdot u^{BC}_{t+h+j|t}\right) + \beta_{i1,h}i^{BC}_{t+h-1|t} + \beta_{i2,h}i^{BC}_{t+h-2|t} + c_h + \varepsilon_{t,h} \tag{D.1}$$

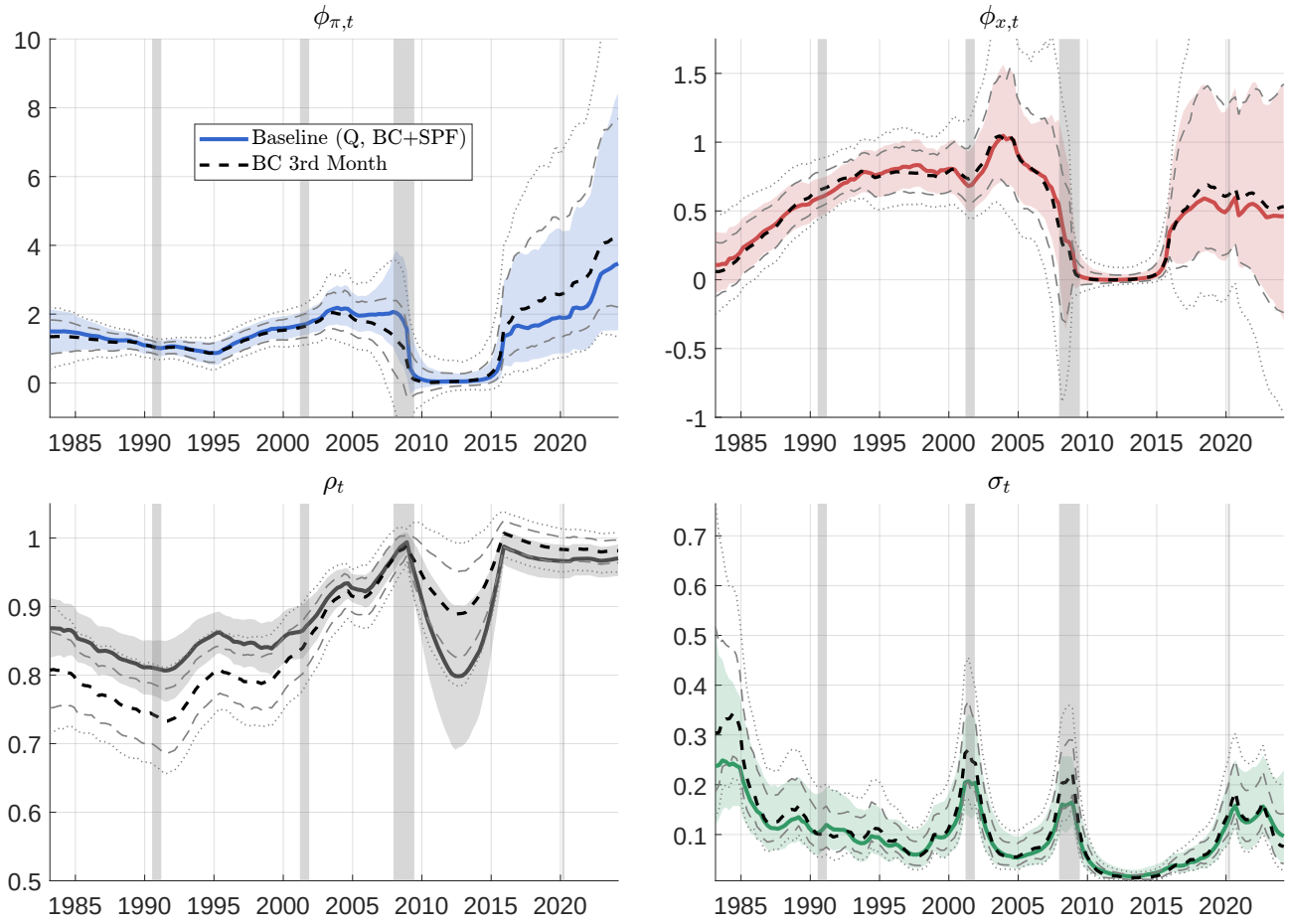


Figure D1: Robustness to Within-Period Signal Extraction: Quarter-End BCEI Only

Note: The figure compares the baseline stacked TVP-SV estimates (BCEI monthly + SPF quarterly with common quarterly $\mathbf{\Theta}_t$, colored bands and solid medians) against an alternative specification using only the third-month BCEI observation of each quarter (black dashed median with gray dotted/dashed credible bands). Both specifications are estimated at quarterly frequency. The four panels report the long-run perceived inflation response $\phi_{\pi,t|t}$, the long-run output gap response $\phi_{x,t|t}$, total inertia ρ_t, and stochastic volatility σ_t.

where $i^{BC}_{t+h|t}$, $\pi^{BC}_{t+h+j|t}$, and $u^{BC}_{t+h+j|t}$ denote the Blue Chip consensus forecasts for the interest rate, inflation, and unemployment rate at horizon h, respectively. The subscript j denotes the policy targeting horizon (e.g., $j \geq 1$ for forward-looking rules).[28] The case $h = 0$ corresponds to the nowcast specification. The unemployment rate is scaled by -2 to interpret the coefficient $\beta_{x,h}$ as the response to the output gap, consistent with a standard Okun's law relationship. Comparing estimates across horizons h allows for an empirical evaluation of the theoretical predictions regarding the covariance bias discussed in Section 3. In the results below, I report estimates for the forward-looking horizon $j = 1$.[29] Results for other horizons are qualitatively

[28] Although the simulation analysis in Section 3 utilized a contemporaneous rule, robustness checks confirm that a similar downward-sloping term structure emerges in alternative empirical specifications with varying targeting horizons j.

[29] Boivin (2005) and Coibion and Gorodnichenko (2011) document that actual monetary policy rules are described better by a forward-looking rule rather than a contemporaneous rule based on statistical tests. I also

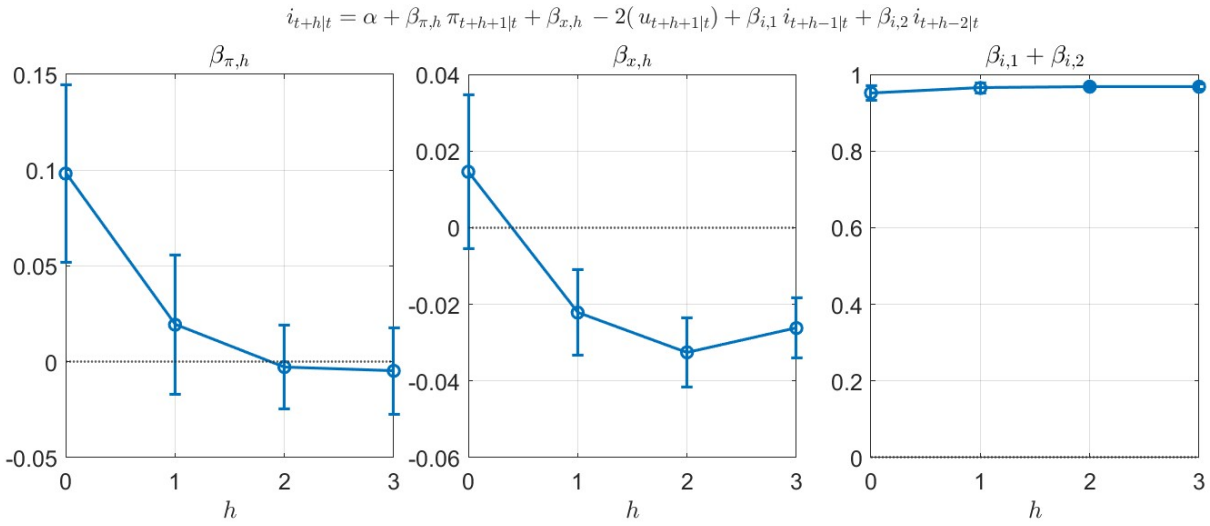


Figure D2: OLS Estimates of Policy Coefficients Across Forecast Horizons (h)

Note: The figure displays the estimated OLS coefficients of equation (D.1) using Blue Chip consensus forecasts at various horizons h.

similar and are reported in the Appendix.

Figure D2 presents the estimation results. A distinct term structure is evident in the OLS coefficients across forecast horizons h. Under the null hypothesis of a constant policy rule, these coefficients should be invariant to h. The observed variation thus leads to a rejection of the constant parameter hypothesis. Moreover, the qualitative pattern of this term structure aligns with the simulation results in Section 3. This suggests that using interest rate *forecasts* to infer the perceptions about the policy rule is not an innocuous exercise.

D.3 OLS vs. GMM Estimation of Perceived Taylor Rule

One assumption in estimating the perceived Taylor rule is that the perceived monetary policy shock does not give rise to a substantial endogeneity bias, as is the case in the conventional Taylor rule estimation. To test the validity of this assumption, I compare OLS estimates with Generalized Method of Moments (GMM) estimates, following the approach of Carvalho, Nechio, and Tristao (2021). If the endogeneity bias is severe, we should observe a statistically meaningful difference between OLS and GMM estimates.

The estimating equation is:

$$i^{BC}_{t|t} = \beta_\pi \pi^{BC}_{t+j|t} + \beta_x \left(-2 \cdot u^{BC}_{t+j|t}\right) + \rho_1 i^{BC}_{t-1|t} + \rho_2 i^{BC}_{t-2|t} + \alpha + e_t \tag{D.2}$$

find similar evidence for the perceived rule as well.

Table D1: Perceived Taylor Rule Estimation: OLS vs. GMM

	Contemporaneous $(\pi_{t\|t},\ u_{t\|t})$		Forward-looking 1 $(\pi_{t+1\|t},\ u_{t\|t})$		Forward-looking 2 $(\pi_{t+1\|t},\ u_{t+1\|t})$		Forward-looking 3 $(\pi_{t+2\|t},\ u_{t+1\|t})$	
	OLS	GMM	OLS	GMM	OLS	GMM	OLS	GMM
ϕ_π	1.95	1.83	2.21	2.02	2.22	2.01	2.18	2.19
	(0.54)	(0.41)	(0.48)	(0.40)	(0.48)	(0.40)	(0.46)	(0.37)
ϕ_x	0.28	0.20	0.26	0.19	0.27	0.17	0.30	0.22
	(0.17)	(0.11)	(0.14)	(0.11)	(0.15)	(0.11)	(0.13)	(0.10)
$\rho = \rho_1 + \rho_2$	0.96	0.97	0.96	0.96	0.96	0.96	0.96	0.96
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
R^2	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99
N	497	497	497	497	497	497	497	497
J-test p-val		0.599		0.467		0.422		0.280

where $i^{BC}_{t|t}$ denotes the Blue Chip consensus interest rate nowcast, $\pi^{BC}_{t+j|t}$ and $u^{BC}_{t+j|t}$ are the consensus inflation and unemployment forecasts at horizon j, and α is a constant term. The unemployment rate is scaled by -2 to interpret β_x as the response to the output gap under Okun's law. I report long-run coefficients derived as $\phi_\pi = \beta_\pi/(1-\rho_1-\rho_2)$ and $\phi_x = \beta_x/(1-\rho_1-\rho_2)$. Standard errors for the short-run coefficients are obtained by the Newey-West standard errors, and standard errors for the long-run coefficients are obtained via the delta method.

For GMM estimation, I follow Clarida, Gali, and Gertler (2000) and Carvalho, Nechio, and Tristao (2021) and use a k-step estimator. The instruments are lagged forecasts in terms of survey months. To avoid overlapping horizons, I use lags with at least three months. In each specification, the number of lags and hence instruments is chosen such that they have sufficiently high first-stage F-statistics ($F > 10$). The Hansen J-test evaluates the validity of the overidentifying restrictions.

Table D1 reports the estimation results across four alternative specifications depending on the targeting horizon j. The OLS and GMM estimates are similar and statistically indistinguishable within a standard error band. The long-run inflation coefficient ϕ_π ranges from 1.83 to 2.22 under GMM, closely tracking the OLS estimates. The output gap coefficient ϕ_x exhibits comparable magnitudes as well. The Hansen J-test p-values range from 0.280 to 0.599, failing to reject the validity of the overidentifying restrictions at conventional significance levels. This indicates that the instruments are valid and that endogeneity due to perceived monetary policy shocks is not a major concern.

Table D2: Properties of Nowcast Errors

	BCEI			
	Month 1	Month 2	Month 3	SPF
Observations	166	166	165	166
Mean $\lvert e_t \rvert$	0.19	0.12	0.08	0.09
Std. dev. $\lvert e_t \rvert$	0.22	0.13	0.07	0.11
AR(1) of e_t	0.27^{***}	0.22^{*}	0.27^{***}	0.32^{**}

Note: e_t denotes nowcast error. The numbers are in annualized percentage points, computed from 1983Q1. AR(1) coefficients are estimated with Newey-West standard errors at 12 lags. ***, ** and * denote significance at the 1%, 5% and 10% levels.

D.4 Properties of Nowcast Errors

Section 4 identifies the perceived rule from the interest rate nowcast. A natural concern is that the nowcast is just a mechanical read-off of the actual rate that has already been observed. This appendix shows that the nowcast errors are persistent and sizable enough to rule out the concern.

The realized target is the quarterly average T-bill, matching the forecast horizon. The nowcast error e_t difference between that realization and the reported $h = 0$ survey nowcast. Because Blue Chip surveys every month, each quarter yields three nowcasts of the same target. These are separately classified as first, second, and third month of a quarter series. The SPF is collected once per quarter and is comparable to the second month survey of BCEI. Persistence is the AR(1) coefficient at quarterly frequency.

Table D2 reports the results. The mean absolute error (MAE) is 0.19 percentage points for a survey taken in the first month of the quarter, falling to 0.12 in the second month and 0.08 in the third. That decline is expected since realization unfolds as it reaches the end of the current quarter. The SPF, collected near the middle of the quarter, sits at 0.09, between the second and third monthly readings. Autocorrelation is around 0.22 to 0.32 across all four series, all statistically significant.

These observations suggest that the nowcast is not a mechanical read-off of the actual rate. First, a nowcast carrying a serially correlated error of this size is not a restatement of an observed data. Second, accuracy improves markedly within the quarter while persistence does not. The average absolute error falls by more than half across the quarter, from 0.19 to 0.08, while the autocorrelation is essentially flat at 0.27, 0.22, and 0.27. This is the pattern one would expect if forecasters are reporting the rate from their own subjective model. Averaging with already-realized rates helps reduce nowcast error, but relying on the subjective model still retains the persistence of the error.

D.5 Comparison with BCFF Microdata Estimates

The baseline estimates use the Blue Chip *consensus* nowcast of the three-month Treasury bill. Two features of that choice invite scrutiny. The consensus is an average across forecasters, so if respondents disagree about the rule, the perceived rule estimated from the average need not equal the average of the perceived rules. And the three-month bill is not the policy rate, so the estimates could in principle be contaminated by a term or risk premium that moves independently of the perceived reaction function.

Both concerns are addressed by re-estimating on individual forecaster data. I use the Blue Chip Financial Forecasts microdata, which reports each respondent's own projections and, crucially, their projection for the *federal funds rate* rather than the Treasury bill. The same TVP specification with stochastic volatility used in Section 4 is applied separately to each forecaster's panel, so no averaging occurs before estimation, and each estimate is identified from one respondent's internally consistent set of projections. The result is a cross-section of perceived rules at each date, summarized by its median and interquartile range, which can be set against the consensus-based posterior median and its 68 per cent credible band.

Figure D3 shows the comparison. The two long-run policy responses track each other closely over most of the sample, and both exceed unity through the pre-crisis period, so neither aggregation nor the substitution of the bill for the funds rate overturns the paper's central empirical claim. The inertia estimates are similarly aligned, with the microdata slightly higher and smoother.

The differences that do appear are modest. The microdata volatility lies somewhat above the consensus volatility, which is the expected consequence of averaging, since idiosyncratic forecaster noise is removed from the consensus before estimation but survives in an individual panel. Otherwise the two sets of estimates are statistically indistinguishable, with the interquartile range of the individual estimates and the credible band of the consensus estimate overlapping throughout the sample.

D.6 Sensitivity to Hyperparameters and Initialization

The estimator of Section 4 conditions on two objects that are themselves estimated in a first stage and then treated as known. The first is the coefficient drift covariance Σ_ν, calibrated from the QLR statistic and the implied Stock and Watson (1998) λ. The second is the initial state prior (β_0, V_0), obtained by OLS on the first 48 months, together with the initialization of the stochastic volatility block. Credible bands computed conditional on these objects understate uncertainty, because they hold fixed quantities the data pinned down only imprecisely. This appendix propagates the sampling uncertainty of both through the entire two-stage procedure.

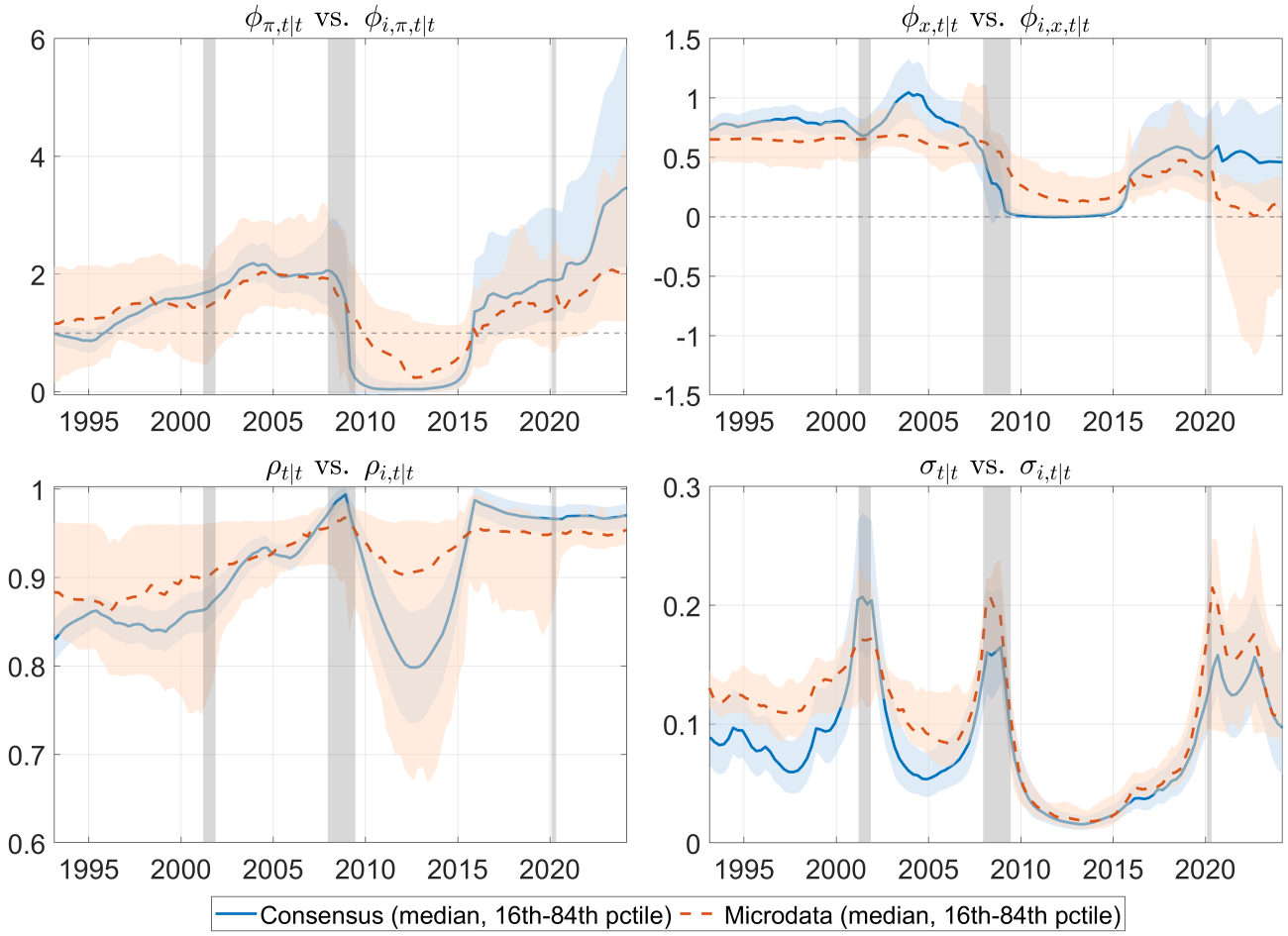


Figure D3: Perceived Rule from Microdata and from the Consensus

Note: Solid blue: consensus-based estimate, posterior median with the 68% credible band, from the baseline specification of Section 4 on the three-month Treasury bill. Dashed red: Blue Chip Financial Forecasts microdata, estimated forecaster by forecaster on each respondent's own federal funds rate projection, shown as the cross-sectional median with the interquartile range. Panels: long-run inflation response, long-run output-gap response, total inertia, and stochastic volatility. Shaded bands are NBER recessions.

Procedure The scheme is a cluster-wild bootstrap with a fixed design and the calendar preserved. Let $\widehat{\beta}_t$ denote the baseline posterior median coefficient path and let $e_{t,j} = y_{t,j} - x'_{t,j}\widehat{\beta}_t$ be the residual of the j-th of the four stacked observations in quarter t. Replication r draws a Rademacher weight $w_t \in \{-1, +1\}$ once per quarter, applies it to all four observations in that quarter, and forms

$$y^*_{t,j} = x'_{t,j}\widehat{\beta}_t + w_t e_{t,j}. \tag{D.3}$$

The whole procedure is then re-run on y^*. The QLR statistic delivers a new λ^* and regime-specific Ω^*, OLS on the first 48 months of the synthetic series delivers (β^*_0, V^*_0) and the volatility initialization, and the same Gibbs sampler follows with the same 2009 and 2016 breaks. Each replication therefore asks what the procedure would have calibrated, and then estimated, had

Table D3: Average Width of the 68% Band, by Source of Uncertainty

	Band width			Ratio to posterior	
	Posterior	Hyperparameter	Total	Hyperparameter	Total
Short-run β_π	0.063	0.039	0.076	0.63	1.22
Long-run ϕ_π^{LR}	0.839	0.388	1.242	0.46	1.48
Long-run ϕ_x^{LR}	0.313	0.128	0.460	0.41	1.47
Total inertia ρ	0.048	0.081	0.058	1.69	1.22

Note: Sample average of the 68% band width. *Posterior* is the baseline band, which treats the first-stage objects as known. *Hyperparameter* holds the data fixed and resamples the calibration and the initialization, 200 replications. *Total* is the cluster-wild bootstrap, which resamples the data as well, 500 replications. Each replication uses $K = 2000$ draws with the first half discarded.

history delivered a statistically equivalent sample.

Three features of the design are deliberate. The regressors are held fixed because they are distinct survey series rather than transformations of the dependent variable, so a fixed design is coherent and avoids having to simulate the joint survey process. The weights are sign flips clustered by quarter, which preserves each quarter's residual magnitudes exactly and so leaves untouched the volatility profile that the stochastic volatility block is fitting, while retaining dependence across the four observations within a quarter. The calendar is preserved because the estimand is a time path, which any resampling that reorders time would destroy. Synthetic nowcasts are censored at zero, since a nominal bill nowcast cannot be negative even though a sign flip can push one there. This binds on 1.5% of synthetic observations.

I report 500 replications with $K = 2000$ draws each. A companion exercise holds the data fixed and resamples only the first-stage calibration and the initialization, which isolates the contribution of the hyperparameters on their own.

Results Table D3 decomposes the average width of the 68% band. Resampling the hyperparameters alone yields dispersion *narrower* than the posterior band for all three policy coefficients, between 0.41 and 0.63 of its width, and wider only for inertia, at 1.69. The first-stage calibration is therefore not the binding source of uncertainty for the coefficients the paper interprets. It is the full bootstrap, which lets the data, the calibration and the estimate move together, that widens the bands, and it does so by 22 to 48%.

The qualitative record of the estimates survives the exercise. Of the 500 replications, 100% keep the pre-2008 long-run inflation response above unity, 95.2% reproduce the fall below unity during the zero lower bound period, and 95.6% return it above unity after 2021. The three facts the paper reads off Figure 4 are thus not artifacts of a particular calibration of the time variation.

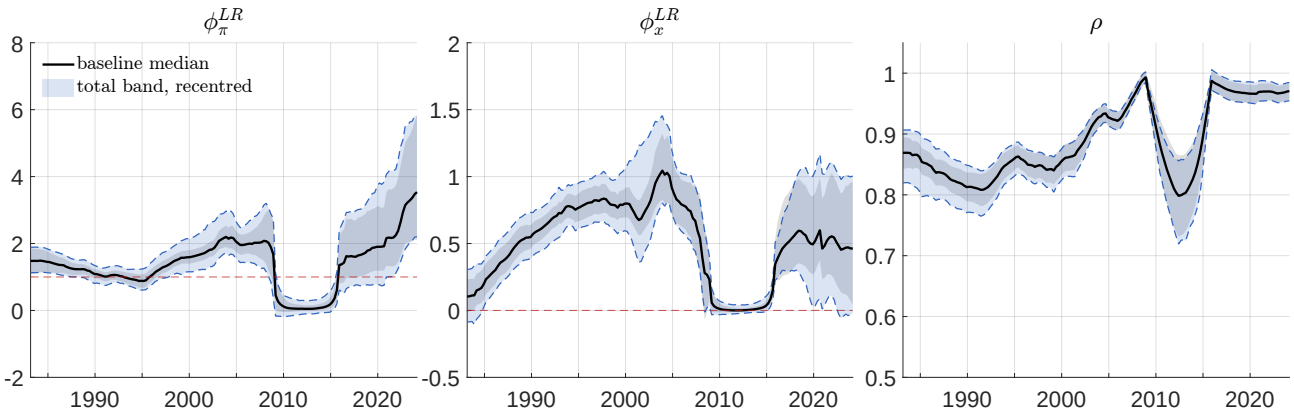


Figure D4: Bootstrap Sensitivity to Hyperparameters and Initialization

Note: Baseline posterior median in black with its 68% band in grey. The blue band is the cluster-wild bootstrap dispersion over 500 replications, recentred on the baseline median. Dashed red lines mark unity for the inflation response and zero for the output gap response.

Figure D4 plots the resulting bands. The bootstrap distribution of a two-stage plug-in estimator need not be centred on the estimate, and here it is not. Splitting the replications into independent halves reproduces the same offset in both, at 5 to 25 times the Monte Carlo standard error of the centre, so it is systematic rather than simulation noise. The figure therefore follows the basic-bootstrap convention and anchors the dispersion on the baseline median,

$$\text{band} = \widehat{\beta}_t + (q^* - c^*)\,, \tag{D.4}$$

where q^* is a bootstrap quantile and c^* the bootstrap centre. Recentring moves location only, so the widths in Table D3 are unaffected by the convention.

Two limitations should be stated alongside the result. The exercise conditions on the baseline median path as the pseudo-truth, so it measures the sensitivity of the procedure around the estimated path rather than sensitivity to misspecification of the model itself. And a wild bootstrap for a time-varying parameter state-space model is a heuristic sensitivity check rather than exact inference, so the widths in Table D3 are best read as evidence on whether the two-stage procedure is fragile, not as corrected confidence bands.

D.7 Shadow-Rate Estimation of the Perceived Rule

The baseline estimation of the perceived rule in Section 4.2.1 uses the survey nowcasts of short-term interest rate as the dependent variable. During the two ZLB episodes (2009-2015, 2020-2022) these nowcasts are censored at the lower bound, which could bias the estimated TVPs. To address this concern, I re-estimate the model with a latent shadow rate substituting for the censored nowcast inside the two ZLB windows.

The procedure closely follows González-Astudillo and Tanvir (2023) and Carriero et al.

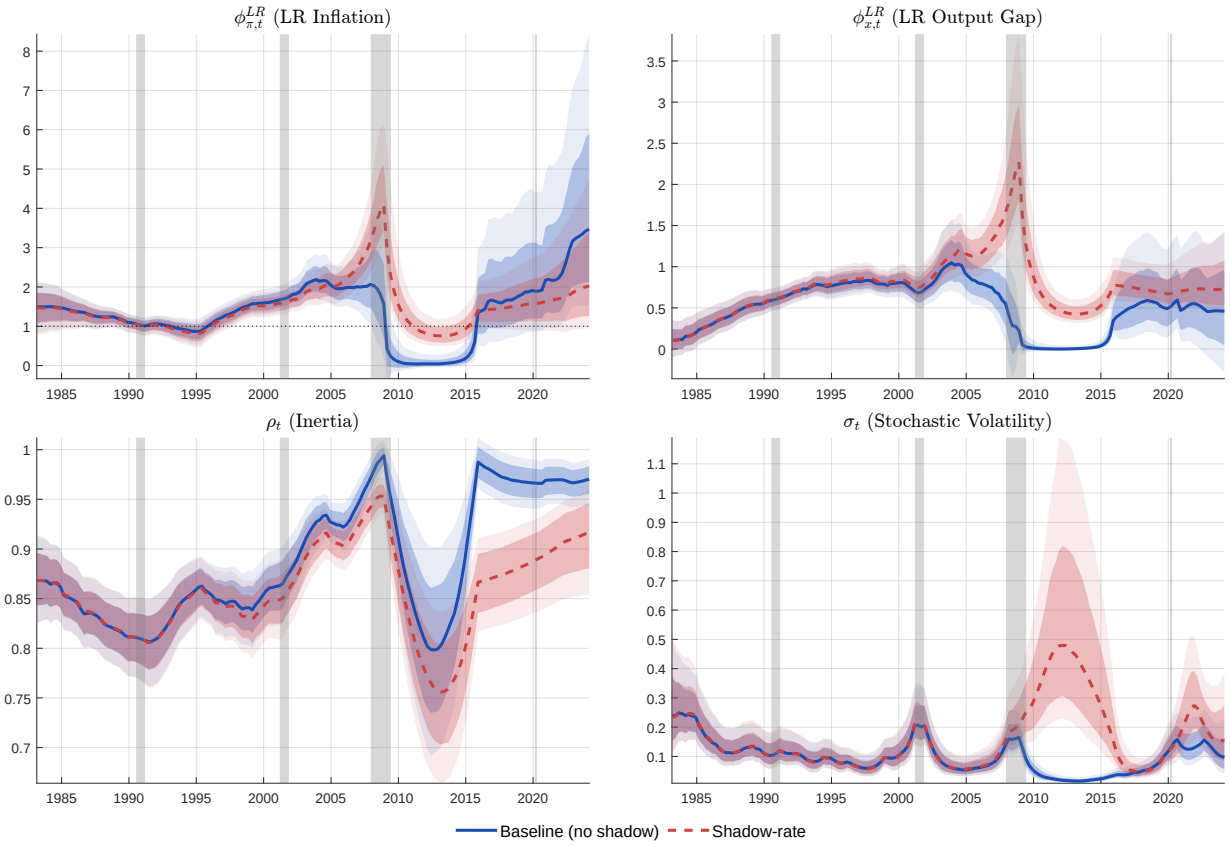


Figure D5: Perceived Rule: Baseline vs. Shadow-Rate Estimation

Note: Posterior medians (solid for the baseline, dashed for the shadow-rate version) and 68% / 90% credible bands for the long-run inflation coefficient $\phi^{LR}_{\pi,t|t}$, long-run output-gap coefficient $\phi^{LR}_{x,t|t}$, total inertia $\rho_{t|t}$, and stochastic volatility σ_t. Blue: baseline (Section 4.2.1); red: shadow-rate estimation.

(2025) and the reader is referred to those papers for full detail. Compared to the baseline, the only change is one additional Gibbs block. Within each ZLB quarter, observations of the nowcast that fall below the ELB threshold (0.25) are replaced each iteration by a draw from a truncated normal whose mean equals the current Taylor-rule fitted value, whose standard deviation equals the stochastic-volatility draw for the quarter, and which is bounded above by the ELB.

Figure D5 compares the resulting posteriors for the long-run coefficients, inertia, and stochastic volatility against the baseline. Outside the first ZLB window the two estimates are essentially indistinguishable. Within the first ZLB and the surrounding global financial crisis, the shadow-rate version produces modestly wider credible bands for $\phi^{LR}_{\pi,t|t}$ and $\phi^{LR}_{x,t|t}$ but the median paths remain close. The second ZLB episode is too short for the imputation to materially shift any coefficient. Figure D6 reports the estimated shadow-rate path, which dips below the ELB during the two windows and coincides with the observed federal funds rate elsewhere by construction. Taken together, these results indicate that the time-varying patterns documented in

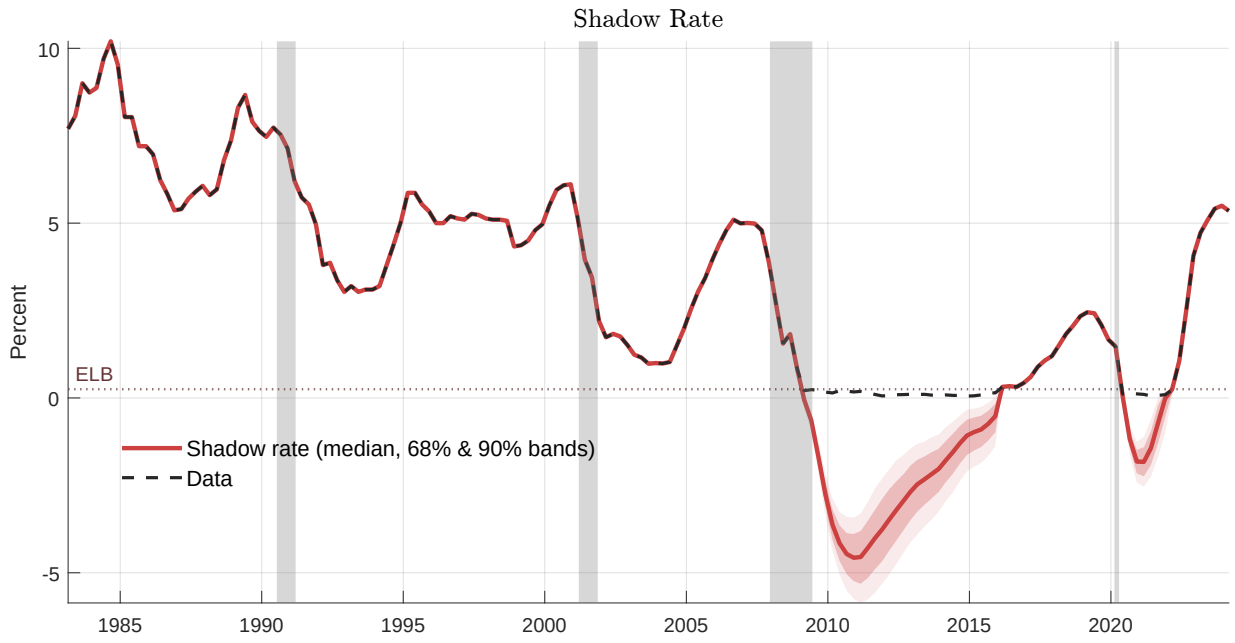


Figure D6: Estimated Shadow Rate of the Perceived Nowcast

Note: Posterior median (red solid) and 68% / 90% credible bands for the latent shadow rate, summarized as the quarterly average of the imputed monthly draws. The dashed black line shows the observed quarterly federal funds rate. Outside the two ZLB windows the shadow rate coincides with the observed rate by construction.

Section 4.2.1—in particular the persistence of the Taylor principle in the pre-ZLB period and the post-ZLB rise in perceived inertia—are not artifacts of ELB censoring.

D.8 Zero Lower Bound Censoring and Shadow Rate

The estimation of the central bank's *actual* reaction function relies on the federal funds target rate as the dependent variable. During the Zero Lower Bound (ZLB) episodes (2009–2015 and 2020–2022), the inability of the federal funds rate to drop below zero creates a censoring problem. Standard estimation might mechanically bias the reaction coefficients ($\phi_{\pi,t}$) downward, as the policy rate appears unresponsive to negative output gaps or below-target inflation.

To address this, I re-estimate the actual policy rule replacing the effective federal funds rate with the shadow rate constructed by Wu and Xia (2016). The shadow rate is unconstrained and can take negative values, providing a better measure of the intended policy stance during unconventional monetary policy periods.

Figure D7 compares the baseline actual rule estimates with those obtained using the shadow rate. While the shadow rate implies a slightly more active policy during the depths of the ZLB, the qualitative trajectory of the actual rule remains similar. Crucially, the major divergence identified in the mid-1990s (pre-ZLB) is unaffected, and the recent post-ZLB dynamics are qualitatively robust. This confirms that the key findings regarding the gap between actual and

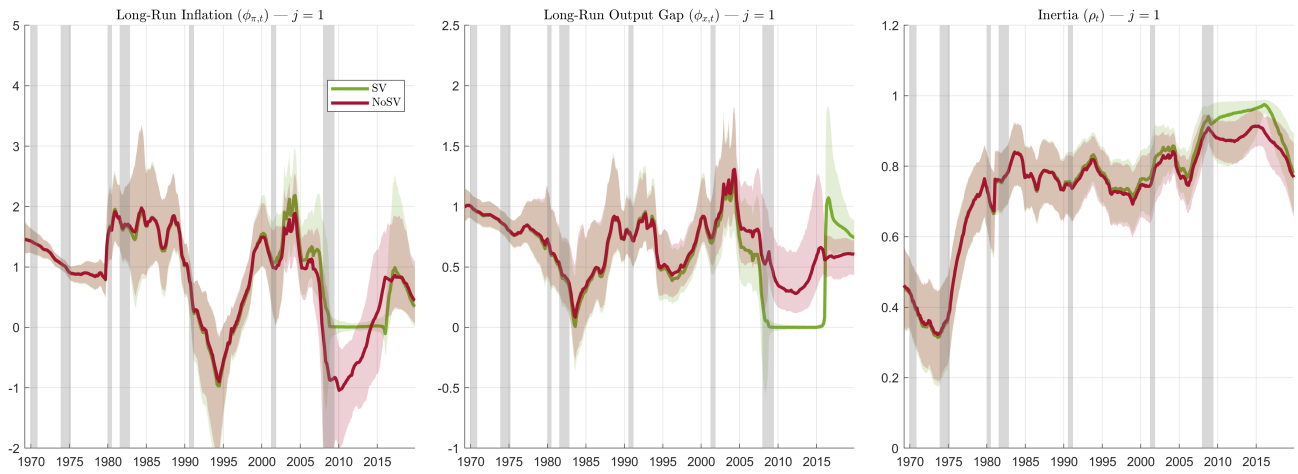


Figure D7: Robustness to ZLB Censoring: Effective Rate vs. Shadow Rate

Note: The figure compares the time-varying estimates of the actual policy rule using the effective federal funds rate (green) versus the shadow rate of Wu and Xia (2016) (red).

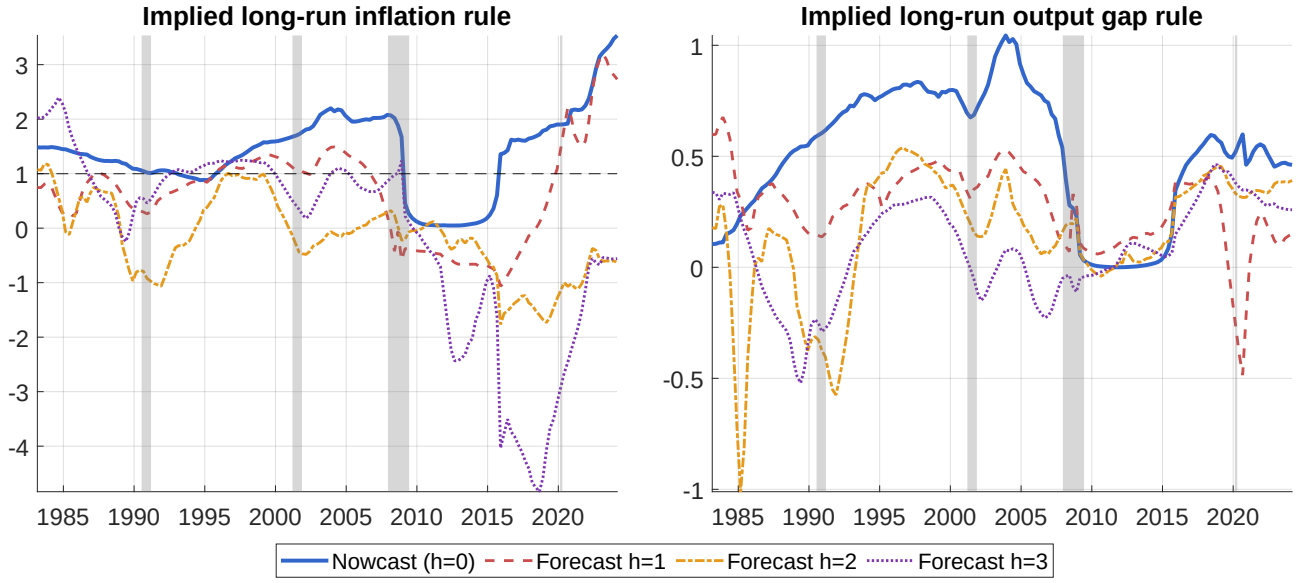


Figure D8: Implied Long-Run Perceived Rules across Forecast Horizons

Note: The figure plots the posterior median of the implied long-run perceived response coefficients to inflation (left) and the output gap (right), estimated from equation (31) at LHS horizons $h \in \{0, 1, 2, 3\}$ with the targeting horizon fixed at $j = 1$. Darker shades correspond to lower h.

perceived rules are not driven by censoring effects at the lower bound.

D.9 Horizon-Specific Implied Long-Run Rules

Section 4 compares the nowcast specification ($h = 0$) with the 2-quarter forecast specification ($h = 2$). This subsection extends that comparison across the full set of horizons $h \in \{0, 1, 2, 3\}$ to confirm that the attenuation is monotone and not driven by the particular choice of $h = 2$.

Figure D8 reports the posterior median paths of $\phi_{\pi,t,h}$ and $\phi_{x,t,h}$ as h increases from 0 to 3. The attenuation is monotone in h: each step up in the LHS horizon shifts both long-run coefficients further toward zero. The h=0 nowcast estimate sits comfortably above the Taylor principle threshold throughout the pre-ZLB sample, while the long-run inflation responses estimated at $h \geq 2$ fall below unity in extended portions of the sample. The pattern is consistent across the full horizon grid, supporting the interpretation in Section 4 that forecast-based regressions systematically understate the public's perceived policy responsiveness.

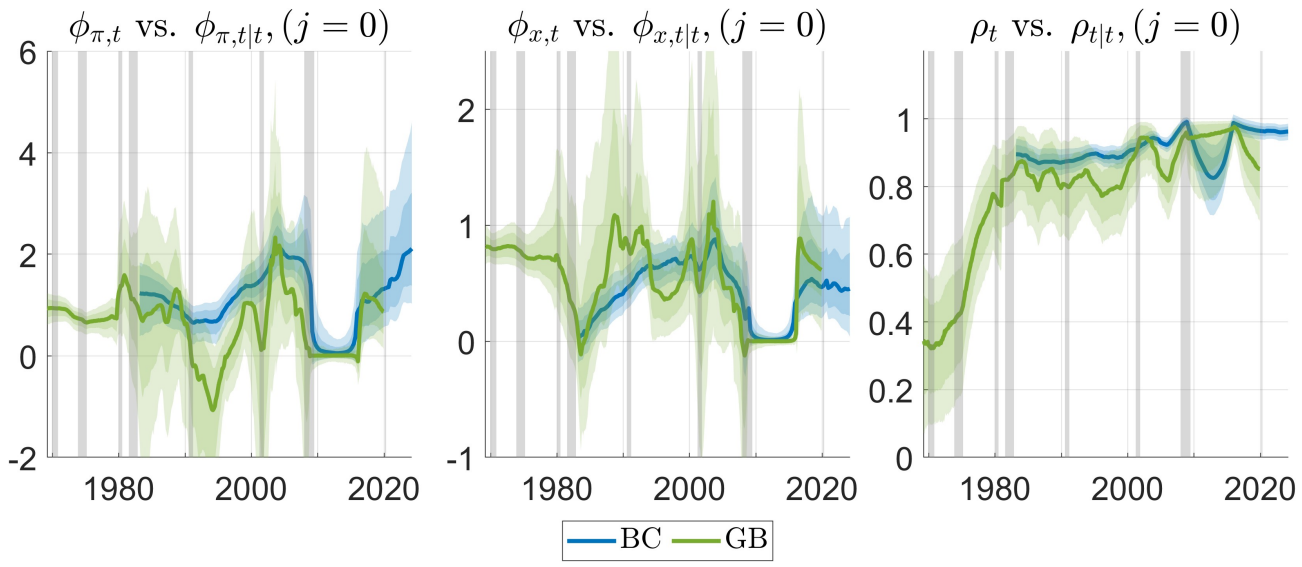
D.10 Forecasting Horizon on the RHS

Section 4 documents a systematic attenuation in the estimated policy coefficients when the interest rate *forecast* (LHS horizon $h > 0$) replaces the nowcast ($h = 0$). A natural concern is that this attenuation is mechanical rather than structural. As the horizon extends, the variance of the right-hand side variables—the inflation and output gap forecasts—declines as they revert toward their long-run means. If the variance of the dependent variable does not decline at the same rate, or if measurement error becomes relatively larger, the coefficients would be driven toward zero for purely statistical reasons.

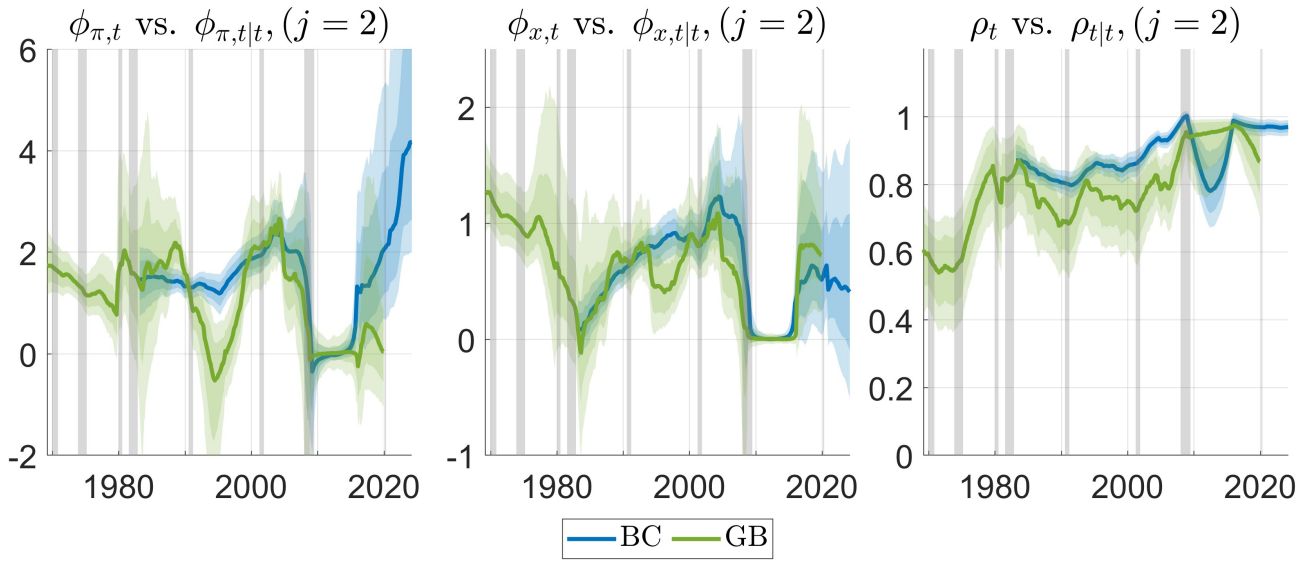
What separates that explanation from the covariance bias of Section 3 is which side of the regression the horizon is extended on, and this appendix extends it on the right-hand side only. The baseline estimation assumes a fixed targeting horizon of $j = 1$ (i.e., the central bank responds to 1-quarter ahead expectations). However, the specific horizon at which market participants believe the central bank targets inflation is an empirical question. If the assumed horizon j is misspecified, the resulting estimates of the perceived policy rule could be biased. To assess this, I re-estimate the perceived rule allowing for alternative targeting horizons while holding the dependent variable at the nowcast, so that the interest rate nowcast is regressed on progressively longer-horizon inflation and unemployment forecasts. To conserve space, I report the results for $j = 0$ (contemporaneous rule) and $j = 2$ (medium-horizon rule).

Figure D9 compares these alternative specifications. The estimated pattern of the actual and perceived coefficients remains qualitatively similar across different values of j. Notably, the divergence during the mid-1990s and the stability of the *perceived* Taylor principle during the Great Moderation are present in all specifications.

Extending the horizon on the right-hand side therefore produces no systematic attenuation: the estimates are robust whether agents are taken to project rates off current or off future inflation. The attenuation is specific to raising the horizon of the *left-hand side* variable, the interest rate expectation itself. That asymmetry is what the mechanical explanation cannot deliver and what the covariance bias of Section 3 predicts: when agents forecast the interest rate ($h > 0$), they internalize the conditional covariance between the future policy parameter and



(a) Estimation with horizon $j = 0$



(b) Estimation with horizon $j = 2$

Figure D9: Robustness to Targeting Horizon (j)

Note: The figure compares the time-varying coefficients of the Perceived Rule (Blue Chip/SPF, blue line) and the Actual Rule (Greenbook, green line) under alternative assumptions for the targeting horizon on the right-hand side (RHS) of the reaction function. Panel (a) displays estimates assuming contemporaneous targeting ($j = 0$), while Panel (b) displays estimates assuming forward-looking targeting ($j = 2$). The solid lines represent the posterior medians, and the shaded regions indicate 68% credible intervals.

future inflation, a term that is absent from the nowcast and present in the forecast regardless of the variance properties of the right-hand side variables.

Table D4: Horizon Gradient under a Common Drift Covariance

	Short-run β_π		Long-run ϕ_π^{LR}	
Horizon	Baseline	Common Σ_ν	Baseline	Common Σ_ν
$h = 0$ (nowcast)	0.129	–	1.391	–
$h = 1$	0.031	0.040	0.668	0.818
$h = 2$	−0.001	−0.011	−0.172	−0.583
$h = 3$	0.018	0.024	−0.126	0.162

Note: Sample averages of the posterior median path. *Baseline* calibrates Σ_ν on each horizon's own data. *Common* Σ_ν imposes the $h = 0$ nowcast calibration at every horizon and changes nothing else. The nowcast row is the common calibration by construction, so it has no separate entry. Every row fixes the targeting horizon at $j = 1$, as in the baseline, so the right-hand side expectations are dated $t + h + 1$. This is the specification used in Figure 5.

D.11 Horizon-Specific Tuning of Hyperparameters

The drift covariance Σ_ν is calibrated separately at each horizon, from the QLR statistic computed on that horizon's own data, and is then conditioned upon. Because the amount of time variation the estimator permits is governed by that object, estimators tuned differently across horizons could in principle generate cross-horizon differences on their own, without any covariance bias. This appendix removes that possibility by holding the tuning fixed.

Procedure The full five-state model is re-estimated at each forecast horizon on that horizon's own data, with a single change. The drift covariance is replaced by the nowcast calibration, so that every horizon is estimated under the $h = 0$ value of Σ_ν, along with the objects derived from it. The initial state prior and the stochastic volatility block are left at each horizon's own baseline values, so the drift covariance is the only difference from the baseline estimate at that horizon. If the horizon gradient is a consequence of horizon-specific tuning, imposing a common Σ_ν should remove it.

Results It does not. Table D4 reports the sample averages. Under a common drift covariance the long-run inflation response still falls from 1.39 at the nowcast to 0.82 at $h = 1$ and to -0.58 at $h = 2$, and the short-run response is still attenuated at every forecast horizon relative to the nowcast. At $h = 2$ the common-Σ_ν estimate is in fact more negative than the baseline, not less.

Figure D10 shows the paths. The common-Σ_ν estimates track the baseline estimates at their own horizon closely and stay far below the nowcast benchmark throughout, in particular over the post-2008 period where the gap is widest. The gradient documented in Section 4 therefore survives the removal of horizon-specific tuning, which is what the covariance interpretation requires and what a pure tuning artifact would not deliver.

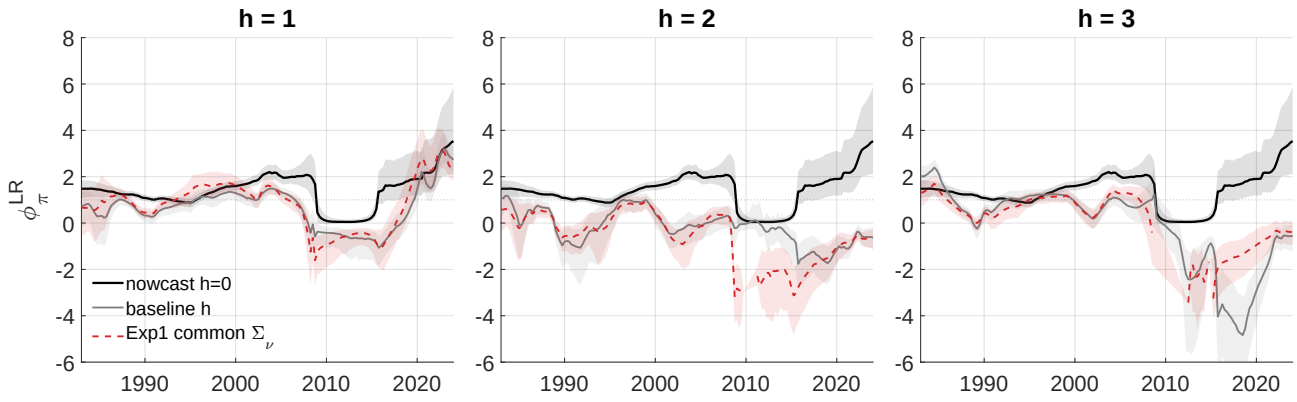


Figure D10: Long-Run Inflation Response under the same hyperparameter

Note: Posterior median long-run inflation response with 68% bands, by forecast horizon. Black is the nowcast benchmark, repeated in every panel. Grey is the baseline estimate at that horizon and red is the same specification estimated under the nowcast drift covariance. The dotted line marks unity.

D.12 Near-Collinear Inertia Regressors

In the interest rate forecast TVP regressions, the dependent variable $i^{BC}_{t+h|t}$ and the two inertia regressors $i^{BC}_{t+h-1|t}$ and $i^{BC}_{t+h-2|t}$ are three points on the same smooth forecasted rate path, so they are nearly collinear. The lag terms may then absorb most of the variation in the dependent variable, leaving $\beta_{\pi,h}$ identified off residual noise. This could in principle generate the observed attenuation on its own.

Procedure To address this issue, I limit the variation explaned by lags by fixing the inertia parameters $(\bar{\rho}_{1,t}, \bar{\rho}_{2,t})$. They are moved to the left-hand side, and the reduced three-state system is estimated on the transformed dependent variable,

$$i^{BC}_{t+h|t} - \bar{\rho}_{1,t}\, i^{BC}_{t+h-1|t} - \bar{\rho}_{2,t}\, i^{BC}_{t+h-2|t} \;=\; \beta_{\pi,h}\, \pi^{BC}_{t+h+1|t} + \beta_{x,h}\left(-2 \cdot u^{BC}_{t+h+1|t}\right) + c_h + \varepsilon_{t,h}. \tag{D.5}$$

With the lags known and subtracted they can no longer compete for the variation, so the near-collinearity channel is shut down by construction. I set $\bar{\rho}_{2,t} = 0$, so total inertia equals $\bar{\rho}_{1,t}$ and the long-run amplification $1/(1-\bar{\rho}_{1,t})$ is a known constant at every date. Two pins are reported: a flat $\bar{\rho}_{1,t} = 0.85$ over the full sample, and a two-step $\bar{\rho}_{1,t} = 0.80$ before 2009Q1 and 0.90 thereafter, which brackets the nowcast inertia path. As in Appendix D.11, the hyperparameters are held at the $h = 0$ nowcast calibration in these runs, so the left-hand side horizon is the only thing that differs across them.

Results Figure D11 reports the paths and the attenuation survives. Averaging the posterior median over the sample, the long-run inflation response under the flat pin is 0.59, 0.38, and 0.41 at $h = 1, 2, 3$, and under the two-step pin 0.59, 0.39, and 0.44, against 1.39 for the nowcast

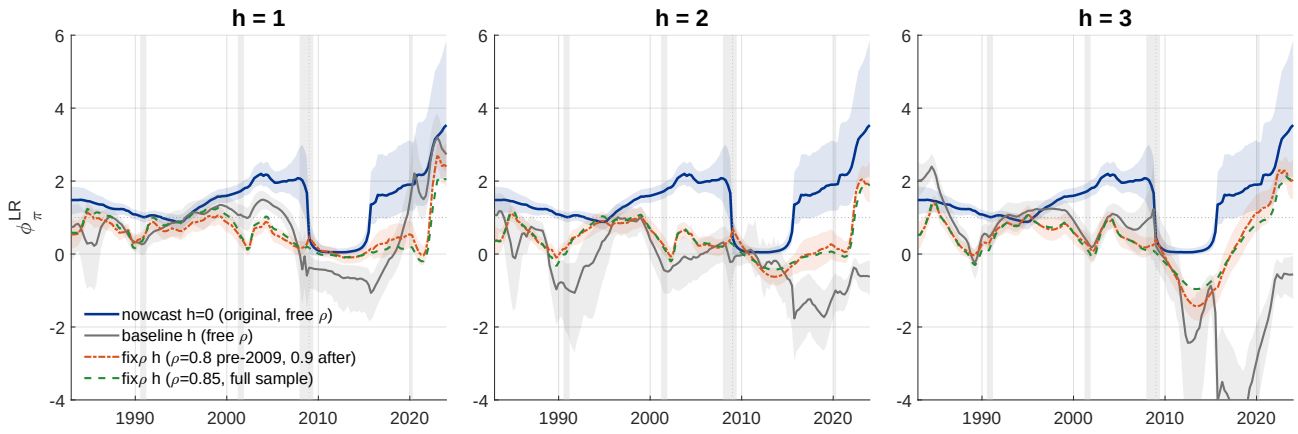


Figure D11: Long-Run Inflation Response with Inertia Fixed

Note: Posterior median long-run inflation response $\phi_\pi^{LR} = \beta_{\pi,h}/(1-\rho)$ by forecast horizon, with 68% bands. Each line is generated as follows. *Navy*: the baseline nowcast ($h = 0$) with inertia freely estimated, repeated unchanged in every panel as the benchmark. *Grey*: the baseline estimate at that horizon, also with inertia freely estimated. *Orange*: equation (D.5) at that horizon with inertia pinned at 0.80 before 2009Q1 and 0.90 after. *Green*: the same, with inertia pinned at 0.85 over the full sample; plotted without a band to keep the panel legible. The two pinned lines therefore differ from the grey line only in that inertia is imposed rather than estimated. Shaded vertical bars are NBER recessions and the dotted line marks unity.

benchmark. The two pins agree to within 0.03 at every horizon, so the result does not depend on where inertia is fixed or on whether it is allowed a break at the ZLB. What the pinning does change is the behavior of the estimates, not their level: the free-inertia paths at $h = 2$ and $h = 3$ wander below zero and, at $h = 3$, reach -4.84, because $1/(1-\rho)$ diverges where the estimated inertia approaches unity. Fixing inertia bounds that amplification and removes the instability, and the pinned paths still sit below unity for most of the post-2008 sample. Removing the near-collinear lag terms therefore does not restore the perceived Taylor principle at forecast horizons.

D.13 Monthly Frequency of Estimation

The baseline perceived rules were estimated at a quarterly frequency by stacking monthly Blue Chip surveys with quarterly SPF data. A potential concern is that this temporal aggregation might mask high-frequency updates in perceptions or artificially smooth transitions within a quarter. To ensure that the quarterly aggregation does not distort the timing or magnitude of the perceived regime shifts, I re-estimate the model at a monthly frequency using only the disjoint BCEI monthly consensus forecasts.

Figure D12 plots the resulting monthly time series of the perceived inflation coefficient against the quarterly baseline. The monthly estimates closely track the quarterly results, capturing the same turning points and structural shifts. This congruence confirms that the quarterly stacked approach efficiently utilizes all available information without inducing much

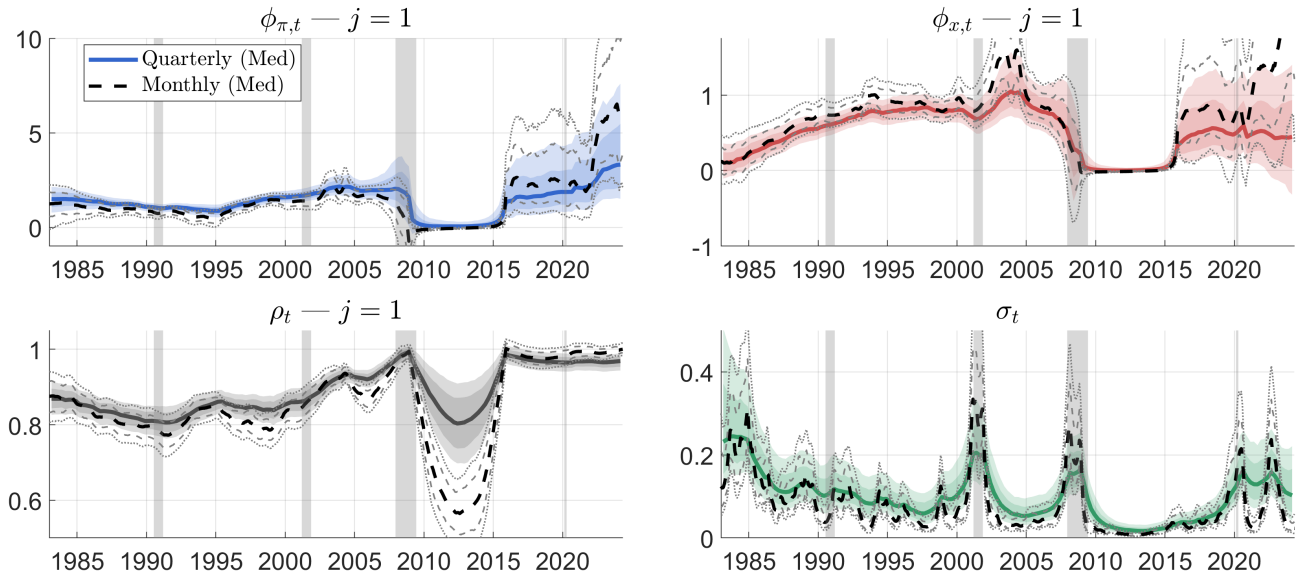


Figure D12: Robustness to Estimation Frequency: Quarterly vs. Monthly

Note: The figure compares the estimated long-run inflation coefficient of the perceived rule obtained from the baseline quarterly stacked model (blue) against the estimates obtained from a purely monthly model using Blue Chip data (red).

aggregation bias.